



JOINT CANADA-UNITED STATES
NATIONAL STANDARD

ANSI/CAN/UL 2743:2025

STANDARD FOR SAFETY

Portable Power Packs



SCC FOREWORD

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UL Standard for Safety for Portable Power Packs, ANSI/CAN/UL 2743

Third Edition, Dated September 24, 2025

Summary of Topics

This new Third Edition of ANSI/CAN/UL 2743 dated September 24, 2025 includes the following changes:

- ***Addition of UPS standard for power pack with AC pass through function; Section [4](#), [5.2](#), [5.42](#), [12.1.3](#), [13.3.2](#), [15.1.1](#), [16.1](#), [16.2](#), [31.1](#), [32.1](#), and [49.1](#)***
- ***Construction clarification for PV input with hazardous voltage level or hazardous energy level; [12.5.1](#)***
- ***Revision/clarification of protective bonding through trace of Printed-wiring board; Section [4](#) and [15.1.12](#)***
- ***Addition of exception to [30.3.3](#) and [30.3.4](#), and Section [4](#)***
- ***Pollution degree for outdoor/indoor use power packs; [31.6](#)***
- ***Clarification of back feed requirements (both construction and test); Section [4](#), [35.2](#), and [71.1](#)***
- ***Revision to the drop test for power packs with wheels for movement; [60.3.2](#)***
- ***Clarification to the requirement for detachable or external battery packs ; [1.1](#), [1.7](#), [1.8](#), Section [4](#), [5.5](#), [5.16](#), [5.20](#), [5.28](#), [5.35](#), [6.5](#), [6.6](#), Section [9](#), [12.1.2](#), [13.2.1.4](#), [30.1.2](#) – [30.1.7](#), Section [30.4](#), [35.1](#), [44.1](#), [50.2.8](#), Section [55.10](#), [55.11.1](#), [55.11.2](#), [55.11.3](#) and [71.2](#)***
- ***Cord and Plug Connection; Section [4](#), [5.14](#), [5.37](#), [7.1.2](#), Section [12.2.1](#), [13.3.2](#), [74.1](#), and [78.2](#)***
- ***Addition of Valve Regulated Nickel and Sodium-ion battery; [1.1](#), [1.8](#), [Table 1.1](#), Section [4](#), [5.33](#), [5.43](#), [5.44](#), [5.45](#), [5.47](#), [5.48](#), [7.1.4](#), [30.1.2](#), [30.2.1](#), [30.3.2](#), [30.3.4](#), [44.6](#), [46.1](#), [70.1](#), [74.1](#), [75.11](#), [77.3](#)***
- ***Addition of requirement for backpacked portable power pack; [1.5](#), [40.1](#), Section [41](#), [Table 42.1](#), [62.1](#), [62.2](#), [77.3](#)***
- ***Peak discharge current rating test; [50.3.1](#), [70.1](#) – [70.4](#), [Figure 70.1](#), [74.1](#)***

The new and revised requirements are substantially in accordance with Proposal(s) on this subject dated October 4, 2024 and May 30, 2025.

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Standard for Portable Power Packs

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This ANSI/UL Standard for Safety consists of the Third Edition.

The most recent designation of ANSI/UL 2743 as an American National Standard (ANSI) occurred on September 24, 2025. ANSI approval for a standard does not include the Cover Page, Transmittal Pages, Title Page, Preface, or SCC Foreword.

This Standard has been designated as a National Standard of Canada (NSC) on September 24, 2025.

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Annex A – Safety Marking Translations (Normative for Canada and Informative for the US)

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Preface

This is the Third Edition of ANSI/CAN/UL 2743, Standard for Safety for Portable Power Packs.

ULSE is accredited by the American National Standards Institute (ANSI) and the Standards Council of Canada (SCC) as a Standards Development Organization (SDO).

This Standard has been developed in compliance with the requirements of ANSI and SCC for accreditation of a Standards Development Organization.

This ANSI/CAN/UL 2743 Standard is under continuous maintenance, whereby each revision is approved in compliance with the requirements of ANSI and SCC for accreditation of a Standards Development Organization. In the event that no revisions are issued for a period of four years from the date of publication, action to revise, reaffirm, or withdraw the standard shall be initiated.

Annex **A** is identified as Normative for Canada and Informative for the US. Informative text is for informational purposes only, and Normative text is considered to be mandatory.

In Canada, there are two official languages, English and French. All safety warnings must be in French and English. Attention is drawn to the possibility that some Canadian authorities may require additional markings and/or installation instructions to be in both official languages.

This Third Edition joint American National Standard and National Standard of Canada is based on, and now supersedes, the Second Edition of ANSI/CAN/UL 2743.

Comments or proposals for revisions on any part of the Standard may be submitted at any time. Proposals should be submitted via a Proposal Request in the Collaborative Standards Development System (CSDS) at <https://csds.ul.com>.

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This Edition of the Standard has been formally approved by the Technical Committee (TC) for Portable Power Packs, TC 2743.

This list represents the TC 2743 membership when the final text in this Standard was balloted. Since that time, changes in the membership may have occurred.

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This Standard is intended to be used for conformity assessment.

The intended primary application of this Standard is stated in its scope. It is important to note that it remains the responsibility of the user of the standard to judge its suitability for this particular application.

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INTRODUCTION

1 Scope

1.1 These requirements cover portable and movable power packs provided with one or more batteries, electrochemical capacitors, or electrochemical capacitor modules, to provide power to different types of appliances or equipment on site. The batteries or electrochemical capacitors shall be internal components of the power pack, external batteries may be connected to the power pack to expand the capacity of the power pack. The power packs are provided with one or more inputs and one or more outputs. For power packs provided with a booster function, the power packs are used for providing a temporary power source to a depleted land vehicle battery, rated 24 V dc maximum, to provide emergency starting power.

1.2 These requirements cover power packs suitable for outdoor use, temporary outdoor use, or indoor use only. Outdoor use packs are intended to be used outdoors with no restrictions. Temporary outdoor use packs are intended to be used outdoors in limited wet conditions and always stored indoors. Indoor use only packs are intended to be stored indoors and used indoors and are not intended to be used outdoors at any time. A power pack with a booster function is not considered indoor use only under any conditions.

1.3 These requirements cover power packs provided with additional systems such as an air compressor (tankless type) for inflating tires or other inflatable items, or with a light to act as warning lights to oncoming traffic, as a flashlight, or the like. These functions are also powered by the internal battery.

1.4 These requirements cover the power pack options such as lights, voltmeters, internal air compressor assemblies, associated gauges, inverters, vehicle adapters, and internal batteries, as well as the charging of the internal batteries, when these options are integral to the power pack.

1.5 These requirements cover the portable power packs integrated or attached in luggage, carrying cases, bags, or backpacks. These requirements do not cover the performance of luggage, carrying cases, bags or backpacks.

1.6 These requirements do not cover wiring or cabling used in the recharging function of electric vehicle recharging equipment.

1.7 These requirements do not cover power banks which are covered in the Standard for Power Banks, UL 2056.

1.8 These requirements do not cover power packs having a capacity exceeding the limits specified in [Table 1.1](#), which are covered by the Standard for Energy Storage Systems and Equipment, UL 9540.

Table 1.1
Portable Power Pack Threshold Quantities

Portable power pack technology	Aggregate capacity	
	kWh	MJ
Lead-acid or valve regulated nickel, all types	70	252
Lithium-ion or sodium-ion, all types	20	72
Electrochemical Capacitors	3	10.8

1.9 These requirements do not cover power packs that are permanently connected to a premise wiring system, or having a function to supply power to a premise wiring system, or have utility interactive function, which are covered by the Standard for Energy Storage Systems and Equipment, UL 9540.

2 Units of Measurement

2.1 The values given in SI (metric) units shall be normative. Any other values are for information only.

2.2 Values stated without parentheses are the requirement. Values within parentheses are considered explanatory or approximate information.

3 Components

3.1 Except as indicated in [3.2](#), a component of a product covered by this Standard shall comply with the requirements for that component.

3.2 A component is not required to comply with a specific requirement that:

- a) Involves a feature or characteristic not required in the application of the component in the product covered by this Standard; or
- b) Is superseded by a requirement in this Standard.

3.3 A component shall be used in accordance with its ratings established for the intended conditions of use.

3.4 Specific components are incomplete in construction features or restricted in performance capabilities. Such components are intended for use only under limited conditions, such as certain temperatures not exceeding specified limits, and shall be used only under those specific conditions.

4 Referenced Publications

4.1 Any undated reference to a code or standard appearing in the requirements of this Standard shall be interpreted as referring to the latest edition of that code or standard.

4.2 Products covered by this Standard shall comply with the referenced installation codes and standards noted in this Section as appropriate for the country where the product is to be used. When the product is intended for use in more than one country, the product shall comply with the installation codes and standards for all countries where it is intended to be used.

4.3 The following publications are referenced in this Standard:

ASTM A90, *Standard Test Method for Weight [Mass] of Coating on Iron and Steel Articles with Zinc or Zinc-Alloy Coatings*

ASTM E230, *Standard Specification and Temperature-Electromotive Force (emf) Tables for Standardized Thermocouples*

ASTM E376, *Standard Practice for Measuring Coating Thickness by Magnetic-Field or Eddy-Current (Electromagnetic) Testing Methods*

ASTM D412, *Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers – Tension*

ASTM B568, *Standard Test Method for Measurement of Coating Thickness by X-Ray Spectrometry*

ASTM A653, *Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process*

CSA C22.2 No. 0.15, *Adhesive Labels*

CSA C22.2 No. 0.17, *Evaluation of Properties of Polymeric Mate*

CSA C22.2 No. 0.2, *Insulation Coordination*

CSA C22.2 No. 0.4, *Bonding of Electrical Equipment*

CSA C22.2 No. 0.23, *General Requirements for Battery-Powered Appliances*

CSA C22.2 No. 42, *General Use Receptacles, Attachment Plugs, and Similar Wiring Devices*

CSA C22.2 No. 66.1, *Low Voltage Transformers – Part 1: General Requirements*

CSA C22.2 No. 66.2, *Low Voltage Transformers – Part 2: General Purpose Transformers*

CSA C22.2 No. 66.3, *Low Voltage Transformers – Part 3: Class 2 and Class 3 Transformers*

CSA C22.2 No. 68, *Motor Operated Appliances (Household and Commercial)*

CSA C22.2 No. 77, *Motors With Inherent Overheating Protection*

CSA C22.2 No. 107.1, *Power Conversion Equipment*

CSA C22.2 No. 107.2, *Battery Chargers*

CSA C22.2 No. 107.3, *Uninterruptible Power Systems*

CSA C22.2 No. 182.3, *Special Use Attachment Plugs, Receptacles, and Connectors*

CSA C22.2 No. 197, *PVC Insulating Tape*

CSA C22.2 No. 127, *Equipment and Lead Wires*

CSA C22.2 No. 210, *Appliance Wiring Material Products*

CSA C22.2 No. 223, *Power Supplies With Extra Low Voltage Class 2 Outputs*

CSA C22.2 No. 60335-2-29, *Household and Similar Electrical Appliances – Safety – Part 2-29: Particular Requirements for Battery Chargers*

CSA C22.2 E60730-1, *Automatic Electrical Controls – Part 1: General Requirements*

CSA C22.2 No. 60950-1, *Information Technology Equipment – Safety – Part 1: General Requirements*

CSA C22.2 No. 62133-2, *Secondary Cells and Batteries Containing Alkaline and Non-Acid Electrolytes – Safety Requirements for Portable Sealed Secondary Cells, and for Batteries Made From Them, for Use in Portable Applications – Part 2: Lithium Systems*

CSA C22.2 No. 62368-1, *Audio/Video, Information and Communication Technology Equipment – Part 1: Safety Requirements*

SAE J1494, *Battery Booster Cables*

UL 62, *Flexible Cords and Cables*

UL 94, *Tests for Flammability of Plastic Materials for Parts in Devices and Appliances*

UL 101, *Leakage Current for Appliances*

UL 498, *Attachment Plugs and Receptacles*

UL 506, *Specialty Transformers*

UL 510, *Polyvinyl Chloride, Polyethylene, and Rubber Insulating Tape*

UL 635, *Insulating Bushings*

UL 746C, *Polymeric Materials – Use in Electrical Equipment Evaluations*

UL 758, *Appliance Wiring Material*

UL 796, *Printed Wiring Boards*

UL 810A, *Electrochemical Capacitors*

UL 817, *Cord Sets and Power-Supply Cords*

UL 840, *Insulation Coordination Including Clearance and Creepage Distances for Electrical Equipment*

UL 943, *Ground-Fault Circuit Interrupters*

UL 969, *Marking and Labeling Systems*

UL 969A, *Marking and Labeling Systems – Flag Labels, Flag Tags, Wrap-Around Labels and Related Products*

UL 1004-3, *Thermally Protected Motors*

UL 1012, *Power Units Other Than Class 2*

UL 1097, *Double Insulation Systems for Use in Electrical Equipment*

UL 1236, *Battery Chargers for Charging Engine-Starter Batteries*

UL 1310, *Class 2 Power Units*

UL 1437, *Electrical Analog Instruments – Panel Board Types*

UL 1450, *Motor-Operated Air Compressors, Vacuum Pumps, and Painting Equipment*

UL 1561, *Dry-Type General Purpose and Power Transformers*

UL 1581, *Reference Standard for Electrical Wires, Cables, and Flexible Cords*

UL 1642, *Lithium Batteries*

UL 1741, *Inverters, Converters, Controllers and Interconnection System Equipment for Use With Distributed Energy Resources*

UL 1778, *Uninterruptible Power Systems*

UL 1839, *Automotive Battery Booster Cables*

UL 1973, *Batteries for Use in Stationary and Motive Auxiliary Power Applications*

UL 1977, *Component Connectors for Use in Data, Signal, Control and Power Applications*

UL 1989, *Valve Regulated or Vented Batteries with Aqueous Electrolytes*

UL 2056, *Power Banks*

UL 2089, *Vehicle Battery Adapters*

UL/ULC 2580, *Batteries for Use In Electric Vehicles*

UL 2595, *General Requirements for Battery-Powered Appliances*

UL 5085-1, *Low Voltage Transformers – Part 1: General Requirements*

UL 5085-2, *Low Voltage Transformers – Part 2: General Purpose Transformers*

UL 5085-3, *Low Voltage Transformers – Part 3: Class 2 and Class 3 Transformers*

UL 9540, *Energy Storage Systems and Equipment*

UL 60335-2-29, *Household and Similar Electrical Appliances – Safety – Part 2-29: Particular Requirements for Battery Chargers*

UL 60730-1, *Automatic Electrical Controls – Part 1: General Requirements*

UL 60950-1, *Information Technology Equipment – Safety – Part 1: General Requirements*

UL 62109-1, *Power Converters for Use in Photovoltaic Power Systems – Part 1: General Requirements*

UL 62133-2, *Secondary Cells and Batteries Containing Alkaline or Other Non-Acid Electrolytes – Safety Requirements for Portable Sealed Secondary Cells, and for Batteries Made From Them, for Use in Portable Applications – Part 2: Lithium Systems*

UL 62368-1, *Audio/Video, Information and Communication Technology Equipment – Part 1: Safety Requirements*

5 Glossary

5.1 For the purpose of this Standard, the following definitions apply:

5.2 AC PASS THROUGH – A branch circuit or generator to which the load is connected when the power conversion portion of the inverter is not supplying power to the load.

5.3 ACCESSIBLE – Able to be contacted by an accessibility probe.

5.4 BATTERY – An electrical storage unit consisting of two or more cells.

5.5 BOOSTER CABLE – The conductors that originate at the power pack's battery and terminate at the clamps that are intended to be temporarily connected to the land vehicle's battery. If the cable is connected through a switch and therefore does not have a continuous path to the exterior of the power pack's enclosure, then the definition shall apply to the portion of cable that exits the power pack's enclosure.

5.6 BOOSTER FUNCTION – When the power packs are used for providing a temporary power source to a depleted land vehicle battery, rated 24 V maximum, to provide emergency starting power.

5.7 CAPACITOR MODULE – A single or multiple series and/or parallel-connected, electrochemical capacitors with associated circuitry.

5.8 CHARGING SYSTEM – Combination of circuitry intended to charge, balance, and/or maintain the state of charge of the battery.

5.9 CLAMP – Devices located at the ends of the power pack's booster cables, and which contain the jaw ends, connectors, and handle ends, which are used to make the temporary connections to the land vehicles battery.

5.10 CLAMP CONNECTORS – Portion of the clamp that contacts the battery terminals.

5.11 CLASS 2 POWER SUPPLY – A source having limited voltage and energy capacity. Requirements for voltage and energy capacity limitations are as indicated in this document.

5.12 CLEARANCE – Shortest distance between two conductive parts, or between conductive parts and the conductive dead metal parts of the enclosure, measured through air.

5.13 CONDUCTORS – Wiring used internally and externally on the power pack.

5.14 CORD SET – A length of flexible cord having an attachment plug at one end and a cord connector at the other end.

5.15 CREEPAGE DISTANCE – Shortest distance between two conductive parts, or between conductive parts and the conductive dead metal parts of the enclosure, measured along the surface of insulating material.

5.16 DETACHABLE BATTERY – Battery which is contained in a separate enclosure from the portable power pack and is intended to be mounted onto the portable power pack and can be removed from the portable power pack without the aid of a general-purpose tool.

NOTE: A battery that is snap-fit onto the power pack is considered as a detachable battery.

5.17 ELECTROCHEMICAL CAPACITOR – An electric energy storage device where electrical charge is stored as a result of non-Faradaic processes at one or both of the electrodes. (A subset of electrochemical capacitors referred to as an "asymmetric electrochemical capacitor" have non-Faradaic processes at one electrode and Faradaic processes at the other electrode.) The unique highly-porous electrode increases

its surface area for holding charge resulting in much larger capacitance and energy density than other types of capacitors. Electrochemical capacitors differ from common electrolytic capacitors in that they store charge at the liquid-solid interface of the electrodes when a potential is applied rather than in a solid dielectric material covering the surfaces of the electrodes. Some other common names for an electrochemical capacitor are “double layer capacitor”, “ultra-capacitor”, “electrochemical double layer capacitor”, “super-capacitor”, and EDLC.

5.18 ENCLOSURE – That part of the product that:

- a) Renders inaccessible all or any parts of the equipment that may otherwise present a risk of electric shock or electric energy hazard; and
- b) Retards propagation of flame initiated by electrical disturbances occurring within.

5.19 EXPOSED – Visible but not necessarily able to be contacted by an accessibility probe.

5.20 EXTERNAL BATTERY – Battery which is contained in a separate enclosure from the portable power pack, is connected to the portable power pack by a cord detachable at one or both ends, and is not intended to be mounted on the portable power pack.

NOTE: A battery pack stacked but not mounted on the power pack, and connected through a power pack through a detachable cord are considered as an external battery.

5.21 FULLY CHARGED BATTERY – A battery charged to the maximum state of charge permitted by the battery charging system intended for use with the appliance and in accordance with the manufacturer's instructions.

5.22 FULLY DISCHARGED BATTERY – A battery that has been discharged at the manufacturer's recommended rate until the battery reaches a total voltage with an average voltage per cell equal to the end discharge voltage for the cell chemistry being used, or as specified by the manufacturer.

5.23 FUNCTIONAL EARTHING – Earthing a point or points in a system or in an installation or in equipment, for purposes other than electrical safety.

5.24 HANDLE END – Portion of the clamp that is held by the user and used to open the jaw ends to allow for the temporary connection to the positive battery terminal and to the land vehicle's grounded chassis.

5.25 HAZARDOUS ENERGY LEVEL – A potential of 2 volts or more and either an available continuous power level of 240 volt-amperes or more, or a reactive energy level of 20 joules or more, between a live part and an adjacent dead metal part or between live parts of different polarity.

5.26 HAZARDOUS VOLTAGE LEVEL – For temporary outdoor and outdoor use applications, hazardous voltage level is the voltage having a value exceeding 30 V dc or 21.2 V peak AC. For indoor use only applications, hazardous voltage level is the voltage having a value exceeding 60 V dc, or 42.4 V peak AC.

5.27 INSULATION – Protective covering of the conductors, cable, and clamps.

5.28 INTERNAL BATTERY – The battery assembled in whole or in part internal to the power pack's enclosure, or permanently fixed onto the power pack, which is used to provide the temporary power output of the power pack.

5.29 JAW END – Portion of the clamp that opens in order to accommodate the land vehicle's battery terminals.

5.30 LAND VEHICLE – An automobile, light duty truck, or other vehicle having a 12 or 24 volt dc electrical system.

5.31 LEAKAGE CURRENT – Electric current which flows through a person upon contact between accessible parts of a power pack and ground or other accessible parts of the power pack.

5.32 LOW VOLTAGE, LIMITED ENERGY (LVLE) – A circuit involving a potential below the hazardous voltage level in [5.26](#) and supplied by a low voltage battery, a Class 2 source, or a combination of an isolating transformer and one or more resistors or a regulating network, or overcurrent protective device that, as a unit, limits the available energy as described in [52.1](#).

5.33 MAXIMUM CHARGING CURRENT – Highest current that cell or battery is permitted to pass during charging for a specified range of temperatures, specified by the cell or battery manufacturer.

5.34 MIDDLE CONNECTOR – A connector located in the middle of the cables that is disconnected prior to the clamps being connected to the vehicle battery and then used to complete the connection once the clamps are connected to the respective vehicle battery terminals.

5.35 PORTABLE POWER PACK – A power pack which can be lifted and moved by a single person without mechanical aids and not permanently connected to an electrical system. It can be either:

a) Hand-held; or

b) Moveable and meets the following criteria:

1) 18 kg (39.7 pounds) or less in mass and is not fixed; or

2) 23 kg (50.7 pounds) or less in mass and provided with wheels, casters, two handles, or other means to facilitate movement by an ordinary person as required to perform its intended use.

5.36 POWER PACK – A portable or movable device that contains an integral or removable battery, or batteries, that when charged are intended to provide power to various outputs of the device, including booster functions for providing temporary emergency power to land vehicles, or power to receptacles and other outputs. All can be further designated as outdoor use, temporary outdoor use, or indoor use only.

5.37 POWER-SUPPLY CORD – A length of flexible cord with an attachment plug at one end and individual cord conductors not terminated in a cord connector at the opposite end.

5.38 PROTECTIVE DEVICE – A device that operates in a specific manner to prevent a hazardous situation under abnormal operation conditions.

5.39 RAINPROOF ENCLOSURE – An enclosure that provides protection to personnel against electric shock or electric energy hazards by preventing a degree of water entering inside of the portable power pack during the Rain Test.

5.40 RAIN-TIGHT ENCLOSURE – An enclosure that completely prevents water entering inside of the portable power pack during the Rain Test.

5.41 REPAIR FACILITY – A building or portion of a building where automotive repairs and maintenance work are conducted on gasoline powered land vehicles, whether only a portion of the land vehicles or all the land vehicles are gasoline powered.

5.42 RISK OF ELECTRIC SHOCK – A risk of electric shock is considered likely to occur at any part if the potential between the part and earth ground or any other accessible part is more than the hazardous voltage level as defined in [5.26](#), and the continuous current flow through a 1500 ohm resistor exceeds 5 milliamperes.

5.43 SODIUM-ION CELLS – Cells that are similar in construction to lithium-ion cells except they utilize sodium as the ion of transport with a positive electrode consisting of a sodium compound, and carbon or similar type anode with an aqueous or non-aqueous electrolyte and with a sodium compound salt dissolved in the electrolyte.

5.44 SPECIFIED OPERATING REGION – Range of permissible operation expressed by lithium-ion or sodium-ion cell parameter limits.

5.45 SPECIFIED OPERATING REGION FOR CHARGING – Conditions during charging in which the lithium-ion or sodium-ion cell operates within its voltage and current range as specified by the cell manufacturer and evaluated in accordance with the Standard in [30.3](#). In general, the permissible range of voltage and current are dependent upon temperature.

5.46 UNINSULATED LIVE PART – A conductive part within the device that is not provided with integral insulation and during intended use has a potential difference with respect to earth ground or any other conductive part.

5.47 UPPER LIMIT CHARGING VOLTAGE – Highest voltage that a lithium-ion or sodium-ion cell is permitted to attain during normal charging for a specified range of temperatures as specified by the cell manufacturer and evaluated in accordance with the Standard in [30.3](#).

5.48 VALVE REGULATED NICKEL BATTERY – An alkaline battery in which the positive active material is nickel oxide and the negative contains cadmium or zinc and the venting of the products of electrolysis is controlled by a reclosing pressure sensitive valve.

CONSTRUCTION

6 General

6.1 If the operation and maintenance of a power pack by the user involves a risk of injury to persons, a risk of electric shock, or a risk of fire, means shall be provided to reduce the risk. When evaluating a power pack, consideration shall be given to reasonably foreseeable misuse of the product.

6.2 Power packs intended for use within a repair facility, and marked as such as indicated in [74.4](#), shall be provided with instructions containing the statement in [79.3](#) and shall be marked as shown in [75.17](#). Power packs that are not intended for use in a repair facility shall be marked in accordance with [75.18](#).

6.3 Outdoor use power packs shall be evaluated for all environmental considerations addressed by this Standard and are intended to be used and stored either outdoors or indoors. Temporary outdoor use power packs shall be evaluated for exposure to rain, shall be marked in accordance with [75.19](#) and [75.20](#), and shall be provided with instructions in accordance with [79.5](#). Indoor use only power packs shall be marked in accordance with [75.21](#) and shall be provided with instructions in accordance with [79.6](#). Indoor use only packs need not comply with the environmental considerations in [7.5](#).

6.4 For power packs not marked in accordance with [75.22](#), the device shall be subjected to the Vibration Test, Section [56](#).

6.5 A portable power pack to be moved and lifted shall not exceed the weight limit specified in [5.35](#). Each individual external battery pack or accessory of the power pack that is intended to be moved and lifted separately shall also comply with the portable or moveable criteria specified in [5.35](#).

6.6 With reference to [6.5](#), the weight of all detachable battery packs shall be included in the total weight of the power pack.

Exception: The weight of some or all detachable battery packs may be excluded from the total weight of the power pack if the power pack is marked to indicate that these battery packs must be removed before lifting. In such cases, each individual detachable battery pack shall also comply with the portable or moveable criteria specified in [5.35](#).

7 Frame and Enclosure

7.1 General

7.1.1 An enclosure of a power pack shall be formed and assembled so that it has the strength and rigidity necessary to resist the abuses to which it may be subjected, without increasing the risk of fire, electric shock, or injury to persons due to total or partial collapse with resulting reduction of spacings, loosening or displacement of parts, or other serious defects.

7.1.2 A power pack shall be provided with an enclosure suitable for the application. The enclosure shall house all live parts that may increase the risk of fire, electric shock, or injury to persons under any condition of use. This requirement does not apply to the cord set, power-supply cord, the output leads, or output terminals.

7.1.3 If an electrical instrument, such as a meter, forms part of the enclosure, the face or the back of the meter, or both together, shall comply with the requirements for enclosures.

7.1.4 The enclosure for a power pack provided with an internal lead acid battery or valve regulated nickel battery shall be ventilated to permit dispersion of gases from the internal battery during normal use and charging. See [7.4](#).

7.1.5 Enclosures for power packs shall comply with the requirements for environmental considerations in [7.5](#).

7.2 Metallic enclosures

7.2.1 Enclosures constructed of metallic materials shall be subjected to the Strength of Enclosure Tests, Section [60](#).

7.2.2 A metallic enclosure shall have a thickness not less than that specified in [Table 7.1](#) and [Table 7.2](#).

7.2.3 Metallic enclosures, if not constructed of a material that is corrosion resistant, shall be provided with corrosion protection in accordance with Corrosion Protection, Section [11](#).

Table 7.1
Minimum Thickness of Sheet Metal for Enclosure Carbon Steel or Stainless Steel

Without supporting frame ^a		With supporting frame or equivalent reinforcing ^a		Minimum thickness uncoated	Minimum thickness coated
Maximum width ^b cm (inches)	Maximum length ^c cm (inches)	Maximum width ^b cm (inches)	Maximum length ^c cm (inches)		
10.2 (4)	Not limited	15.9 (6.25)	Not limited	0.51 (0.020)	0.58 (0.023)
12.1 (4.75)	14.6 (5.75)	17.1 (6.75)	21.0 (8.25)		
15.2 (6)	Not limited	24.1 (9.5)	Not limited	0.66 (0.026)	0.74 (0.029)
17.8 (7)	22.2 (8.75)	25.4 (10)	31.8 (12.5)		
20.4 (8)	Not limited	30.5 (12)	Not limited	0.81 (0.032)	0.86 (0.034)
22.9 (9)	29.2 (11.5)	33.0 (13)	40.6 (16)		
31.8 (12.5)	Not limited	49.5 (19.5)	Not limited	1.07 (0.042)	1.14 (0.045)
35.6 (14)	45.7 (18)	53.3 (21)	63.5 (25)		
45.7 (18)	Not limited	68.6 (27)	Not limited	1.34 (0.053)	1.42 (0.056)
50.8 (20)	63.5 (25)	73.7 (29)	91.4 (36)		
55.9 (22)	Not limited	83.8 (33)	Not limited	1.53 (0.060)	1.61 (0.063)
63.5 (25)	78.7 (31)	89.0 (35)	109.2 (43)		
63.5 (25)	Not limited	99.1 (39)	Not limited	1.70 (0.067)	1.78 (0.070)
73.7 (29)	91.4 (36)	104.1 (41)	129.5 (51)		

^a A supporting frame is a structure of angle or channel or a folded rigid section of sheet metal which is rigidly attached to and has the same outside dimensions as the enclosure surface and which has torsional rigidity to resist the bending moments which are capable of being applied via the enclosure surface when it is deflected. A construction that is determined to have equivalent reinforcing can be accomplished by constructions that produce a structure which is as rigid as one built with a frame of angles or channels. Constructions determined to be without a supporting frame include:

- 1) A single sheet with single formed flanges (formed edges);
- 2) A single sheet which is corrugated or ribbed; and
- 3) An enclosure surface loosely attached to a frame, for example, with spring clips.

^b The width is the smaller dimension of a rectangular sheet metal piece which is part of an enclosure. Adjacent surfaces of an enclosure can have supports in common and be made of a single sheet.

^c For panels which are not supported along one side, for example, side panels of boxes, the length of the unsupported side shall be limited to the dimensions specified unless the side in question is provided with a flange at least 12.7 mm (1/2 inch) wide.

Table 7.2
Minimum Thickness of Sheet Metal for Enclosures Aluminum, Copper, or Brass

Without supporting frame ^a		With supporting frame or equivalent reinforcing ^a		Minimum thickness
Maximum width ^b cm (inches)	Maximum length ^c cm (inches)	Maximum width ^b cm (inches)	Maximum length ^c cm (inches)	
7.6 (3)	Not limited	17.8 (7)	Not limited	0.58 ^d (0.023)
8.9 (3.5)	10.2 (4)	21.7 (8.5)	24.1 (9.5)	
10.2 (4)	Not limited	25.4 (10)	Not limited	0.74 (0.029)
12.7 (5)	15.2 (6)	26.7 (10.5)	34.2 (13.5)	
15.2 (6)	Not limited	35.6 (14)	Not limited	0.91 (0.036)
16.5 (6.5)	20.4 (8)	38.1 (15)	45.7 (18)	

Table 7.2 Continued on Next Page

Table 7.2 Continued

Without supporting frame ^a		With supporting frame or equivalent reinforcing ^a		Minimum thickness mm (inches)
Maximum width ^b cm (inches)	Maximum length ^c cm (inches)	Maximum width ^b cm (inches)	Maximum length ^c cm (inches)	
20.4 (8)	Not limited	48.3 (19)	Not limited	1.14 (0.045)
24.1 (9.5)	29.2 (11.5)	53.3 (21)	63.5 (25)	
30.5 (12)	Not limited	71.1 (28)	Not limited	1.47 (0.058)
35.6 (14)	40.6 (16)	76.2 (30)	94 (37)	
45.7 (18)	Not limited	106.7 (42)	Not limited	1.91 (0.075)
50.8 (20)	63.4 (25)	114.3 (45)	139.7 (55)	
63.4 (25)	Not limited	152.4 (60)	Not limited	2.41 (0.095)
73.7 (29)	91.4 (36)	162.6 (64)	198.1 (78)	

^a A supporting frame is a structure of angle or channel or a folded rigid section of sheet metal which is rigidly attached to and has the same outside dimensions as the enclosure surface and which has torsional rigidity to resist the bending moments which are capable of being applied via the enclosure surface when it is deflected. Construction that is determined to have equivalent reinforcing can be accomplished by constructions that produce a structure which is as rigid as one built with a frame of angles or channels. Construction determined to be without supporting frame includes:

- 1) Single sheet with single formed flanges (formed edges);
- 2) A single sheet which is corrugated or ribbed; and
- 3) An enclosure surface loosely attached to a frame, for example, with spring clips.

^b The width is the smaller dimension of a rectangular sheet metal piece which is part of an enclosure. Adjacent surfaces of an enclosure can have supports in common and be made of a single sheet.

^c For panels which are not supported along one side, for example, side panels of boxes, the length of the unsupported side shall be limited to the dimensions specified unless the side in question is provided with a flange at least 12.7 mm (1/2 inch) wide.

7.3 Nonmetallic enclosures

7.3.1 In addition to the performance tests specified in this Standard, the factors to be considered when evaluating the suitability of a polymeric enclosure include the requirements in [7.3.2](#) – [7.3.4](#).

7.3.2 The enclosure material shall have a minimum flame rating of V-1 in accordance with [8.1](#).

Exception: If a metal or minimum V-1 rated sub-enclosure houses all insulated or uninsulated live parts and internal wiring that involve a risk of fire, then the outer polymeric enclosure may be classed HB or better.

7.3.3 The enclosure material shall have a minimum Relative Thermal Index (RTI) value that exceeds the maximum temperature observed on the material during operation but no less than 80 °C (176 °F) for packs intended to be stored in the trunk or passenger compartment of a vehicle. If an RTI value that exceeds the maximum temperature observed on the material during operation, but is lower than 80 °C (176 °F), is used, the power pack shall be marked in accordance with [75.22](#) and provided with instructions as shown in [79.7](#).

7.3.4 A conductive coating applied to a nonmetallic surface such as the inside surface of an enclosure, shall be evaluated in accordance with UL 746C, and CSA C22.2 No. 0.17, unless it can be determined that flaking or peeling of the coating does not result in a reduction of spacings or the bridging of live parts that may result in a risk of fire, electric shock, or injury to persons.

7.3.5 An adhesive used to secure parts of an enclosure shall comply with the requirements in UL 746C and CSA C22.2 No. 0.17.

7.3.6 Enclosures of molded or formed nonmetallic materials shall be constructed so that any shrinkage or distortion of the material over time will not allow for the user to contact live parts at hazardous voltage or hazardous energy levels. Compliance is determined by the Mold Stress Test, Section [61](#).

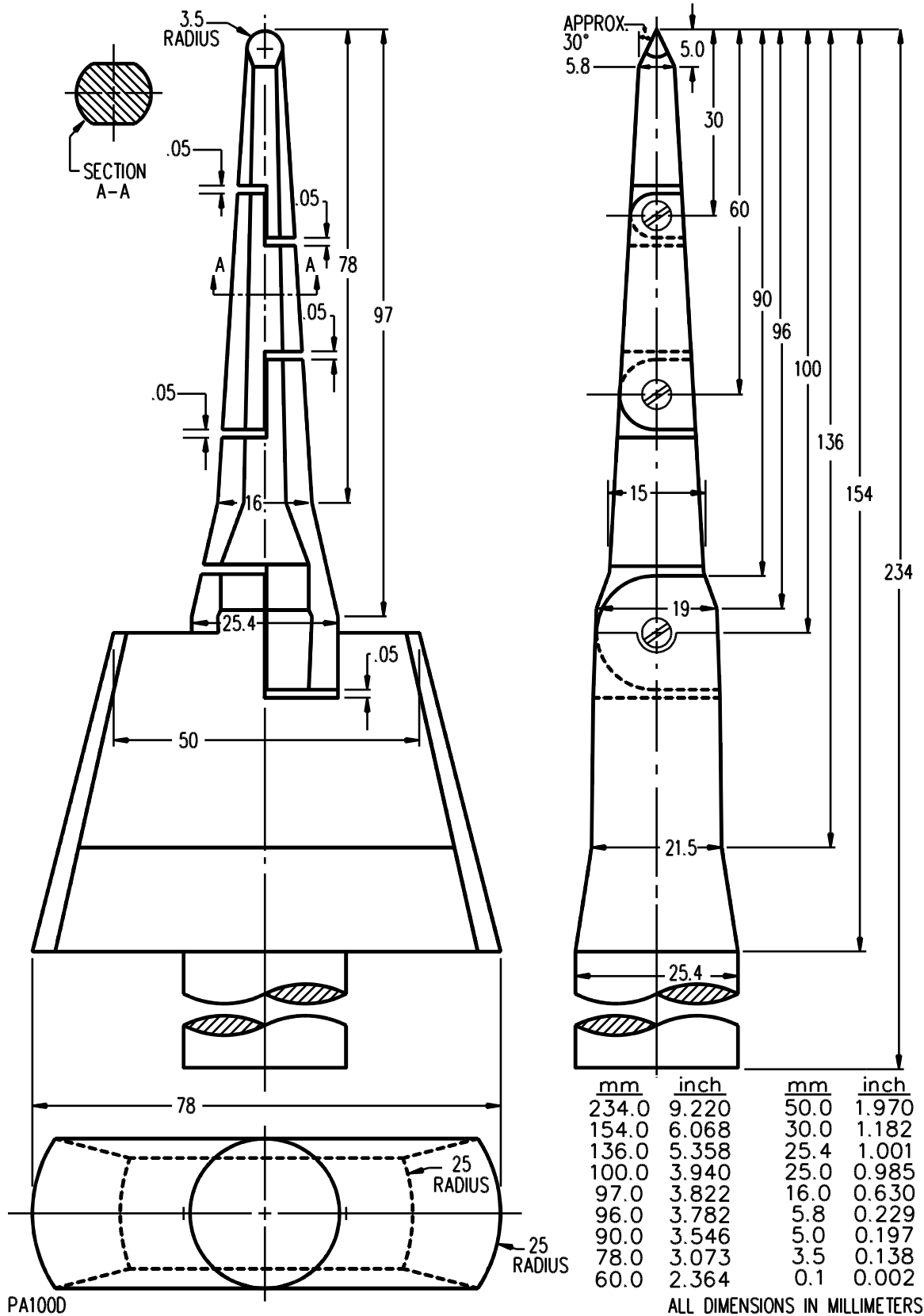
7.3.7 Enclosures constructed of nonmetallic materials shall be subjected to the Strength of Enclosure Tests, Section [60](#).

7.4 Openings in enclosures

7.4.1 Power pack enclosures shall not allow the entrance of water in accordance with [7.5](#).

7.4.2 A probe as illustrated in [Figure 7.1](#), when inserted through an opening, shall not touch any uninsulated live part that can cause electric shock.

Figure 7.1
Articulate Probe with Web Stop



7.4.3 Thermoplastic covering an opening for user servicing, such as replacement of a pilot lamp, and that reduces the risk of unintentional contact with a live part involving a risk of electric shock shall be evaluated as an enclosure. It shall be reliably retained in place.

7.4.4 An uninsulated live part at hazardous voltage or hazardous energy levels shall be located or enclosed so that protection against unintentional contact is provided.

7.4.5 A door or cover that provides access to a live part at hazardous voltage or hazardous energy levels shall be securely held in place so that it can be opened or removed only by using a tool.

Exception: A door or cover that provides access to a live part that is not at hazardous voltage or hazardous energy levels shall be securely held in place, but need not be secured so that it is necessary to use a tool to open or remove it.

7.4.6 The door or cover of an enclosure shall be hinged or attached in an equivalent manner if it provides access to an overload-protective device, the functioning of which requires renewal, or if it is necessary to open the cover in connection with the operation of the protective device. A door or cover providing access to a fuseholder shall be tight-fitting and shall be positively held closed.

Exception: A hinged cover is not required if the only overload-protective devices enclosed are:

- a) Connected in control circuits, provided the protective devices and the circuit loads are within the same enclosure;*
- b) Rated 2 amperes or less for loads not exceeding 100 volt-amperes;*
- c) Extractor fuses having an integral enclosure; or*
- d) Fuses connected in a low-voltage limited energy circuit.*

7.4.7 The operating handle of a circuit breaker, an operating button of a manually operable protector, the capped portion of an extractor-type fuseholder, or a similar part may project outside the enclosure.

7.5 Environmental considerations

7.5.1 The enclosure of an outdoor use or temporary outdoor use power pack shall be constructed to exclude a beating rain in accordance with [7.5.2](#).

7.5.2 All outdoor use or temporary outdoor use power packs shall be subjected to the Rain Test, Section [65](#).

7.5.3 A gasket employed in a power pack in order to comply with [7.5.1](#) shall be tested in accordance with Accelerated Aging of Gaskets, Sealing Compounds, and Adhesives, Section [67](#).

7.5.4 A nonmetallic enclosure for an outdoor use power pack shall be judged on the basis of the effect of exposure to ultraviolet light and water in accordance with the applicable tests in UL 746C, CSA C22.2 No. 0.17. Temporary outdoor use power packs need not comply with this requirement.

8 Flammability of Materials

8.1 Nonmetallic materials used for enclosures shall have a minimum flammability rating of V-1 in accordance with the requirements in UL 94 and CSA C22.2 No. 0.17. As an alternative, finished enclosures may be tested in accordance with the 20 mm end-product flame test in UL 746C and CSA

C22.2 No. 0.17. Metallic materials used for enclosures are considered to comply without further evaluation, except magnesium shall not be used for enclosure materials.

8.2 Nonmetallic materials used for internal parts within the overall enclosure shall be rated V-2 minimum.

Exception No. 1: The internal insulating system of components where component requirements exist need not comply with this requirement.

Exception No. 2: A small part, gasket, or other nonmetallic part that is located such that it cannot propagate flame from one area to another within the equipment, and is not located in close proximity to uninsulated live parts, is not required to comply with this requirement.

8.3 Nonmetallic materials located outside the enclosure, and not used to complete the enclosure, are considered decorative parts. These parts shall be rated HB minimum.

8.4 Printed-wiring board materials shall be rated V-1 minimum.

8.5 For the requirements outlined in [8.2](#) – [8.4](#), the flammability rating of the material shall be provided as part of the material rating or the flammability rating may be determined in accordance with UL 94 and CSA C22.2 No. 0.17.

9 Wirings and Terminals

9.1 Wiring shall be insulated and acceptable for the purpose, when considered with respect to temperature, voltage, and the conditions of service to which the wiring is likely to be subjected within the equipment.

9.2 Wiring shall be secured and routed away from sharp edges or parts exceeding insulation. Openings in compartments through which insulated wiring is routed shall be smooth and well-rounded or provided with protective insulating bushings or grommet to prevent abrasion. Wiring connections between various parts of a power pack and accessories shall be routed and secured to prevent the potential for short circuit conditions to occur.

9.3 An external terminal shall be designed to prevent inadvertent shorting or inadvertent misalignment. The power pack shall be designed to prevent connecting or disconnecting an external terminal (including the terminal of a detachable) under load conditions unless the output complies with Section [51](#) or the terminal has been evaluated for disconnect current in its component evaluation.

9.4 External non-detachable cords and leads that are accessible in the end use installation shall be provided with strain relief that prevents strain to internal conductors under pull and push-back conditions. Compliance is determined by the tests of [59.1](#) and [59.2](#).

9.5 External flexible cables or cords used on the power pack with separable battery packs and interconnection cords shall be rated for their anticipated service including voltage, current, temperature and environment. External flexible cables or cords used on the power pack with separable battery packs and interconnection cords shall comply with [12.2.1.6](#) or shall be additionally provided a covering or sleeving for the intended applications.

10 Assembly

10.1 An uninsulated live part shall be secured to the base or surface so that it is prevented from rotating or shifting in position as the result of normal stresses, if such movement results in a reduction of spacings below the minimum values indicated in Spacings, Section [31](#).

10.2 A component such as a control switch, lampholder, or connector, shall be mounted securely and shall be prevented from turning by means other than friction between surfaces.

10.3 With reference to [10.2](#), a switch is not required to be mounted as described in [10.2](#), if all of the following conditions are met:

- a) The switch is a plunger or other type that does not tend to rotate when operated. A toggle switch is considered to be subject to forces that tend to turn the switch;
- b) The means for mounting the switch makes it unlikely that operation of the switch will loosen it;
- c) Spacings are not reduced below the minimum values indicated in Spacings, Section [31](#), if the switch rotates; and
- d) Operation of the switch is by mechanical means rather than by direct contact by persons.

10.4 With reference to [10.2](#), a lampholder of the type in which the lamp cannot be replaced, such as a sealed neon pilot or indicator light, is not required to be mounted in accordance with [10.2](#), if rotation cannot reduce spacings below the minimum values indicated in Spacings, Section [31](#).

10.5 A small stem mounted device having a single hole mounting means may be prevented from rotating by a properly applied lock washer.

11 Corrosion Protection

11.1 Metal shall be used in combinations that are galvanically compatible.

11.2 Hinges and other attachments shall be resistant to corrosion.

11.3 The requirements in [11.5](#) – [11.15](#) do not contemplate corrosion that might be caused by exposure to the earth or other corrosive agents.

11.4 The requirements in [11.5](#) – [11.15](#) do not apply to a part, such as a decorative grill, that is not required to form a part of the enclosure.

11.5 A metallic enclosure shall be protected against corrosion as specified in [11.6](#) – [11.15](#).

11.6 Copper, bronze, brass containing not less than 80 % copper, or stainless steel may be used without additional protection against corrosion. Sheet, extruded, or cast aluminum, die-cast zinc, and other metals shall be of a grade or alloy known to be resistant to atmospheric corrosion, shall be subjected to tests, or shall be additionally protected against corrosion.

11.7 An enclosure of cast iron or malleable iron at least 3.2 mm (1/8 inch) thick shall be protected against corrosion by:

- a) A 0.0038 mm (0.00015 inch) thick coating of zinc, cadmium, or the equivalent on the outside surface, and a visible coating of such metal on the inside surface, or
- b) One coat of an organic finish of the epoxy or alkyd-resin type or other outdoor paint on each surface.

11.8 Unless suitability of a paint can be determined by consideration of its composition, corrosion tests are required.

11.9 An enclosure of sheet steel less than 3.20 mm (0.126 inch) thick if zinc-coated or 3.12 mm (0.123 inch) thick if uncoated shall be protected against corrosion by one of the following means or by other metallic or nonmetallic coatings that have been found to give equivalent protection as described in [11.12](#):

- a) Hot-dipped, mill galvanized, sheet steel conforming with the coating designation G90 in the Weight (Mass) of Coating Requirements table in ASTM A653, with not less than 40 % of the zinc on any side, based on the minimum single-spot-test requirement in this ASTM specification. The weight of zinc coating may be determined by any method; however, in case of question, the weight of coating shall be established in accordance with ASTM A90.
- b) A zinc coating, other than that provided on hot-dipped, mill-galvanized sheet steel, uniformly applied to an average thickness of not less than 0.015 mm (0.00061 inch) on each surface with a minimum thickness of 0.014 mm (0.00054 inch). An annealed coating shall also comply with [11.14](#).
- c) A zinc coating conforming with [11.10](#) (a) or (b), with one coat of an organic finish of the epoxy or alkyd-resin type or other outdoor paint on each surface applied after forming. See [11.8](#).

11.10 An enclosure of zinc coated steel 3.20 mm (0.126 inch) thick or thicker or an enclosure of uncoated sheet steel 3.12 mm (0.123 inch) thick or thicker shall be protected against corrosion by one of the following means or by other metallic or nonmetallic coatings that have been shown to give equivalent protection as described in [11.12](#):

- a) Hot-dipped, mill galvanized, sheet steel conforming with the coating designation G60 or A60 in the Weight (Mass) of Coating Requirements table in ASTM A653, with not less than 40 % of the zinc on any side, based on the minimum single-spot-test requirement in this ASTM specification. The weight of zinc coating may be determined by any method; however, in case of question, the weight of coating shall be established in accordance with ASTM A90.
- b) A zinc coating, other than that provided on hot-dipped, mill-galvanized sheet steel, uniformly applied to an average thickness of not less than 0.010 mm (0.00041 inch) on each surface with a minimum thickness of 0.009 mm (0.00034 inch).
- c) Two coats of an organic finish of the epoxy or alkyd-resin type or other outdoor paint on each surface. See [11.8](#).
- d) Any one of the means specified in [11.9](#).

11.11 The requirements in [11.10](#) also apply to an enclosure of zinc-coated sheet steel 1.42 mm (0.056 inch) thick or thicker and an enclosure of uncoated sheet steel 1.35 mm (0.053 inch) thick or thicker if the enclosure is intended to be mounted within and protected from direct exposure to weather by the enclosure of other equipment. Such an enclosure shall not be marked rainproof or raintight.

11.12 With reference to [11.9](#) – [11.11](#), other finishes, including paints, metallic finishes, and combinations of the two, may be accepted if comparative tests with galvanized sheet steel – without annealing, wiping, or other surface treatment – conforming with [11.9\(a\)](#) or [11.10\(a\)](#), as applicable, indicate that they provide equivalent protection. Among the factors that are to be taken into consideration when judging the suitability of such coating systems are exposure to salt spray, moist carbon dioxide-sulfur dioxide-air mixtures, moist hydrogen sulfide-air mixtures, ultraviolet light, and water.

11.13 If tests are required, test specimens of a finish as described in [11.7](#), [11.9\(c\)](#), [11.10\(c\)](#), or [11.12](#) are to be consistent with the finish that is to be used in production with respect to the base metal, cleaning, or pretreatment method, application method, number of coats, curing method, thickness, and the like.

11.14 A hot-dipped, mill-galvanized A60 (alloyed) coating or an annealed zinc coating that is bent or similarly formed after annealing and that is not otherwise required to be painted, shall be painted in the bent or formed area if the bending or forming process has damaged the zinc coating, except that such an

area on the inside surface of an enclosure that is not exposed to water during the water spray test need not be painted. The zinc coating is considered to be damaged if flaking or cracking of the zinc coating at the outside radius of the bent or formed section is visible at 25-power magnification. Simple sheared or cut edges and punched holes are not considered to be formed.

11.15 If verification of the thickness of coatings in [11.9](#) and [11.10](#) is required, the thickness shall be established by the Metallic Coating Thickness Test, Section [68](#).

12 Supply Connections

12.1 General

12.1.1 Power packs are intended to be connected to the power supply circuit in order to recharge the internal battery, electrochemical capacitors, or electrochemical capacitor modules. Additionally, outputs of the power pack may be powered when the pack is connected to the supply circuit.

12.1.2 Power packs shall be provided with one or more of the following input options:

- a) Cord and plug connection, rated at 240 V ac maximum, 60 Hz;
- b) DC voltage input connector intended for connection to the output of an external power supply;
- c) Photovoltaic panel or device with inverter;
- d) Vehicle adapter, rated 24 V dc maximum;
- e) USB input port (can also act as output port);
- f) Cables for charging from a vehicle battery rated up to 24 V dc maximum; or
- g) DC voltage input connector intended for connection to the output of other external DC source, such as detachable or external battery pack.

12.1.3 Power packs with AC pass through function shall additionally comply with the applicable requirements of the UL 1778 / CSA C22.2 No. 107.3. As a minimum, the following sections shall be considered:

- a) Backfeed Protection and the Annex for Backfeed Protection Test;
- b) Provisions for Earthing and Bonding and the Annex for Earthing and Bonding;
- c) Overcurrent and Earth Fault Protection in Primary Circuits;
- d) Abnormal Operating and Fault Conditions;
- e) The Annex for Additional Component Requirements;
- f) The Annex for Harmonic Distortion Test; and
- g) For lead-acid batteries, the Annex for Ventilation of Battery Compartments.

NOTE: Devices employing EPS (Emergency Power Supply) design in which the alternative power transfer typically occurs through bypass relay or similar component to AC output and automatically transfers to inverter mode with loss of AC mains supply are also covered under [12.1.3](#).

12.2 Cord and plug connection

12.2.1 General

12.2.1.1 For cord and plug connected devices, the power pack shall be connected to the supply circuit by means of one of the following:

- a) A cord set or power-supply cord in accordance with UL 817; or
- b) A power-supply cord consisting of an attachment plug in accordance with UL 498, assembled to a flexible cord in accordance with UL 62.

12.2.1.2 For devices intended for use with a cord set, the cord set need not be provided with the device when the power pack is marked in accordance with [75.17](#) or [75.18](#) and the instructions are in accordance with [78.2](#).

12.2.1.3 For power packs intended for use in a repair facility with a cord set, the cord set shall be provided with a means to insure that the cord set cannot be inadvertently detached during use.

Exception: A device intended for use in a repair facility may be provided with a cord set without a means to prevent inadvertent detachment, when the power pack is marked in accordance with [75.17](#).

12.2.1.4 A cord set or power-supply cord in accordance with [12.2.1.1](#) shall have a voltage rating not less than the rated voltage of the equipment and shall have an ampacity not less than the current rating of the equipment or the actual current measured during the Power Input Test, Section [45](#), whichever is greater.

12.2.1.5 The ampacity of an attachment plug for a portable power pack intended to be continuously loaded for 3 hours or more shall not be less than 125 % of the input rating of the portable power pack.

12.2.1.6 A flexible cord shall be a hard usage or extra-hard usage type and be rated for oil-resistance. A flexible cord intended to be used in an outdoor or temporary outdoor application shall be additionally rated for a wet application, indicated by a suffix "W".

Exception: A flexible cord that connects between the external batteries and the power pack shall comply with [9.5](#).

NOTE: Examples of a flexible cord for hard usage or extra-hard usage type and rated for oil-resistance are Type SO, SJO, SJEO, and SJTO. Examples of a flexible cord for use in an outdoor or temporary outdoor application are SJOW or SJTW.

12.2.1.7 The length of a power-supply cord shall not be less than 1.8 m (6 feet) as measured from the face of the attachment plug to the point of entry to the device. The length of a cord set shall not be less than 1.8 m (6 feet) as measured from the face of the attachment plug to the face of the load fitting.

12.2.1.8 If a product incorporates a disconnecting means, such as an appliance inlet, the arrangement shall be such that no hazardous live parts will be exposed under normal conditions.

12.2.1.9 A permanently attached flexible cord shall exit the enclosure in an area that is not in close proximity to a hot spot or moving part.

12.2.2 Strain relief

12.2.2.1 Strain relief shall be provided on a non-detachable flexible supply cord and output wiring to reduce the risk of mechanical stress being transmitted to terminals, splices, or interior wiring. See the Direct-pull strain relief test, [59.1](#). A knot in the flexible cord is not considered a form of strain relief.

12.2.2.2 A metal strain relief clamp or band provided in accordance with [12.2.2.1](#) shall be provided with auxiliary insulation over the cord if damage to the cord insulation results when the strain relief test is conducted without auxiliary insulation.

12.2.2.3 Means shall be provided to prevent a flexible supply cord and output wiring from being pushed into the equipment through the cord entry hole if such displacement would:

- a) Result in mechanical damage to the cord;
- b) Expose the cord to a temperature other than that for which it is rated; or
- c) Reduce spacings, such as to a metal strain relief clamp, below the minimum values indicated in Spacings, Section [31](#).

To determine compliance, the flexible supply cord and output wiring shall be tested in accordance with the Push-back strain relief test, [59.2](#).

12.2.3 Bushings

12.2.3.1 At the point where a non-detachable flexible cord passes through an opening in the enclosure, there shall be a bushing or the equivalent that is secured in place, and that has a smooth, well rounded surface against which the cord may bear. An insulating bushing shall be provided, if the enclosure is of metal, or if the construction is such that the cord may be subjected to strain or motion. A bushing that complies with the applicable requirements in UL 635, is considered to comply with this requirement.

12.2.3.2 A hole in porcelain, phenolic composition, or other non-conducting material, having a smooth, rounded surface, is considered to be equivalent to a bushing.

12.2.3.3 A bushing of the same material as, and molded integrally with, a supply cord is acceptable if the built-up section is not less than 1.6 mm (1/16 inch) thick at the point where the flexible cord passes through the enclosure.

12.2.3.4 At a point of flexure, no additional flexible cords or wires shall be routed through a bushing or opening with the power supply cord.

12.3 External power supplies

12.3.1 As an alternative to flexible cord connections, the power pack may be provided with a DC voltage rated input connector intended for connection to the output of an external power supply.

12.3.2 The output rating of the external power supply shall be equal to or greater than the input rating of the power pack; and the output of the external power supply shall be a power source in accordance with any of the following:

- a) UL 62368-1 / CSA C22.2 No. 62368-1;
- b) UL 60335-2-29 / CSA C22.2 No. 60335-2-29;
- c) A Class 2 power source in accordance with UL 1310 and CSA C22.2 No. 223;
- d) A power source other than Class 2 in accordance with UL 1012 and CSA C22.2 No. 107.2; or
- e) UL 60950-1 / CSA C22.2 No. 60950-1.

12.4 Vehicle adapters

12.4.1 A power pack intended for connection to an automobile cigar lighter receptacle shall be provided with a vehicle adapter that complies with the enclosure and input contacts requirements in UL 2089 and CSA C22.2 No. 107.1. The connector plug shall incorporate a fuse or other protective device having a current rating not greater than 15 A.

Exception: The protective device may be provided in the output cord of the vehicle adapter not more than 127 mm (5 inches) from the vehicle adapter enclosure.

12.4.2 With reference to [12.4.1](#), if the fuse is user replaceable, the vehicle adapter shall be marked in accordance with [75.10](#). This marking shall appear adjacent to the fuseholder.

12.5 Photovoltaic panels

12.5.1 Power packs provided with a photovoltaic panel as an input for charging the internal battery, and with rated voltage exceeding hazardous voltage levels as defined in [5.26](#) or energy exceeding hazardous energy levels as defined in [5.25](#), shall be provided with an inverter and/or a charge controller and shall be in accordance with the applicable requirements in UL 1741; or with the applicable requirements in UL 62109-1.

13 Output Connections

13.1 General

13.1.1 Power packs shall be provided with one or more outputs that terminate in a suitable connector or other connection means in accordance with the requirements in this Standard.

13.1.2 The output of the power pack shall include at least one of the output options shown in (a) – (d). Additional outputs are not restricted.

- a) Booster cable assemblies, see [13.2](#);
- b) Receptacles, see [13.3](#);
- c) DC Connectors, see [13.4](#);
- d) Vehicle adapter sockets, see [13.5](#);
- e) USB Connectors, see [13.4](#);
- f) Induction power transmitters, see [13.6](#).

13.1.3 If two or more bare parts (one of which may be earthed) of an output can be accessed and bridged by the test probe as illustrated in [Figure 7.1](#) in a straight position without appreciable force, the maximum available power from the output shall not be more than 240 VA. Compliance shall be confirmed by the Energy Hazard Measurement Test, Section [51](#).

Exception: A USB output need not comply with [13.1.3](#), and instead it shall comply with LVLE circuit requirements.

13.1.4 A USB output shall be a LVLE circuit and comply with the LVLE Circuit Test in Section [52](#).

13.2 Booster cable assemblies

13.2.1 General

13.2.1.1 Booster cable assemblies shall be detachable or non-detachable assemblies. Detachable assemblies shall comply with [13.2.1.2](#) and the applicable requirements in [13.2.2](#) and [13.2.3](#). Non-detachable assemblies shall comply with [13.2.1.3](#) and the applicable requirements in [13.2.2](#) and [13.2.3](#).

13.2.1.2 For detachable booster cable assemblies, the connector shall be in accordance with UL 1977 and CSA C22.2 No. 182.3, or the connector shall be in accordance with UL 1839 (middle connectors). The connector shall have a voltage and current rating that is the same as or higher than the output rating of the booster cable assembly.

13.2.1.3 For non-detachable booster cable assemblies, the cable shall be provided with strain relief in accordance with [12.2.2](#) and bushings shall be provided in accordance with [12.2.3](#).

13.2.1.4 The power pack with a booster function shall be provided with a means of reverse polarity protection to prevent reverse connect to a starter battery in the vehicle. Compliance is determined by the Reverse Polarity of Booster Cables Test, [55.3](#).

13.2.2 Cables

13.2.2.1 The cable provided with the booster cable assembly shall comply with the applicable requirements for cables in UL 1839.

13.2.2.2 As an alternative to [13.2.2.1](#), the cable can be shown to comply with the requirements in [13.2.2.3](#) – [13.2.2.7](#), when the power pack is provided with a protection system for the cable assembly. The protection system shall provide output limiting such that a maximum current cannot be exceeded under normal or abnormal conditions and the maximum temperature of the cable insulation shall not be exceeded during the Normal Temperature Test, Section [50](#).

13.2.2.3 The conductors used within the cables shall:

- a) Comply with the requirements in UL 758 and either CSA C22.2 No. 127, or CSA C22.2 No. 210; and
- b) Be suitably sized based on the rating of the output current associated with the booster function.

13.2.2.4 The final cable assembly shall be subjected to the Cold Bend Test, Section [72](#).

13.2.2.5 At all points where the cable enters the enclosure of the device or any other enclosure, excluding the cable connection to the clamp, the connection shall be subjected to a Strain Relief Test as described in Section [59](#).

13.2.2.6 The conductors of a booster cable set employing parallel conductors shall be separated at each end to allow the jaw ends of the clamps to span a minimum distance of 920 mm (3 feet). For power packs provided with a safety circuit that prevents energy from being available at the booster clamps prior to connection and immediately upon disconnection, such that arcing and sparking is prevented at the battery terminals, the 920-mm minimum distance can be reduced. The safety circuit shall be subject to the requirements in Section [43](#), Safety Circuits and Control Circuits.

13.2.2.7 The cable shall be subjected to the Power Pack Ampacity Test in Section [70](#).

13.2.3 Clamps

13.2.3.1 The clamp provided with the booster cable assembly shall comply with the applicable requirements for clamps in UL 1839.

13.2.3.2 As an alternative to [13.2.3.1](#), the cable can be shown to comply with the requirements in [13.2.3.3](#) – [13.2.3.6](#), when the power pack is provided with a protection system for the cable assembly. The protection system shall provide output limiting such that a maximum current cannot be exceeded under normal or abnormal conditions and the maximum temperature of the cable insulation shall not be exceeded during the Normal Temperature Test, Section [50](#).

13.2.3.3 The clamp bodies shall be fabricated from nonmetallic materials and these materials shall have a flame rating of HB minimum.

13.2.3.4 The clamps shall be subjected to the Cold Drop Test using a conditioning temperature equal to the manufacturer's lower ambient temperature rating. See [73.2](#).

13.2.3.5 The clamps shall be subjected to the Dielectric Voltage-Withstand Test of [73.3](#).

13.2.3.6 The clamp shall be subjected to the Secureness Test of [73.4](#).

13.3 Receptacles

13.3.1 Receptacles provided as an output on power packs shall be of a NEMA type receptacle that is in accordance with UL 498 and CSA C22.2 No. 42. Each AC output power circuit shall be provided with overcurrent protection for all ungrounded conductors in accordance with Section 31, Output Alternating Current Power Circuits, of UL 1012. Except as indicated in [13.3.2](#), receptacles shall be rated 120 Vac, 15 A maximum, and shall consist of a double blade, ungrounded configuration.

13.3.2 Receptacles other than as indicated in [13.3.1](#), may be acceptable provided the criteria in either (a) – (c) is met:

a) The following criteria is met:

1) The receptacles including their ground terminals are isolated from any of the AC mains input circuits by double insulation in accordance with Section [16](#), Double Insulation Products; and

2) The portable power pack is marked per [75.24](#) and provided with instructions per [77.3\(q\)](#).

b) For portable power packs with inverter output which the grounded conductor (Neutral) was bonded to any ground terminal, the receptacles shall have the output circuits protected by integral ground-fault circuit-interrupters intended for the current involved, GFCI protection provided integral to the power pack for all receptacle outlets. The GFCIs shall be in accordance with UL 943 and shall interrupt the output supply circuit in the event of a ground fault of receptacles; or

c) For power packs with AC pass through function, compliance is determined per the applicable requirements of UL 1778 / CSA C22.2 No. 107.3. See [12.1.3](#).

13.3.3 Markings shall be provided that indicate the maximum rating of the receptacle to the user. This marking shall be permanent.

13.4 DC output connectors and USB connectors

13.4.1 Power packs provided with USB connections as an output shall incorporate connectors that are in accordance with UL 1977 and CSA C22.2 No. 182.3.

13.4.2 DC output connectors provided as an output shall incorporate connectors that are in accordance with UL 1977 and CSA C22.2 No. 182.3.

13.5 Vehicle adapter sockets

13.5.1 Adapter sockets provided as an output on a power pack, for connecting products with a standardized vehicle adapter connector, shall comply with [13.5.2](#).

13.5.2 The cable, if provided, shall be suitable for the voltage and ampacity of the output rating. The cable shall be in accordance with UL 758 and either CSA C22.2 No. 127 or CSA C22.2 No. 210.

13.6 Induction power transmitter

13.6.1 The induction power transmitter provided as an output on a power pack shall comply with [13.6.2](#).

13.6.2 The maximum power input to any single transmitting coil RF conversion circuit, with or without a single component fault of short or open circuit within power pack, shall not exceed 240 VA when the power pack is tested in accordance with the Energy Hazard Measurement Test, Section [51](#).

14 LVLE Circuits

14.1 Except as indicated in [14.2](#), an LVLE circuit (see [5.32](#)), including associated electronic components on printed-wiring boards, that does not extend out of the unit is not required to be evaluated.

14.2 Safety circuits shall comply with the applicable requirements in this Standard. Printed wiring boards, insulated wire, and motors shall be types that are required for the application. See Section [18](#), Internal Wiring, Section [27](#), Printed-Wiring Boards, and [21.2](#), Motors and thermal protection.

14.3 A LVLE circuit shall comply with the maximum voltage and current limits as specified in [52.1](#).

14.4 A LVLE circuit shall be separated from a hazardous voltage circuit by an isolated transformer. The maximum current shall be limited through one or more resistors, a regulating network, or an overcurrent protective device (such as a fuse or nonadjustable manual-reset protective device).

14.5 In a circuit of the type described in [14.4](#), the secondary winding of the transformer, the fuse or circuit protective device, or the regulating network, and all wiring up to the point at which the current and voltage are limited, shall be evaluated to the applicable requirements in this Standard.

15 Grounding

15.1 General

15.1.1 A product shall have a provision for grounding in accordance with [15.1.2](#) – [15.2.2](#) that is reliably connected when the unit is connected to the mains power source. The means for grounding may be contained within a separate or integral power supply that has been shown to comply with the applicable requirements for power supplies. In this case, no further evaluation of grounding is needed.

Exception No. 1: Double insulated products are not required to comply with Section [15](#). See Double Insulated Products, Section [16](#).

Exception No. 2: For power packs with AC pass through function, compliance can be alternatively determined per the applicable requirements of UL 1778 / CSA C22.2 No. 107.3, including Provisions for Earthing and Bonding and the Annex for Earthing and Bonding.

15.1.2 All exposed dead metal parts that are likely to become energized shall be reliably connected to the means for grounding.

15.1.3 The equipment grounding conductor of the power supply cord may be used for grounding.

15.1.4 The grounding conductor of a supply cord shall be secured to the metallic enclosure of the product by means of a separate screw or other equivalent means, intended for that purpose only, that is not likely to be removed during any servicing operation not involving the power supply cord. Solder alone shall not be used for securing the grounding conductor. Servicing as mentioned in this requirement include repair of the product by a qualified service person.

Exception: Products with no exposed dead metal parts need not comply with [15.1.4](#).

15.1.5 The grounding conductor and the dead metal parts shall be conductively connected to the grounding member of an attachment plug. The grounding member shall be fixed.

15.1.6 A separable connection, such as that provided by an attachment plug and a mating connector or receptacle, shall be such that the equipment grounding connection is made before and broken after the supply conductors.

15.1.7 The grounding terminal, intended solely for the connection of an equipment grounding conductor, shall be capable of securing a conductor of the size necessary for the application. A connection device that depends on solder alone shall not be provided for connecting the equipment grounding conductor.

15.1.8 A wire binding screw or pressure wire connector intended for the connection of an equipment grounding conductor shall be located so that it is unlikely to be removed during servicing of the power pack.

15.1.9 A screw used to secure the grounding conductor to the frame shall engage the metal by at least two full threads. The metal thickness shall not be less than 1.27 mm (0.050 inch) thick. The metal may be extruded to increase the effective thickness. Only the supply cord grounding conductor shall be secured by the grounding screw.

15.1.10 A stud and nut combination used to secure the grounding conductor to the frame shall be secured to the frame by welding the stud in place. The ground conductor shall be connected first and be in contact with the frame and secured in place by a dedicated nut and lock washer. Other bonding jumpers may be connected to the stud, but they shall be connected above the main ground connection and secured by separate nut and lock washers.

15.1.11 With reference to the requirement in [15.1.2](#), the following dead metal parts are not considered likely to become energized:

a) A small metal part, such as an adhesive attached foil marking, a screw, a handle, and the like, that is:

1) On the exterior of the enclosure and separated from all electrical components by grounded metal; or

2) Electrically isolated from all electrical components.

b) A panel, cover, or other metal part that is isolated from all electrical components by a barrier of vulcanized fiber, varnished cloth, phenolic composition, or other moisture resistant insulating material not less than 0.80 mm (1/32 inch) thick and securely mounted in place; and

c) A panel, cover, or other metal part that does not enclose uninsulated live parts and is electrically isolated from other electrical components.

15.1.12 The main grounding path shall not include a trace on a printed-wiring board.

Exception No. 1: Traces on printed-wiring board serving as protective bonding conductors shall be evaluated in accordance with the Impedance Test, and Limited Short-Circuit Test in CSA C22.2 No. 0.4.

Exception No. 2: Traces on printed-wiring board serving as protective bonding conductors can alternatively be evaluated in accordance with the Bonding Conductor Test in UL 1012.

15.2 Grounding identification

15.2.1 The surface of the insulation on a grounding conductor of a flexible cord shall be green with or without one or more yellow stripes, and no other lead shall be so identified.

15.2.2 The unit shall be marked at the point of grounding connection to the enclosure or frame with the grounding symbol  (IEC 60417 No. 5019).

16 Double Insulated Products

16.1 A device may be provided with a system of double insulation that complies with the requirements in UL 1097, in lieu of a means for grounding.

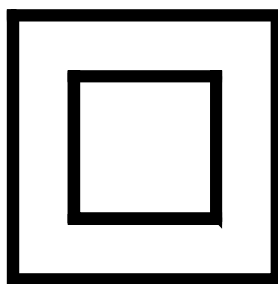
Exception: For power packs with AC pass through function, compliance can be alternatively determined per the applicable requirements of UL 1778 / CSA C22.2 No. 107.3.

16.2 With reference to [16.1](#), hazardous parts shall be insulated from accessible conductive parts and accessible low voltage circuits by double insulation. The spacing requirement, Dielectric Voltage-Withstand Test, and Leakage Current Test requirements in UL 1097 shall be applied to the system of double insulation instead of the requirements in this Standard. In addition, the system of double insulation shall comply with the requirements in this Standard.

Exception: For power packs with AC pass through function, compliance can be alternatively determined per the applicable requirements of UL 1778 / CSA C22.2 No. 107.3.

16.3 Except as indicated in [16.4](#), a product shall be marked as being a suitable double insulated product, see [Figure 16.1](#). All such products shall not be provided with a means for grounding.

Figure 16.1
Double Insulation Marking



16.4 If an earthing conductor in the mains supply cord for a power pack is only used for establishing functional earthing, and the power pack is provided with a system of double insulation, then the power pack shall bear the symbol , IEC 60417-6092.

16.5 With reference to [16.4](#), wiring terminals to be used only for the connection of functional earthing shall be marked with the symbol , IEC 60417-5018. These terminals shall not be marked with the symbol , IEC 60417-5017, or with the symbol , IEC 60417-5019, unless these symbols are provided on a component (for example, a terminal block) or subassembly.

16.6 With reference to [16.4](#), the power pack inlet, if used, shall comply with the creepage distance and clearance requirements for double insulation or reinforced insulation.

17 Current Carrying Parts

17.1 A current carrying part shall be of silver, copper, a copper alloy, stainless steel, or other material suitable for the application. Ordinary unplated iron or steel shall not be used for current carrying parts.

18 Internal Wiring

18.1 Mechanical protection

18.1.1 Wiring and connections between parts of equipment shall be protected or enclosed so that the conductor insulation is not exposed to contact with any rough, sharp, or moving part.

18.1.2 Insulated wiring accessible through an opening in an enclosure is considered to be protected as required in [18.1.1](#) if the opening complies with Openings in enclosures, [7.4](#). Internal wiring not so protected may be acceptable if it is so secured within the enclosure that it is not likely to be subjected to stress or mechanical damage.

18.1.3 An opening in the frame or enclosure through which insulated wires pass shall be provided with a smooth, well rounded bushing or shall have smooth, well rounded surfaces upon which the wires may lie.

18.1.4 Insulated wires, complying with the requirements for Separation of Circuits, Section [19](#), may be bunched and passed through a single smooth opening in a wall within the enclosure.

18.2 Wiring insulation

18.2.1 The internal wiring of power packs shall be of a type rated for the application, when considered with respect to the temperature and voltage involved, with respect to its exposure to oil and grease, and with respect to other conditions of service to which it is subjected.

18.2.2 The insulation of wiring used in a high voltage circuit shall have a flame retardant rating of VW-1 in accordance with the requirements in UL 1581.

18.2.3 Sleeving, tape, and tubing used for insulation on wires shall be rated for the voltage and temperature involved.

18.3 Splices and connections

18.3.1 Splices and connections shall be mechanically secure and provide electrical contact. A soldered connection shall be made mechanically secure before being soldered.

18.3.2 Equipment subjected to vibration shall be provided with lock washers or other means to mechanically secure wire binding screws and nuts. A twist on type connector shall be additionally secured to the wires by means of at least two layers of tape. Tape used for this means shall be evaluated for its intended application and comply with UL 510 and CSA C22.2 No. 197.

18.3.3 A splice shall be provided with insulation equivalent to that of the wires connected if spacing between the splice and other metal parts is not maintained.

18.3.4 The insulation on a splice may consist of a minimum of two layers of tape when the voltage involved is less than 250 volts. When evaluating the splice insulation consideration is to be given to such factors as its dielectric properties, heat resistance, and moisture resistance. Tape shall not be wrapped over a sharp edge.

18.3.5 If internal wiring is stranded, loose strands of wire shall not contact other uninsulated hazardous live parts of opposite polarity or dead metal parts. At a wire binding screw, this may be accomplished by using upturned lugs, a cupped washer, barriers, or other means to hold the wires under the head of the screw. Other means of retaining the loose stranded internal wiring in position are use of a pressure terminal connector, soldering lug, or crimped eyelet.

18.3.6 Aluminum conductors, insulated or uninsulated, used as internal wiring, such as for interconnection between current carrying parts, shall be terminated at each end by a method that is acceptable for the combination of metals involved at the connection point.

18.3.7 With reference to [18.3.6](#), a wire binding screw or a pressure terminal connector used as a terminating device shall be rated for use with aluminum under the condition involved – for example, temperature, heat cycling, vibration, and the like.

18.3.8 Insulation of internal wiring consisting of coated fabric, thermoplastic, or other types of tubing is to be considered with respect to electrical, mechanical, and flammability properties of the material.

19 Separation of Circuits

19.1 Conductors of different circuits operating at different potentials shall be separated or segregated from each other unless each conductor is provided with insulation rated for the highest potential involved.

19.2 An insulated conductor shall be positioned so that it cannot contact an uninsulated live part of a different circuit.

19.3 Segregation of insulated conductors may be accomplished by clamping, routing, or a means that maintains permanent separation from insulated and uninsulated live parts and from conductor of a different circuit.

19.4 A barrier used to separate or segregate internal wiring shall have the mechanical strength for the application, and it shall be reliably held in place.

19.5 Hazardous output circuits shall be isolated from DC low-voltage input circuit and DC low-voltage output circuits that may be user accessible by double insulation in accordance with Section [16](#), Double Insulation Products.

20 Insulating Materials

20.1 Insulating materials, used in circuits other than low-voltage, limited-energy circuits, shall be porcelain, phenolic composition, or other similar material, and shall be evaluated in accordance with UL 746C and CSA C22.2 No. 0.17.

20.2 Ordinary vulcanized fiber may be used for insulating bushings, washers, separators, and barriers, but shall not be used as the sole support for uninsulated hazardous live parts if shrinkage, current leakage, or warpage is likely to result in a risk of fire, electric shock, or injury to persons.

20.3 A thermoplastic material shall not be used for the sole support of hazardous live parts unless it complies with all of the following:

- a) Mechanical strength;
- b) Rigidity;
- c) Resistance to heat;
- d) Resistance to flame propagation;
- e) Dielectric withstand; and
- f) Resistance to abnormal heat.

21 Compressors

21.1 General

21.1.1 Air compressor assemblies provided with power packs, and housed within the overall power pack enclosure, shall comply with the requirements in [21.2](#) and [21.3](#). Air compressors supplied with the power pack, but as a separate unit shall be evaluated in accordance with UL 1450 and CSA C22.2 No. 68.

21.2 Motors and thermal protection

21.2.1 A motor shall be acceptable for the application, and shall be capable of handling the maximum normal load of the power pack without creating a risk of fire, electric shock, or injury to persons.

21.2.2 A motor winding shall resist the absorption of moisture. Film coated wire used for motor windings is not required to be additionally treated to prevent moisture absorption. Fiber slot liners, cloth coil wraps,

and similar moisture absorptive materials are to be provided with impregnation or otherwise treated to prevent moisture absorption.

21.2.3 A motor shall be provided with a thermal protector as described in [21.2.4](#).

21.2.4 With reference to [21.2.3](#), thermal protection shall be evaluated in accordance with UL 1004-3 and CSA C22.2 No. 77.

Exception No. 1: A thermally protected motor which drives a fully enclosed oil-less and tankless air compressor, or inflator, evaluated in accordance with UL 1450 and CSA C22.2 No. 68.

Exception No. 2: A motor intended to move air only, by means of an air moving fan that is integrally attached, keyed, or otherwise fixed to the motor, is required to have locked rotor protection only.

Exception No. 3: A linear style pump is required to have locked rotor protection only.

21.2.5 The thermal protector shall have a current and voltage rating not less than the load that it controls.

21.3 Parts subject to pressure

21.3.1 A part of the power pack that is subject to pressure during normal or anticipated abnormal operation shall withstand, without rupture, a pressure corresponding to five times the maximum pressure that can be developed by the system.

21.3.2 In the event that a test is required to determine whether a part complies with the requirement in [21.3.1](#), two samples of the power pack are to be subjected to the Hydrostatic Strength Test, Section [64](#). Prior to the test, parts molded of polymeric material are to be conditioned in an air circulating oven for 7 hours at a temperature of 70 °C (158 °F) or 10 °C (18 °F) higher than the maximum temperature measured on the part under normal load, whichever is greater. The samples are to be removed from the oven and allowed to cool to room temperature prior to the test.

22 Capacitors and Electrochemical Capacitor Modules

22.1 Capacitors

22.1.1 A capacitor connected from one side of the line to the enclosure of a product shall have a capacitance rating of not more than 0.10 µF, or the power pack shall be subjected to the Capacitor Discharge Test, Section [48](#).

22.1.2 If a product employs a combination consisting of a rectifier and an electrolytic capacitor, no risk of fire, electric shock, or injury to persons shall result when either the rectifier or the capacitor is short-circuited.

22.1.3 Under both normal and abnormal conditions of use, a capacitor employing a liquid dielectric medium more flammable than askarel shall not expel the dielectric medium when tested in accordance with the performance requirements of this Standard.

22.2 Electrochemical capacitor modules

22.2.1 Electrochemical capacitor modules are allowed in power packs to provide the booster cable output. The electrochemical capacitor module shall comply with the requirements in UL 810A as well as the requirements in [22.2.2](#) – [22.2.4](#).

22.2.2 Electrochemical capacitor modules shall be provided with a means to monitor the voltage at the input of the module and shall shut down the power to the module if the voltage exceeds the voltage rating of the module. Any appropriate means is allowed, but any software or programmable components associated with this protection means shall be evaluated for reliability.

22.2.3 Electrochemical capacitor modules shall have temperatures of the module continuously monitored during any defined operating condition and the module shall be shut down if the temperature exceed the rated temperature of the module or individual capacitor in the module. Any appropriate means is allowed, but any software or programmable components associated with this protection means shall be evaluated for reliability.

22.2.4 Means shall be provided such that the capacitor module shall not be allowed to discharge while the charging voltage is applied. Any attempt to jump start a vehicle while the power pack is plugged in shall be prohibited by the power pack itself. Any appropriate means is allowed, but any software or programmable components associated with this protection means shall be evaluated for reliability.

23 Resistors

23.1 The assembly of a power resistor, such as a wire wound type requiring a separate support, shall be reliable. The resistor shall be prevented from loosening or rotating by a means other than friction between surfaces.

23.2 An assembly employing lock washers may be considered to comply with [23.1](#).

24 Lampholders

24.1 The screw shell of a lampholder shall be connected to a conductor that is intended to be connected to the grounded conductor of the power supply circuit.

24.2 A lampholder shall be designed or installed so that uninsulated hazardous live parts, other than a screw shell, are not exposed to inadvertent contact by persons removing or replacing the lamp in normal service.

24.3 If the lampholder is supplied from an LVLE circuit, the requirements in [24.1](#) and [24.2](#) do not apply.

25 Transformers

25.1 Transformers provided as part of a power pack, shall comply with one of the following:

- a) UL 506;
- b) UL 1561; or
- c) UL 5085-1 and CSA C22.2 No. 66.1, and one of the following:
 - 1) UL 5085-2 and CSA C22.2 No. 66.2; or
 - 2) UL 5085-3 and CSA C22.2 No. 66.3.

26 Switches and Controls

26.1 A switch or other control device shall be suitable for the application and shall have current and voltage ratings not less than those of the circuit that it controls when the power pack is operated as intended.

26.2 A primary circuit switch that controls an inductive load having a power factor less than 75 %, such as a transformer, and that does not have an inductive rating, shall be rated not less than twice the full load current rating of the load, or the switch shall be investigated for this application.

26.3 A switch or other control device not having an inductive rating that is connected in a transformer secondary circuit shall comply with the Normal Temperature Test, Section [50](#), and with the Overload of Switches and Controls Test of [58.2](#).

26.4 Unless rated for the application, a switch or other device that controls a motor and is not interlocked so that it will not break the locked rotor motor current shall be subjected to the Overload of Switches and Controls Test of [58.2](#), based on the locked rotor current of the motor.

26.5 A switch that controls a tungsten-filament lamp shall have a tungsten-filament lamp current rating not less than the maximum current it will control.

Exception: A switch having a tungsten-filament lamp current rating less than the maximum current it will control but rated 3 amperes or more may be used to control a 15-watt or smaller lamp.

26.6 A switch shall not disconnect the ground conductor of a circuit unless the switch simultaneously disconnects all other conductors.

26.7 If unintentional operation of a switch results in a risk of injury to persons, the actuator of the switch shall be located or guarded so that such operation is unlikely. The actuator of a switch may be guarded by recessing, ribs, barriers, or the like.

27 Printed-Wiring Boards

27.1 A printed-wiring board shall comply with the requirements in UL 796 and have a minimum flammability classification of V-1.

27.2 A resistor, capacitor, inductor, or other part that is mounted on a printed-wiring board to form a printed wiring assembly shall be secured so that it cannot be displaced to cause a risk of fire or electric shock by a force likely to be exerted on it during assembly, normal operation, or servicing of the power pack.

27.3 Consideration is to be given to a barrier or partition that is part of the power pack assembly and that provided mechanical protection and electrical insulation for a component connected to a printed-wiring board.

28 Interlocks

28.1 An interlock required to reduce the risk of electric shock or injury to persons shall comply with [28.2](#) – [28.6](#).

28.2 The interlock device shall not be defeated readily without:

- a) Damaging the equipment;
- b) Making wiring connections or alterations;
- c) Using other than ordinary tools; or
- d) Using materials other than those readily available. Adhesive tape, string, or conventional extension cord sets are identified as readily available.

28.3 The interlock device shall be such that during normal operation and user servicing:

- a) The interlock is not defeated by improper disassembly, for example removal of the wrong screws during removal of the cover;
- b) The cover in which the interlock is mounted shall not be rotated by its own weight about the interlock axis perpendicular to the cover during any stage of its removal or replacement, if such rotation gives access to a hazardous live part, or damages the interlock or cover;
- c) The act of removal or replacement of the interlocked cover shall not subject the user to unintentional contact with hazardous live parts;
- d) The interlocked cover is not capable of being readily misapplied to result in a risk of electric shock; and
- e) The equipment is marked in accordance with [75.16](#).

28.4 If two momentary contact switches must be operated to energize the power pack, the arrangement shall be spaced from each other and from hazardous live parts so that, if the means are operated simultaneously by one individual, contact with hazardous live parts shall not occur.

28.5 With reference to [28.3\(c\)](#), parts that are recessed more than 64 mm (2-1/2 inches) from the edge of the cabinet opening, normally in the plane of the cover, are excluded when determining that the act of removal or replacement of a cover will subject the user to unintentional contact with hazardous live parts.

28.6 An interlock shall comply with Overload of Interlocks in [58.4](#).

29 Overload Protection Devices

29.1 An overcurrent or thermal protective device shall be suitable for the application.

29.2 An automatic reset device used to comply with [29.1](#) shall be cycled through 200 operations. At the end of the 200 operations, the device shall be able to perform its intended function with no additional risk of fire, electric shock, or injury to persons. See Overload of protection devices, [58.3](#).

29.3 A fuse involving a risk of electric shock shall be inaccessible:

- a) To the user from outside the enclosure; and
- b) To the user during any user servicing.

29.4 A fuse that can be serviced by the user shall be secured in a fuseholder that is constructed and installed such that no uninsulated hazardous live parts will be accessible to contact by persons removing or replacing the fuse. The power pack shall be marked in accordance with [75.10](#). This marking shall be adjacent to the fuse.

29.5 The screw shell of a plug fuseholder and the accessible contact of an extractor type fuseholder shall be connected to the load.

30 Battery

30.1 General

30.1.1 Connections to the terminals of the internal battery shall be secure. The connection means shall not allow the connection to loosen during normal operation to the point where the connection is lost, or where arcing or overheating may occur.

30.1.2 Internal batteries provided as part of power packs shall be in accordance with [30.2](#) or [30.3](#). Detachable batteries are considered a form of internal batteries and shall comply with all the requirements of this Standard for internal batteries unless otherwise required. External batteries provided as accessories of power packs to expand the capacity of the power packs shall be in accordance with [30.4](#).

30.1.3 If a battery pack is removable or can be connected to the power pack by the user, the battery pack shall be keyed or provided with a means to prevent inadvertent reverse polarity connections when replacing the battery pack. If the battery pack is not provided with this means of prevention, then the power pack shall be subjected to the test in [55.10](#).

30.1.4 For power packs intended for use in a repair facility, a detachable or an external battery pack shall be provided with a means to ensure that the battery pack cannot be inadvertently detached during use.

30.1.5 A battery provided for the power pack shall be protected from short circuit. Compliance is determined by Output Connections Short Circuit Test, [55.2](#).

30.1.6 A battery provided for the power pack shall be protected from overcharge during operation. Compliance is determined by the Overcharging Test, [55.9](#).

30.1.7 A lithium-ion power pack with a sum of rated energy larger than 1 kWh, including all its internal, detachable, and external batteries, shall be subjected to the Thermal propagation test, [55.11](#).

30.2 Internal lead acid or valve regulated nickel batteries

30.2.1 A lead acid or valve regulated nickel battery shall comply with the requirements for portable application in UL 1989 or UL 1973.

30.3 Internal lithium-ion batteries or sodium-ion batteries

30.3.1 A lithium-ion battery cell shall comply with the requirements in UL 1642, or UL 62133-2 / CSA C22.2 No. 62133-2, or the cell requirements in UL/ULC 2580 or UL 1973.

30.3.2 A Sodium-ion battery cell shall comply with the cell requirements in UL/ULC 2580 or UL 1973.

30.3.3 The battery shall be subjected to the Normal Charging Operation Test, Section [46](#).

Exception: This test can be waived for power packs which employ battery packs and/or an associated charging system that complies with UL 2595 / CSA C22.2 No. 0.23.

30.3.4 The power pack shall be subjected to the Lithium/Sodium Charging System Test, Section [47](#).

Exception: This test can be waived for power packs which employ battery packs and/or an associated charging system that complies with UL 2595 / CSA C22.2 No. 0.23.

30.3.5 A lithium-ion power pack with rated energy larger than 1 kWh shall be subjected to the Thermal propagation test in [55.11](#).

30.4 External batteries

30.4.1 An external battery used to expand the power capacity of the power pack shall comply with UL 1973.

Exception: An external battery with capacity less than 1 kWh may alternatively comply with the battery requirements in UL 2595 / CSA C22.2 No.0.23.

30.4.2 An external battery shall be included in the evaluation of the power pack. Tests related to operating conditions that interact with the external battery pack shall be conducted on the power pack with the external battery pack connected.

31 Spacings

31.1 Spacings between hazardous live parts of opposite polarity, and between hazardous live and dead metal parts shall not be less than specified in [Table 31.1](#). If an uninsulated hazardous live part is not rigidly secured in position by means other than friction between surfaces, or if a movable dead metal part is in proximity to an uninsulated hazardous live part, the construction shall be such that at least the minimum spacings are maintained. See [31.2](#).

Exception No. 1: Spacings between traces on a printed-wiring board need not comply with [Table 31.1](#) if the printed-wiring board complies with the Printed-wiring board abnormal test, [55.6](#). The requirements specified in [55.6](#) do not substitute for the minimum required spacings between the printed-wiring board foils and dead metal parts or the spacings between the primary and secondary board as specified in [Table 31.1](#).

Exception No. 2: This requirement does not apply to spacings complying with [31.5](#) – [31.9](#).

Exception No. 3: This requirement does not apply to spacings complying with [31.10](#) and [31.11](#).

Exception No. 4: For power packs with AC pass through function, compliance can be determined per the applicable requirements of UL 1778 / CSA C22.2 No. 107.3.

Table 31.1
Minimum Spacings

Potential involved, volts	Minimum spacing, mm (inch)	
	Through air	Over surface
50 or less	1.6 (1/16)	1.6 (1/16)
51 – 150	3.2 (1/8)	6.4 (1/4)
151 – 300	6.4 (1/4)	9.5 (3/8)

31.2 The spacing requirements in [Table 31.1](#) do not necessarily apply to the inherent spacings of a component or assembly of a power pack. This includes switches, fuses, transformers, attachment plugs, and similar components or assemblies. Such spacings shall comply with the requirements for that component or assembly in question, where actual spacing requirements exist.

31.3 In a low voltage, limited energy secondary circuit, spacings between live parts of opposite polarity and between a live part and dead metal that is grounded are not specified.

31.4 Epoxy or equivalent material may be used to reduce spacings when all of the following are met:

- a) Spacings of a minimum 0.8 mm (1/32 inch) are maintained prior to application of the encapsulant;
- b) There are no significant voids in the encapsulant;
- c) The encapsulant is a minimum 0.8 mm (1/32 inch) thick;
- d) The area of reduced spacing, with the encapsulant applied, complies with the Dielectric Voltage Withstand Test, Section [53](#); and
- e) The encapsulant temperature during the Normal Temperature Test, Section [50](#), does not exceed 65 °C (117 °F) rise [based on an assumed operating ambient rating of 25 °C (77 °F)] or 90 °C (194 °F) limit (when tested at an ambient rating of greater than 25 °C).

Exception: When the encapsulant is suitable for use at a higher operating temperature, the temperature shall not exceed the material temperature rating.

31.5 As an alternative to the spacing requirements of [Table 31.1](#), the spacing requirements in UL 840 and CSA C22.2 No. 0.2, may be used. The spacing requirements of UL 840 and CSA C22.2 No. 0.2 are not to be used for spacings to a dead metal enclosure. In determining the pollution degree and overvoltage category, the end use application is to be considered and may modify those characteristics given in [31.6](#) and [31.7](#).

31.6 Power packs are considered to be used in a pollution degree 3 environment for outdoor or temporary outdoor use, and pollution degree 2 environment for indoor use. Hermetically sealed or encapsulated enclosures, or coated printing-wiring boards in compliance with the Printed Wiring Board Coating Performance Test of UL 840 and CSA C22.2 No. 0.2 are considered pollution degree 1.

31.7 It is anticipated the equipment will be rated overvoltage category II or overvoltage category I as defined in UL 840 and CSA C22.2 No. 0.2.

31.8 In order to apply clearance B spacings, control of overvoltage shall be achieved by providing an overvoltage device or system as an integral part of the product.

31.9 For the purpose of power packs covered by these requirements, all printed-wiring boards are considered to have a minimum comparative tracking index of 100 without further evaluation.

31.10 An insulating liner or barrier of material such as vulcanized fiber or thermoplastic employed in lieu of required spacings shall not be less than 0.71 mm (0.028 inch) thick and shall be so located or of such material that it is not adversely affected by arcing.

Exception: Vulcanized fiber not less than 0.33 mm (0.013 inch) thick or mica not less than 0.165 mm (0.0065 inch) thick may be used:

- a) In conjunction with an air spacing of not less than 50 % of the minimum through air spacing;*
- b) Between a heat sink and a metal mounting surface, including the enclosure, of an isolated secondary circuit rated 50 Vrms or less.*

31.11 Insulating material having a thickness less than that specified in [31.10](#) may be used if, upon evaluation, it is found to comply with the requirements for the application, and has a dielectric breakdown strength of not less than 5000 volts in the thickness used for [31.10](#) – 2500 volts in the thickness used for the Exception to [31.10](#) – as determined by Tests on Insulating Materials, Section [66](#).

31.12 Film coated wire is regarded as an uninsulated live part when spacings are being considered.

32 Inverters

32.1 Inverters provided as part of the power pack shall be shown to comply with the applicable requirements in this Standard. See [32.2](#).

Exception No. 1: Inverters that comply with UL 1012 and CSA C22.2 No. 107.1, comply without further evaluation.

Exception No. 2: Inverters that comply with UL 1741, comply without further evaluation.

Exception No. 3: For power packs with AC pass through function, compliance can be determined per the applicable requirements of UL 1778 / CSA C22.2 No. 107.3.

32.2 With reference to [32.1](#), specific attention should be given to:

- a) Printed-Wiring Boards, Section [48](#);
- b) Spacings, Section [31](#);
- c) Normal Temperature Test, Section [50](#);
- d) Dielectric Voltage Withstand Test, Section [53](#); and
- e) Abnormal Operation Tests, Section [55](#).

Other requirements shall also apply as applicable.

33 Charging Functions

33.1 Specialized packs that provide a charging function while connected to the source of supply that is intended to charge the external battery through the pack's booster cable assembly, or other output connection, shall have the charging circuits evaluated in accordance with the applicable requirements in UL 1236 and CSA C22.2 No. 107.2.

PROTECTION AGAINST INJURY TO PERSONS

34 General

34.1 If the operation and maintenance of a power pack by the user involves a risk of injury to persons, means shall be provided to reduce the risk.

34.2 When judging a power pack with respect to the requirement in [34.1](#), consideration shall be given to reasonably foreseeable misuse of the power pack.

34.3 Whether a guard, a release, an interlock, or the like is required, and whether such a device meets the intent of the requirement, shall be determined from an evaluation of the complete power pack, its operating characteristics, and the likelihood of a risk of injury to persons resulting from a cause other than gross negligence. The evaluation shall include consideration of the results of breakdown or malfunction of any component, but not more than one component at a time, unless one event contributes to another. If the evaluation shows that breakdown or malfunction of a particular component can result in a risk of injury to persons, that component is to be investigated for reliability.

34.4 Portable power pack with rated capacity larger than 1 kWh shall be marked in accordance with [75.23](#).

35 Back Feed Protection

35.1 Power packs shall be provided with back feed protection such that no current is passed through the device from the battery to the power supply input connections. Constructional features shall be provided to prevent this under normal operation and single fault condition. As an alternate means of determining compliance, the measurements in the Back Feed Test, Section [71](#), can be performed.

35.2 Power packs that employ the AC pass through function shall comply with the Annex for Backfeed Protection Test of UL 1778 / CSA C22.2 No. 107.3.

36 Sharp Edges

36.1 An enclosure, an opening, a frame, a guard, a knob, a handle, or the like, shall not be sufficiently sharp to cause a risk of injury to persons in normal maintenance or use.

37 Strength of Enclosure

37.1 The enclosure shall be tested in accordance with the Strength of Enclosure Tests, Section [60](#).

38 Attachments

38.1 A functional attachment that is made available or recommended by the manufacturer for use with the basic power pack shall be included in the evaluation of the power pack. Unless the manufacturer recommends the use of two or more attachments at the same time, only one attachment at a time is to be evaluated with the product.

39 Stability

39.1 A power pack shall be tested as described in Stability Test, Section [63](#). The product shall not overturn as a result of the tests nor shall there be any risk of injury that develops due to the test.

Exception: A product with a mass equal to or less than 7 kg (15.4 pounds) is not required to be subjected to the Stability Test. The mass of the product is the total weight of a product which includes the weight of removable batteries and other removable parts.

40 Strength of Handles

40.1 A handle, shoulder harness, belt harness, or other supporting means used to support or carry a product shall be subjected to the Strength of Handles Test, Section [62](#).

41 Design of Shoulder Harnesses

41.1 A portable power pack may be provided with a single shoulder harness, double shoulder harness or belt harness for supporting a portable power pack on the body of the operator. A single shoulder harness or belt harness shall not be permitted for a total mass of portable power pack of 7.5 kg (16.5 pounds) or more. The harness shall be adjustable to the size of the operator and its operation shall be in accordance with the instruction provided by the manufacturer. The harness need not be reusable after the release system has been activated.

41.2 A shoulder harnesses shall be:

- a) Designed in a way for easy removal; or
- b) Equipped with a quick release mechanism that ensures that the portable power pack or the separable battery pack(s) can be removed or released quickly from the operator.

41.3 The quick release mechanism shall be positioned either at the connection between the portable power pack and harness or between the harness and operator. The quick release mechanism shall only allow separation by deliberate action of the operator. The quick release mechanism shall be designed to open while under the weight of the portable power pack. It shall require the use of only one hand and have no more than two release points.

NOTE: An example of a release point is a buckle that requires squeezing between a thumb and finger before releasing, e.g. side release buckles.

41.4 A double shoulder harness shall be considered to be designed in a way for easy removal, if the left and right shoulder straps are not connected to each other in front of the operator's body. If the straps to connect between the left and right shoulder straps are provided, it is also considered to be designed in a way for easy removal when the straps connecting between the left and right shoulder straps can be released under the load of the battery pack by using one hand and no more than two release points. The release mechanism shall only allow separation by deliberate action of the operator.

41.5 The requirements in [41.2](#) – [41.4](#) shall be evaluated using the heaviest portable power pack with maximum load weight installed identified in the instruction provided by the manufacturer.

42 Surface Temperatures

42.1 During the Normal Temperature Test, Section [50](#), the temperature of a surface that may be contacted by the user shall not be more than the value specified in [Table 42.1](#). If the test is conducted at a room temperature of other than 25 °C (77 °F), the results are to be corrected to that temperature.

Table 42.1
Maximum Surface Temperatures

Location	Composition of surface ^a			
	Metal		Nonmetallic	
	°C	(°F)	°C	(°F)
Handles, shoulder harness, belt harness, or knobs that are grasped for lifting, carrying, or holding	50	(122)	60	(140)
Handles or knobs that are contacted but do not involve lifting, carrying, or holding; and other surfaces subject to contact and user maintenance	60	(140)	85	(185)
Surfaces subject to casual contact but not required to be contacted to operate the device	70	(158)	95	(203)
^a A handle, knob, or similar device made of a material other than metal that is plated or clad with metal having a thickness of 0.127 mm (0.005 inch) or less is judged as a nonmetallic part.				

43 Safety Circuits and Control Circuits

43.1 Circuits that are provided to limit outputs, switch outputs, control operational functions and the like, are considered safety circuits or control circuits, if their failure to provide their intended function will result in a hazardous condition or a risk of fire, shock, or injury to the user. Circuits that are classified as safety circuits or control circuits shall be evaluated to the applicable requirements in UL 60730-1 and CSA C22.2 E60730-1, and be allowed to remain in the device to provide protection during testing. A circuit not

evaluated to UL 60730-1 or CSA C22.2 E60730-1 shall have its components subjected to single fault condition or bypassed during testing.

PERFORMANCE

44 General

44.1 For tests involving the overall device, complete samples of the power pack are required. The battery provided in the power pack shall be fully charged or fully discharged as indicated in the individual test. Unless otherwise stated, all tests are to be conducted at the rated input voltage and frequency for the power pack. For all electrical tests requiring connection to the branch circuit, the branch circuit shall be protected by a branch circuit protective device having a rating equal to the smallest rated receptacle to which the power pack could be connected. For tests where the battery is to deliver output power, the tests shall be performed with the output loaded to its maximum rating, unless otherwise specified.

44.2 Some tests may result in fire or explosion. It is therefore important that personnel be protected from the flying fragments, explosive force, sudden release of heat, chemical burns, intense light, and noise that may result from such explosions. The test area is to be well ventilated to protect personnel from possible harmful fumes or gases.

44.3 Measurements of cell voltages during the tests of lithium-ion systems shall be made using a single pole resistive-capacitive low pass filter with a cut-off frequency of 5 kHz $\pm$ 500 Hz. To determine if maximum charging voltages have been exceeded, the peak value of the voltage measured after this network shall be used. The measurement shall have a measurement tolerance within ± 1 %.

44.4 The tests are made in a draught-free location and at an ambient temperature of 20 $\pm$ 5 °C (68 $\pm$ 9 °F), unless otherwise specified. All test samples shall be stabilized to the ambient temperature prior to the test. If the temperature attained by any part is limited by a temperature sensitive device, or is influenced by temperature, the room temperature is, in case of doubt, maintained at 23 $\pm$ 2 °C (73.4 $\pm$ 3.6 °F).

44.5 Currents measured during battery charging shall be average current with the averaging period of 1 – 5 seconds.

44.6 The end of discharge voltages are 1.75 V/cell for lead acid batteries, 1.0 V/cell for valve regulated nickel batteries, 1.5 V/cell for sodium-ion batteries, and 2.5 V/cell for lithium-ion batteries, unless the cell manufacturer specifies a different voltage.

45 Power Input Test

45.1 The current input to a power pack shall not exceed 110 % of the marked current rating of the power pack, when the power pack is operated under the conditions of maximum normal load.

45.2 Maximum normal load shall consist of the maximum current draw while the power pack is operating in all possible modes. For example, this may include running an air compressor, while a light is on, and the internal battery is charging. Any load that can be operated at the same time shall be operated in order to obtain the maximum normal load.

46 Normal Charging Operation Test

46.1 Charging a lithium-ion or a sodium-ion battery under normal conditions shall not exceed the specified operating region for charging of the cell.

46.2 Compliance with [46.1](#) is checked by the following tests in [46.3](#) – [46.7](#).

46.3 The battery is charged in accordance with the charging system instructions starting with a fully discharged battery. Testing is carried out at an ambient temperature of $20 \pm 5^\circ\text{C}$ ($68 \pm 9^\circ\text{F}$) and:

- a) If the power pack is recommended to be operated at a minimum temperature lower than 4°C (39.2°F), the test is also conducted at that minimum temperature plus 0/minus 5°C (plus 0/minus 9°F); or
- b) If the appliance is recommended to be operated at a maximum temperature greater than 40°C (104°F), the test is also conducted at that maximum temperature plus 5/minus 0°C (plus 9/minus 0°F).

46.4 All individual cell voltages, temperatures and the charging current (or multiple current measurements as in the case of parallel configurations unless analysis makes this unnecessary) are monitored. Cells shall not experience conditions outside their specified operating region for charging (e.g., limits of voltage and current). Below is an example result of such analyses.

The charging current for each branch of a parallel connection would not need to be monitored, if the maximum deliverable current of the charger did not exceed the maximum charging current of a single cell.

46.5 For batteries employing series configurations, the test is to be repeated with a deliberately imbalanced battery. The imbalance is to be introduced into a fully discharged battery by charging one cell to approximately 50 % of full charge.

46.6 With reference to [46.5](#), if it can be demonstrated through testing and/or design evaluation that an imbalance less than 50 % would actually occur in normal use, then this lower imbalance may be used. The following provides examples of testing and design:

- a) An example for testing is repeated charging and discharging a battery in accordance with the manufacturer's instructions until its capacity has decreased to 80 % of the rated capacity, using the imbalance at the end of the test.
- b) Those designs that employ circuitry intended for maintaining balance between cells in the battery pack. Systems with a small number of cells in series may be shown to exhibit limited imbalance in practice, if the product ceases to operate with a battery prepared with a smaller initial imbalance.

46.7 Battery systems intended for use with power packs which may be left on, such as flashlights and fans shall additionally be tested with their battery discharged by allowing the appliance to remain "on" for at least 12 hours prior to recharging.

47 Lithium/Sodium Charging System Test

47.1 A sample of the power pack subjected to this test shall be considered to comply with this test provided none of the following has occurred:

- a) There has been no explosion during this test;
- b) No charring or burning of the medical gauze, cheesecloth or tissue paper has resulted; and
- c) The cells shall not have exceeded the upper limit charging voltage by more than 150 mV or, if they have, then the charging system shall be permanently disabled from recharging the battery.

47.2 With reference to [47.1\(b\)](#), charring is defined as a blackening of the medical gauze or cheesecloth caused by combustion. Discoloration of the medical gauze or cheesecloth caused by smoke is acceptable.

Charring or igniting of the tissue paper, cheesecloth, or medical gauze from the shorting means in not considered a failure.

47.3 With reference to [47.1\(c\)](#), to determine if recharging is disabled, the battery shall be discharged by using the power pack to approximately a 50 % charge, followed by an attempt to recharge the battery normally. There shall be no charging current after 10 minutes or after 25 % of the nominal capacity has been delivered, whichever occurs first.

47.4 A sample of the power pack is to be placed on a soft wood surface covered by two layers of tissue paper. The sample is then to be covered by one layer of untreated 100 % cotton medical gauze or cheesecloth. The power pack is to be operated as specified in the operating instructions in each of the abnormal conditions described in (a) – (d). The cumulative stress resulting from successive tests on electronic circuits or the battery is to be avoided. Additional samples may be used as necessary. As result of testing, there shall be no evidence of damage to the cell vent.

a) Components in the charging circuit are faulted as described in [47.6](#), one at a time, if the outcome of such a fault is uncertain based upon analysis. For each fault condition introduced, the state of the battery before charging is as follows:

- 1) A series configured battery is to have a deliberate imbalance. The imbalance is to be introduced into a fully discharged battery by charging one cell to approximately 50 % of full charge;
- 2) If the test of Section [46](#), Normal Charging Operation Test, is conducted with an imbalance of less than 50 %, a series configured battery is to have a deliberate imbalance as established in [47.6](#); or
- 3) A single cell or parallel only configuration battery is to be fully discharged.

b) If the test of Section [46](#), Normal Charging Operation Test, is conducted with an imbalance of less than 50 % due to the function of circuit(s), and if a single fault of any component within that circuit(s) is shown to result in the loss of that function, then a series configured battery is to be charged with a deliberate imbalance. The imbalance is to be introduced into a fully discharged battery by charging one cell to approximately 50 % of full charge.

c) For a battery with a series configuration, all cells are to be at approximately 50 % charge except for one which is shorted. The battery is to then be charged.

d) With a fully charged battery connected to the charger, a short is to be introduced to the charging system across a component or between adjacent PCB tracks at a location expected to produce the most unfavorable results to evaluate the effect of back-feed from the battery. For a charger with a cord that connects to the battery, the short is to be introduced at the point likely to produce the most adverse effects. The resistance of the short shall not exceed 10 mΩ.

47.5 During the tests of [47.4](#), each cell voltage is to be continuously monitored to determine if it has exceeded the limit conditions. Venting of the cells is to be permitted. The test is to be continued until the sample under test experiences a failure as described in [47.1](#), returns to room temperature or, if neither of these, until at least 7 hours or twice the normal charging period has elapsed, whichever is longer.

47.6 Fault conditions for components as required by [47.4](#) shall be as follows:

- a) Open-circuit at the terminal of any component;
- b) Short-circuit of capacitors, unless they are certified EMI capacitors;
- c) Short-circuit of any two terminals of an electronic component, other than a monolithic integrated circuit. This fault is not applied between the two circuits of an optocoupler;

d) Failure of triacs in the diode mode; and

e) Failure of a monolithic integrated circuit or other circuits that cannot be assessed by the fault conditions (a) – (d). All possible output signals are considered under fault conditions within the integrated circuit. Components such as thyristors and triacs are not subjected to this fault condition.

48 Capacitor Discharge Test

48.1 A power pack provided with filtering capacitors, or other primary capacitors, rated in excess of 0.10 μF and connected between one side of the line and ground, shall be subjected to the Capacitor Discharge Test.

48.2 The device is to be connected to a supply source of rated voltage at 60 Hz. The output is to be connected to a suitable load such that rated current is drawn from the output of the device. A storage oscilloscope is to be connected across the point of disconnection of the supply.

48.3 The device is to be connected to the source of supply and energized. The power is to then be removed and the resulting discharge curve for the stored charge on capacitors is to be measured and captured on the oscilloscope. The value of the stored charge shall decay to less than 37 % of its initial value within 1 second.

48.4 The test is to be repeated with all switches in all possible positions and combinations.

49 Leakage Current Test

49.1 A power pack with hazardous voltage circuits shall be tested in accordance with [49.2](#) – [49.7](#). Leakage current shall not be more than 0.5 mA.

Exception No. 1: The leakage current of a power pack provided with no AC supply shall be measured with the measurement instrument connected between the output ground terminal and any user accessible conductive parts. In addition, there shall be no risk of electric shock between the hazardous circuit (input or output) to output ground terminal and/or any user accessible conductive parts, see [5.42](#).

Exception No. 2: For power packs with AC pass through function, leakage current in a protective bonding conductor shall not be more than 3.5 mA as determined per the applicable requirements of UL 1778 / CSA C22.2 No. 107.3.

49.2 All accessible conductive parts are to be tested for leakage currents. Leakage currents from these parts are to be measured to the grounded supply conductor individually as well as collectively if simultaneously accessible, and from one part to another if simultaneously accessible. A part is considered to be accessible unless it is guarded by an enclosure that has been evaluated for protection against the risk of electric shock. Conductive parts are considered to be simultaneously accessible if they can be readily contacted by one or both hands of a person at the same time. These measurements do not apply to terminals operating at voltages that are not considered to involve a risk of electric shock. If all accessible conductive parts are bonded together and connected to the grounding conductor of the power supply cord, the leakage current can be measured between the grounding conductor and the grounded supply conductor.

49.3 If a conductive part other than metal is used for an enclosure or part of an enclosure, leakage current is to be measured using a metal foil with an area of 10 by 20 cm (4 by 8 inches) in contact with the surface. If the conductive surface has an area less than 10 by 20 cm (4 by 8 inches), the metal foil is to be the same size as the surface. The metal foil is to conform to the shape of the surface but is not to remain in place long enough to affect the temperature of the product.

49.4 A typical measurement circuit for leakage current with the ground connection open is illustrated in [Figure 49.1](#). The measurement instrument is defined in [Figure 49.2](#). The meter that is actually used for a measurement need only indicate the same numerical value for a particular measurement as would the defined instrument; it need not have all the attributes of the defined instrument. Over the frequency range 20 Hz to 1 MHz with sinusoidal currents, the performance of the instrument is to be as follows:

- The measured ratio V_1/I_1 with sinusoidal voltages is to be as close as feasible to the ratio V_1/I_1 calculated with the resistance and capacitance values of the measurement instrument shown in [Figure 49.2](#).
- The measured ratio V_3/I_1 with sinusoidal voltages is to be as close as feasible to the ratio V_3/I_1 calculated with the resistance and capacitance values of the measurement instrument shown in [Figure 49.2](#). V_3 is to be measured by the meter M in the measuring instrument. The reading of meter M in RMS volts can be converted to MIU by dividing the reading by 500 ohms and then multiplying the quotient by 1,000. The mathematic equivalent is to multiply the RMS voltage reading by 2.

Figure 49.1
Typical Leakage Current Measurement Circuit

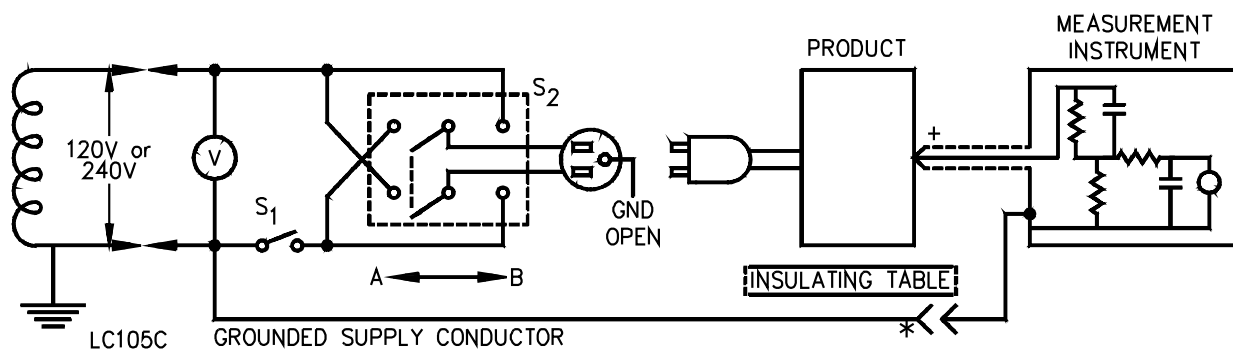
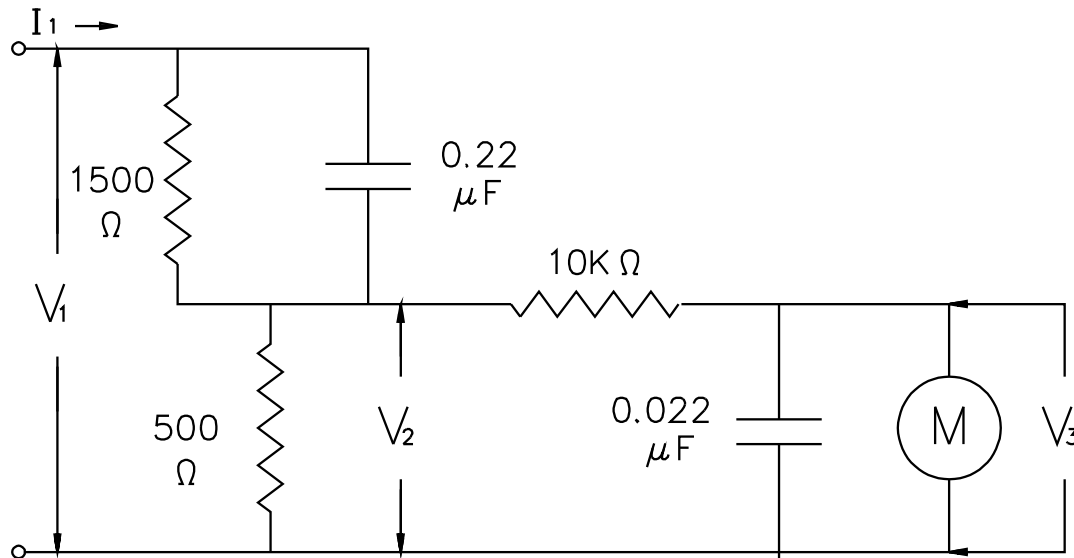


Figure 49.2
Measurement Instrument For Leakage Current



S3263B

Note – Detailed specifications and guidance for the calibration of this instrument are given in UL 101.

49.5 Unless the measurement instrument is being used to measure leakage current from one part of the power pack to another, it is to be connected between accessible parts and the supply conductor connected to ground.

49.6 A sample of a product is to be tested for leakage current starting with the as received condition – the as received condition being without prior energization, except as may occur as part of the production line testing. The supply voltage is to be adjusted to rated voltage.

49.7 The test sequence is to be as follows, with reference to the [Figure 49.1](#) measurement circuit:

- a) With switch S_1 open, the power pack is to be connected to the measurement circuit. Leakage current is to be measured using both positions of switch S_2 , and with the product switching devices in all their normal operating positions.
- b) Switch S_1 is then to be closed, energizing the product. Within 5 seconds, the leakage current is to be measured using both positions of switch S_2 and with the product switching devices in all their normal operating positions.
- c) Leakage current is to be monitored until thermal stabilization. Both positions of switch S_2 are to be used in determining this measurement. Thermal stabilization is to be obtained by operation as in the Normal Temperature Test, Section [50](#).
- d) The leakage current is also to be monitored with switch S_1 open while the product is at operating temperature and while cooling.

49.8 Normally a sample will be subjected to the entire leakage current test, as specified in [49.7](#), without interruption for other tests. With the concurrence of all those concerned, the leakage current test may be interrupted to conduct other nondestructive tests.

50 Normal Temperature Test

50.1 General

50.1.1 A power pack, when tested under the conditions of maximum normal load as described in [50.2](#) shall not reach a temperature high enough to cause a risk of fire, to damage any materials used, or exceed the temperature rises specified in [Table 50.1](#). See Section [42](#), Surface Temperatures.

50.1.2 A thermal or overload protective device shall not open the circuit during the Temperature Test.

50.1.3 All values of temperature rise in [Table 50.1](#) are based on an assumed ambient temperature of 40 °C (104 °F).

Table 50.1
Maximum Temperature Rises

Material and components	°C	(°F)
A. COMPONENTS		
1. Capacitors:		
(a) Electrolytic	40	(72)
(b) Other types ^a	65	(117)
2. Fuses ^b	65	(117)
3. Relay, solenoid, and coils (except transformers) with		
(a) Class 105(A) insulated systems		
Thermocouple method	65	(117)
Resistance method	85	(153)
(b) Class 130(B) insulation systems		
Thermocouple method	85	(153)
4. Sealing Compound ^c	40	(104)
5. Transformers		
(a) Class 105(A) insulation systems:		
Thermocouple method	55	(99)
Resistance method	60	(108)
(b) Class 120(E) insulation systems:		
Thermocouple method	70	(126)
Resistance method	75	(135)
(c) Class 130(B) insulation systems:		
Thermocouple method	80	(176)
Resistance method	85	(185)
(d) Class 155(F) insulation systems:		
Thermocouple method	110	(198)

Table 50.1 Continued on Next Page

Table 50.1 Continued

Material and components	°C	(°F)
Resistance method	115	(207)
(e) Class 180(H) insulation systems:		
Thermocouple method	125	(225)
Resistance method	135	(243)
B. CONDUCTORS		
1. Rubber- or thermoplastic insulated wires and cords ^b	35	(63)
C. ELECTRICAL INSULATION – GENERAL		
1. Fiber employed as electrical insulation	65	(117)
2. Phenolic composition employed as electrical insulation or as a part the deterioration of which could result in a risk of fire or electric shock ^b		
(a) Laminated	100	(180)
(b) Molded	125	(225)
3. Varnished-cloth insulation	60	(108)
^a A capacitor is to be judged on the basis of its marked temperature limit when it operates at a temperature rise of more than 65 °C (117 °F). ^b A fuse that has been investigated, and found to comply with requirements for use at a higher temperature is able to be used at that temperature. ^c The temperature rise limit for sealing compounds indicates the maximum rise that may be observed on a sealing compound. However, the maximum temperature of the sealing compound shall not exceed the melting point of the sealing compound regardless of overall temperature rise.		

50.1.4 For this test, the test voltage shall be the same as the rated voltage of the power pack.

50.1.5 A power pack shall be operated for a duration that is reasonable in comparison to its normal use. For example, when charging the internal battery, the duration of the charging function is complete when the battery is fully charged. See [50.2](#).

50.1.6 With reference to tests that are to be continued until thermal stabilization is obtained, thermal stabilization is considered to exist when three successive readings taken at intervals of 10 % of the previously elapsed test duration, but not less than 5 minute intervals, indicate no change in temperature.

50.1.7 Coil winding temperatures are to be measured by thermocouples or by using the change-of-resistance method.

50.1.8 Thermocouples are to consist of wires not larger than 24 AWG (0.21 mm²) and not smaller than 30 AWG (0.05 mm²). Whenever reference temperature measurements by thermocouples are necessary, thermocouples consisting of 30 AWG iron and constantan wire and a potentiometer-type instrument are to be used. The thermocouple wire is to conform with the requirements specified in the Tolerances on Initial Values of EMF versus Temperature tables in ASTM E230.

50.1.9 When using the change-of-resistance method, the windings are to be at room temperature at the start of the test, and the temperature rise of a winding is to be calculated using the formula:

$$\Delta t = \frac{R}{r} (k + t_1) - (k + t_2)$$

in which:

Δt = the temperature rise in degrees C;

R = the resistance of the coil in ohms at the end of the test;

r = the resistance of the coil in ohms at the beginning of the test;

t_1 = the initial room temperature in degrees C at the time " r " is measured

t_2 = the room temperature in degrees C at the end of the test;

k = 234.5 for copper and 225.0 for electrical conductor grade (EC) aluminum; values for the constant for other conductors are to be determined.

50.2 Maximum normal load

50.2.1 Maximum normal load is considered to be the load that approximates as closely as possible the most severe conditions of normal use. It is not a deliberate overload except as the conditions of actual use are likely to be somewhat more severe than the maximum load conditions that are recommended by the manufacturer of the power pack. A product having features not contemplated in these test procedures may be tested as necessary to meet the intent of these requirements.

50.2.2 A power pack may consist of separate components or assemblies such as lights, air compressors, internal battery, and the like. Each of these assemblies will create a load for the product. All functions that can be performed, and all assemblies that can be operated, at that same time are to be operated in order to create the maximum load. This would include turning on all lights, charging the internal battery, running the air compressor, and the like.

50.2.3 For a power pack that contains provisions for eliminating specific functions when others are in process the maximum normal load shall take this into account. In addition, the means for controlling such load balancing shall be investigated for reliability.

50.2.4 The unit shall be designed to minimize the allowable inrush current from the vehicle battery when jump starting. The unit shall be able to accept the maximum allowable inrush current from the clamp leads for at least 10 seconds with no damage to the power pack.

50.2.5 If the unit is provided with an inverter with AC output, the inverter should be able to operate if clamps are applied to external DC source.

50.2.6 A 12 volt DC output (cigarette lighter) shall be tested with 15A output to temperature stabilization, without risk of damage to unit.

50.2.7 Air compressors shall be tested under the following conditions with the internal battery fully charged:

- a) Locked-rotor;
- b) By operation with output under rated load based on the marked duty cycle provided by the manufacturer. If no marked duty cycle is provided, then the air compressor shall be tested by filling a 10-gallon air tank, with a duty cycle of 5 minutes on, 1 minute off; and
- c) Continuous running unit with end of air-fill hose sealed.

For the condition in (c), the pressure shall be monitored and utilized for application of Section [64](#), Hydrostatic Strength Test, applied to the compressor and all hoses subject to pressure.

50.2.8 For power packs that allow outputs to be powered while the pack is connected to the source of supply, the test shall be performed with all allowable outputs loaded until thermal stabilization. For packs

where the output can only be powered by the battery, the test duration shall be determined based on the batteries capability of maintaining the supply current as it discharges.

50.3 Power pack ampacity temperature test

50.3.1 When conducting the Power Pack Ampacity Test of Section 70, the Power Pack Ampacity Temperature Test is also to be conducted. The Power Pack Ampacity Test is to be conducted for a duration of 25 seconds, or whatever duration is allowed by the inherent functionality of the battery pack. If the power pack can be discharged again after a specified time interval, the discharge shall be continued until the power pack can no longer output at maximum discharge or peak discharge current. At the end of duration the output of the power pack is to be turned "off". During this time, the temperatures of the battery and all internal affected components including cables are to be recorded and shall be within the limits specified in Table 50.1. Additionally, the temperatures of the clamps are to be monitored from the start of the test and continuing through 2 minutes after the current is turned "off". At no time during this test shall the temperatures exceed those in Table 50.2:

Table 50.2
Maximum Surface Temperature Rises on Clamp

Clamp Location (portion adjacent to live-jaw)	Composition of surface			
	Metallic		Nonmetallic	
	°C	(°F)	°C	(°F)
Area between pivot and jaw-ends of clamp	45	(81)	70	(126)
Handle portion of clamp area between pivot and hand ends	25	(45)	35	(63)
NOTE – Maximum surface temperature rise of handle ends is based on UL Casualty Requirements for carrying and holding handles SAE J1494 specifies 66 °C (150 °F) max.				

51 Energy Hazard Measurement Test

51.1 When tested as specified in 51.2 – 51.5, the maximum energy measured in any output shall not exceed 240 VA, and the power input to any single transmitting coil RF conversion circuit measured shall not exceed 240 VA.

51.2 The power pack is to be fully charged before the test.

Exception No. 1: The power pack may be fully charged and supplied by one of its specified power sources if it can result in a higher output energy at the measured output or a higher input power to the transmitting coil.

Exception No. 2: The power pack may be a lower SOC state and supplied by one of its specified power sources if it can result in a higher output energy at the measured output or a higher input power to the transmitting coil.

51.3 For each of the conductive output connectors, an adjustable resistor is to be connected across the power terminals of the connector. The current through the resistor and voltage across the resistor are to be monitored using suitable meters. The resistance is to be adjusted to obtain maximum energy at the output.

51.4 For any inductive output, an adjustable simulated load is to be used which incorporates an induction receiver appropriate to the induction power transmitter in the power pack. The simulated load shall be capable of providing a load demand to the transmitting coil greater than the maximum input of 240 VA. The

input power to the transmitting coil is to be monitored while the simulated load is adjusted to result in the maximum power to the transmitting coil.

51.5 The test in [51.4](#) is to be repeated with a single fault condition (open or short) on any component which can result in a higher input power to the transmitting coil.

52 LVLE Circuit Test

52.1 A control circuit that is considered an LVLE circuit shall comply with the maximum voltage and current limits in [Table 52.1](#) when tested as specified in [52.2](#) – [52.7](#):

Table 52.1
Voltage and Current Limits for LVLE Circuits

Application	Open circuit voltage, V	Maximum current ^b
Outdoor use or temporary outdoor use	30 volts DC or 21.2 volts peak or less	8
Indoor use	30 volts DC or 42.2 volts peak or less	8
Indoor use	More than 30 but not greater than 60 volts dc	150/ V ^a

^a V is the maximum output voltage of the power pack with all load disconnected and energized in accordance with [52.2](#).

^b The current is able to exceed the specified value when the circuit includes an overcurrent protection device as described in [52.4](#) and [52.5](#).

52.2 The power pack shall be fully charged before testing, and then the maximum open circuit voltage is to be measured with all loading circuits disconnected.

Exception 1: The power pack should be fully charged and supplied by one of its specified power sources if it can result in a higher output voltage or current.

Exception 2: The power pack should be a lower SOC state and supplied by one of its specified power sources if it can result in a higher output voltage or current.

52.3 The maximum current is to be measured under any condition of loading including short circuit using a resistor that is to be continuously readjusted during the 1-minute period to maintain maximum load current.

52.4 If a fuse or circuit-protective device is used to limit the current, the fuse shall be rated or set at not more than the values specified in [Table 52.2](#). In this case, the maximum current is not required to be measured.

Table 52.2
Rating for Secondary Fuse or Circuit Protector

Circuit voltage, V rms	Maximum overcurrent protection, amperes
20 or less	5
More than 20 and not greater than 60	100/V ^a

^a V is the maximum output voltage of the power pack with all load disconnected and energized in accordance with [52.2](#).

52.5 If a fuse or circuit-protective device connected to the primary of a transformer is used to limit the secondary circuit current in accordance with [52.1](#), the fuse shall comply with [52.4](#) and additionally operate

to open the circuit in not more than the time indicated in [Table 52.3](#) when the transformer is delivering the specified secondary current.

Table 52.3
Maximum Time to Open for Overcurrent Protective Device

Maximum transformer open circuit secondary voltage (Vmax) volts	Secondary test current amperes	Maximum time for overcurrent protective device to open minutes
20 or less	10	2
20 or less	6.75	60
Over 20	200/Vmax	2
Over 20	135/Vmax	60

52.6 To determine whether a fuse or circuit protective device complies with the requirement in [Table 52.3](#), the transformer is to deliver the test current to a resistance load with the primary connected to a circuit as described in [52.2](#). During the 2-minute test, the load is to be adjusted continuously to maintain the required test current. During the 60-minute test, the load is to be adjusted once after 15 minutes of operation, and the test is to be continued without further adjustment. Where the fuse or circuit protective device is used to protect more than one secondary winding or taps, each winding or partial winding is to be tested with the remaining windings delivering rated load if applicable.

52.7 If a regulating network is used to limit the voltage or current in accordance with [52.1](#), the performance testing is to be conducted both under normal operating conditions and after any simulated single fault in the regulating network (open circuit or short circuit of component), excluding a resistor.

53 Dielectric Voltage Withstand Test

53.1 While in a well-heated condition, a power pack shall withstand for 1 minute without breakdown the application of a 60 Hz essentially sinusoidal potential of:

- a) 1000 volts plus twice the maximum rated voltage between:
 - 1) The primary circuit and dead metal parts;
 - 2) The primary and secondary circuits;
 - 3) Secondary circuits operating above 50 V and dead metal parts; and
 - 4) Secondary circuits operating above 50 V and secondary circuits operating below 50 V.
- b) 500 volts between:
 - 1) Secondary circuits operating below 50 volts and dead metal parts, and
 - 2) Secondary circuits operating below 50 volts and other secondary circuits operating below 50 V.

Exception: The ac potentials shown above may be replaced with a dc potential equal to 1.414 times the ac potential for this test.

53.2 To determine if the power pack complies with the requirements in [53.1](#), the power pack is to be tested using a 500 volt-ampere or larger capacity transformer, the output voltage of which can be varied. The applied potential is to be increased from zero until the required test level is reached, and is to be held

at that level for 1 minute. The increase in applied potential is to be at a substantially uniform rate as rapid as is consistent with correct identification of its value by a voltmeter.

54 Leakage Current Following Humidity Conditioning

54.1 A power pack with hazardous voltage circuits shall comply with the requirements for leakage current in Section 49, Leakage Current Test, following exposure for 48 hours to air having a relative humidity of $88 \pm 2\%$ at a temperature of $32 \pm 2^\circ\text{C}$ ($90 \pm 4^\circ\text{F}$).

54.2 To determine whether a product complies with the requirement in 54.1, a sample of the power pack is to be heated to a temperature just above 34°C (93°F) to reduce the likelihood of condensation of moisture during conditioning. The heated sample is to be placed in the humidity chamber and is to remain for 48 hours under the conditions specified in 54.1.

54.3 Following the conditioning, the sample is to be tested as described in the Leakage Current Test, Section 49, while either in the humidity chamber or immediately after removal of the conditioned sample from the humidity chamber. For each test condition, the maximum leakage current is to be recorded and the test is to be discontinued when the leakage current stabilizes or decreases.

55 Abnormal Operation Tests

55.1 General

55.1.1 A power pack shall not emit flame or molten metal or become a risk of fire, electric shock, or injury to persons when subjected to the tests specified in 55.2 – 55.10. Separate samples are to be used for conducting each test, unless using a sample for more than one test is agreeable to all concerned.

55.1.2 Following each test, the Dielectric Voltage Withstand Test specified in Section 53 is to be conducted.

55.1.3 With reference to 55.1.1, a risk of fire, electric shock, or injury to persons exists when:

- a) Flame, burning oil, or molten metal is emitted from the enclosure of the power pack as evidenced by ignition, glowing, or charring of the cheesecloth or tissue paper; or
- b) The insulation breaks down when tested in accordance with 55.1.2 or hazardous live parts are made accessible.

55.1.4 During these tests, the device is to be placed on a softwood surface covered with a white tissue paper and a single layer of cheesecloth is to be draped loosely over the entire enclosure. The cheesecloth is to be untreated cotton cloth running $26 - 28 \text{ m}^2/\text{kg}$ ($14 - 15$ square yards per pound), and having, for each 6.45 cm^2 (1 inch^2), a count of 32 threads in one direction and 28 in the other direction.

Exception: When it is impractical to drape cheesecloth over the entire enclosure, the cheesecloth is required to be placed only over all ventilation openings.

55.1.5 The supply circuit is to have branch circuit overcurrent protection, the size of which equals 125 % of the input current rating (20 ampere minimum), except where this value does not correspond with the standard rating of a fuse or circuit breaker, the next higher standard device rating shall be used. The rated voltage and frequency are to be used for this test.

55.1.6 Each test is to be continued until further change as a result of the test condition is reduced significantly. When an automatically reset protector functions during a test, the test is to be continued for 7 hours. When a manual reset protector functions during a test, the test is to be continued until the protector

is operated for 10 cycles using the minimum resetting time, and not faster than 10 cycles of operation per minute. The following are examples of acceptable test terminations:

- a) Opening or shorting of one or more components such as capacitors, diodes, resistors, solid state devices, printed wiring board traces, or similar devices.
- b) Opening of the intended branch circuit overcurrent protection device described in [55.1.5](#).
- c) Opening of an internal fuse.

55.1.7 During the abnormal tests specified in [55.2](#), [55.3](#), [55.9](#), and [55.10](#) for the battery, control circuits used to limit the output or input shall be evaluated in accordance with Section [43](#), Safety Circuits and Control Circuits. Fuses and other protective devices determined reliable are to remain in the device to protect the battery. A control circuit not evaluated to UL 60730-1 or CSA C22.2 E60730-1 shall have its components subjected to single fault condition or bypassed during testing.

55.2 Output connections short circuit test

55.2.1 The power pack is to be tested as described in [55.2.2](#). The device shall comply with the requirements in [55.1.1](#).

55.2.2 With reference to [55.2.1](#), fuses and other protective devices provided as part of the power pack are to remain in the circuit. The output connections of the power pack are to be short-circuited at the connector with a resistance not exceeding 10mΩ and the power pack connected to a source of supply adjusted to its rated voltage. The test is to be continued until the internal protection opens or constant temperatures are obtained. When an automatically reset protector or a manually reset protector ends the test, the test is to be continued as indicated in [55.1.6](#).

55.2.3 If the power pack is provided with an output cable (e.g. booster cable), then the short shall also occur at the end of the cable in order to judge that cables ability to withstand the fault for the duration.

55.2.4 If the power pack is provided with an on/off switch which is used to control an output, the test shall be conducted with that switch opened and closed.

55.3 Reverse polarity of booster cables

55.3.1 A power pack with a booster function is to be tested in accordance with [55.3.2](#). The power pack shall comply with the requirements in [55.1.1](#).

55.3.2 With reference to [55.3.1](#), fuses and other protective devices provided as part of the power pack are to remain in the device. The output booster cable connections of the power pack are to be reverse connected to a fully charged battery. The test is to be continued until the internal protection opens or constant temperatures are obtained. When an automatically reset protector or a manually reset protector ends the test, the test is to be continued as indicated in [55.1.6](#).

55.3.3 If the power pack is provided with a switch that controls the booster cable output function, the test shall be conducted with that switch opened and closed.

55.4 Component faults

55.4.1 A component, such as a capacitor, diode, solid state device, resistor, or similar component, connected in the power pack are to be short- or open-circuited, any two terminals one at a time, during any condition of operation including start-up. This test is not required where circuit analysis indicates that no

other component or portion of the circuit is overloaded. At the end of each component fault condition, the power pack shall comply with the requirement in [55.1.1](#).

55.5 Relay and solenoid burnout

55.5.1 An electromagnetic relay or a solenoid having an open coil construction is to be tested by blocking the armature or the plunger in the de-energized position. The test shall be continued until constant temperatures are obtained or for 7 hours maximum. The test results shall comply with [55.1.1](#).

55.6 Printed-wiring board abnormal test

55.6.1 To comply with Exception No. 1 to [31.1](#), a printed-wiring board is to be tested as described in [55.6.2](#) – [55.6.5](#).

55.6.2 During this test, if a printed-wiring board trace opens, the gap is to be electrically shorted and the test continued until ultimate results occur. This procedure applies to each occurrence. If the circuit is interrupted by the opening of a component other than described in [55.6.3](#), the test is to be repeated two more times using new components as necessary.

55.6.3 Operation of an overcurrent protective device other than the branch circuit overcurrent protective device is allowed.

55.6.4 A sample of the power pack employing a printed-wiring board is to be connected to its rated supply voltage. A foil trace is to be short-circuited to each of its adjacent traces that do not have the spacing specified in [Table 31.1](#), one at a time.

55.6.5 The test is to be continued for 1 hour or until one of the conditions described in [55.1.3](#) occurs. However, if at the end of 1 hour no condition described in [55.1.3](#) occurs, but indications are that such a condition may eventually occur, the test is to be continued until ultimate results are obtained (usually 7 hours).

55.7 Disconnected fan test

55.7.1 A device having forced ventilation is to be operated with the fan disconnected. For a device having more than one fan, the test is to be conducted with each fan disconnected, one at a time, or with two or more fans disconnected, if they are controlled or powered by the same connection. If part of the circuitry senses a disconnected fan and shuts down the unit, the circuitry shall be bypassed to allow operation with the fans disconnected or the circuitry shall be evaluated for suitability of this protective function.

55.8 Blocked ventilation test

55.8.1 A power pack provided with ventilation openings for allowing air flow to cool internal components shall be subjected to this test. The vents are to be blocked, and the unit is to be operated at maximum normal load, as defined in [50.2](#). The test is to continue until constant temperatures are obtained or for 7 hours maximum. During the test, the power pack shall comply with [55.1.1](#).

55.9 Overcharging test

55.9.1 Power packs shall withstand abusive overcharging without risk of fire or explosion when tested in accordance with [55.9.2](#).

55.9.2 The power pack is to be placed on a soft wood surface covered by two layers of tissue paper and one layer of untreated 100 % cotton medical gauze. The battery is to be charged at a rate as described in (a) or (b) for 1.25 hours:

- a) For a universal charging port, the maximum charging voltage and 3 times the maximum charging current for the power pack, with or without a single fault condition in the charging protection circuit; or
- b) For a non-universal charging port, the maximum voltage and current output that is available from an external charger or internal charging circuit provided for the battery with or without a single fault condition in the charging protection circuit.

There shall be no explosion and no charring or burning of the gauze or tissue paper. Charring is defined as a blackening of the gauze caused by combustion. Discoloration of the gauze caused by smoke is acceptable. Venting of the cells is acceptable.

55.10 Battery reverse polarity test

55.10.1 For power packs with removable internal, detachable, or external batteries that are not keyed or otherwise prevented from being connected incorrectly, the test of [55.10.2](#) is to be carried out and the pack shall comply with [55.1.1](#).

55.10.2 During this test, fuses and other protective devices provided as part of the power pack are to remain in the device. Two samples are to be tested. The first sample is to be tested by connecting a fully discharged battery in reverse polarity and then attempting to charge the battery with a normal charging cycle. The second sample is to be tested by connecting a fully charged battery in reverse polarity and then attempting to use the pack under the conditions of maximum normal load. In both cases, the test is to be continued until the internal protection opens or constant temperatures are obtained. When an automatically reset protector or a manually reset protector ends the test, the test is to be continued as indicated in [55.1.6](#).

55.11 Thermal propagation test

55.11.1 A lithium-ion power pack with a sum of rated energy larger than 1 kWh, accounting for all its internal, detachable and external batteries, shall be tested in accordance with Single Cell Failure Design Tolerance Test in UL 1973.

55.11.2 During the test, detachable or external batteries or attachments are to be installed or detached as intended to represent the worst-case condition. Batteries are to be fully charged before testing.

55.11.3 The single cell selected for failure during the test shall represent the worst-case scenario, taking into account its location in the internal battery, as well as in the detachable or external battery, if applicable. If it is difficult to determine the worst-case scenario, the Thermal Propagation Test shall be repeated to failure by a single cell at a different location.

Exception: Individual detachable or external batteries that have been subjected to an equivalent Thermal Propagation Test do not need to have one of its cells failed in the test.

55.11.4 As a result of testing, there shall be no fire propagating from the DUT or explosion of the DUT.

56 Vibration Test

56.1 Cells shall not catch fire nor explode during or immediately following the Vibration Test.

56.2 The Vibration Test shall consist of vibration for 4 hours at a frequency of 22 cycles per second with a displacement of 6.4 mm (1/4 inch) in a vertical plane. The unit is to be mounted as intended during the test.

57 Ground Continuity

57.1 The resistance between the point of connection of the equipment-grounding means at or within the product and any other point in the grounding circuit of the product shall not be more than 0.1Ω as determined by an ohmmeter or other equivalent means.

57.2 If unacceptable results are observed with an ohmmeter, a low voltage current source can be employed. With the low voltage current source, an alternating current of 25 A (for 15 A maximum rated products) from a power supply of 12 V or less is to be passed for one minute minimum. The current shall be passed from the point of connection of the equipment grounding means to the metal part in the grounding circuit under test. The resulting drop in potential is to be measured between the two points.

58 Overload Tests

58.1 General

58.1.1 The overload tests in [58.2](#) – [58.4](#) are required as applicable.

58.2 Overload of switches and controls test

58.2.1 Unless known to be evaluated for the application, a switch or other control device shall perform acceptably when subjected to an overload test consisting of 50 cycles of operation making and breaking the load. For this test, the load may be the actual load applied in the product, or if this is unknown, the load shall be based on the rating of the switch or control device. As a result of this overload test there shall be no electrical or mechanical breakdown of the device, undue burning or pitting of the contacts, or opening of the fuse in the grounding connections.

58.2.2 To determine whether a switch or other control device complies with the requirements in [58.2.1](#), the power pack is to be connected to a supply circuit of rated frequency and the maximum rated voltage (or a fully charged power pack). The load for the device under test is to be the same as that which it is intended to control in regular service. During the test, accessible dead metal parts of the power pack are to be connected to ground through a 3-ampere plug fuse. The device is to be operated at a rate of not more than 10 cycles per minute, except that a faster rate of operation may be employed if agreeable to those concerned.

58.3 Overload of protection devices

58.3.1 An overload protective device is to be cycled through 200 operations of make and break action. The device is to be connected to the voltage present in the end product and the current load is to be adjusted to be equal to the maximum load seen by the device in the end product. The device is to be cycled at 10 cycles per minute, or faster if all involved agree. At the end of the 200 cycles, the device shall be able to operate as intended.

58.4 Overload of interlocks

58.4.1 The contact of the safety interlock switch or relay is to be subjected to an overload test consisting of 50 cycles of operation at the rate of 6 to 10 cycles per minute, making and breaking 150 % of the current imposed in the application, except that where a contact switches a motor load, the test is to be conducted with the rotor of the motor in a locked condition. After the test, the switch or relay shall still be functional.

59 Strain Relief Test

59.1 Direct-pull strain relief test

59.1.1 The strain relief means provided for a supply cord or a multi-conductor output cord where the interconnection of outputs exceeds LVLE shall withstand the force described in [59.1.3](#) for 1 minute with the connections within the power pack disconnected. At the point of disconnection of the conductors, there shall be no movement of the cord as to indicate that stress on the connections would have resulted from the pull force. There shall be no breakage of the cord or deformation of its anchoring surface.

59.1.2 The output wiring of a unit other than one having a multi-conductor cord as specified in [59.1.1](#) shall withstand the force described in [59.1.4](#) applied for 1 minute. The results are considered acceptable if, with the output wiring connected internally, movement of the cord does not result in:

- a) A reduction of spacings to primary or dead metal parts;
- b) Damage to the transformer or enclosure; or
- c) Interruption of the output wiring.

59.1.3 A 15.9-kg (35-pound) weight is to be suspended from the cord and supported by the power pack so that the strain relief means is stressed from any angle the construction of the power pack permits.

59.1.4 For units employing a flexible output cord, or a multi-conductor cord where the interconnection of outputs does not exceed LVLE, a 20 pounds-force (89 N) is to be applied to the cord and supported by the unit so that the strain relief means is stressed from the most severe angle that the construction of the unit permits. For units employing output wiring consisting of separate leads, a 10 pounds-force (44 N) is to be applied to each lead.

59.2 Push-back strain relief test

59.2.1 To determine compliance with [12.2.2.3](#), a product shall be tested in accordance with [59.2.2](#) without occurrence of any of the conditions specified in [12.2.2.3](#) (a) – (c).

59.2.2 The supply cord or lead is to be held 25.4 mm (1 inch) from the point where the cord or lead emerges from the product and is then to be pushed back into the product. When a removable bushing which extends further than 25.4 mm (1 inch) is present, it is to be removed prior to the test. When the bushing is an integral part of the cord, the test is to be carried out by holding the bushing. The cord or lead is to be pushed back into the product in 25.4-mm (1-inch) increments until the cord buckles or the force to push the cord into the product exceed 26.7 N (6 pounds-force). The supply cord or lead within the product is to be manipulated to determine compliance with [12.2.2.3](#).

60 Strength of Enclosure Tests

60.1 General

60.1.1 A power pack shall be tested as described in [60.2](#) and [60.3](#). Following these tests, the power pack shall:

- a) Not permit a probe, as illustrated in [Figure 7.1](#), to contact an uninsulated live part that may involve a risk of electric shock;
- b) Comply with the Dielectric Voltage Withstand Test, Section [53](#), with the potential applied between live parts and accessible dead metal parts; and

- c) Not have a permanent distortion of a metallic enclosure that reduces spacings below the minimum values, as specified in Spacings, Section [31](#).

60.2 Impact test

60.2.1 Two samples of the power pack are to be subjected to the Impact Test. The first sample is to be tested in the as-received condition. The second sample is to be conditioned in a cold chamber at 0 °C (32 °F), or the minimum operating temperature specified by the manufacturer if lower than 0 °C (32 °F), for 4 hours. The sample is to be removed from the cold chamber and immediately subjected to the impact described in [60.2.2](#). During handling of the sample, gloves are to be worn to minimize heat transfer.

60.2.2 An enclosure, guard, or cover is to be subjected to an impact of 6.78 N-m (5 foot-pounds) on any surface that may be subjected to an impact during intended use. The impact is to be produced by dropping a steel sphere, 50.8 mm (2 inches) in diameter and weighing approximately 0.535 kg (1.18 pounds), from a height of 1.30 m (51 inches). For surfaces other than the top, the steel sphere is to be suspended by a cord and allowed to swing as a pendulum dropping through a vertical distance of 1.30 m (51 inches). The impact shall be applied one time to each surface that is exposed to a blow during any condition of intended use.

60.3 Drop test

60.3.1 Two samples of the power pack are to be subjected to the Drop Test. The first sample is to be tested in the as-received condition. The second sample is to be conditioned in a cold chamber at 0 °C (32 °F), or the minimum operating temperature specified by the manufacturer if lower than 0 °C (32 °F), for 4 hours. The sample is to be removed from the cold chamber and immediately subjected to the impact described in [60.3.2](#). During handling of the sample, gloves are to be worn to minimize heat transfer. Following the test, the power pack shall be in accordance with [60.1.1](#) (a) – (c).

60.3.2 The sample of a portable power pack weighing equal to or less than 18 kg (39.7 pounds) is to be dropped three times from a height of 0.9 m (3 feet) to strike a concrete surface in the positions most likely to produce adverse results. The sample of a movable power pack weighing more than 18 kg (39.7 pounds) and to be equipped with handle(s) is to be dropped one time from a height of 0.9 m (3 feet) to strike a concrete surface in the position most likely to produce adverse results.

Exception: The sample of a movable power pack weighing more than 25 kg (55.1 pounds) and to be equipped with wheel(s) for ease of movement is to be dropped one time from a height of 0.5 m (1.6 feet) to strike a concrete surface in the intended use position to simulate a sudden drop from vehicles.

61 Mold Stress Test

61.1 One sample, consisting of the complete equipment or the complete enclosure, is to be subjected to this test.

61.2 The sample is to be placed in an air circulating oven at a temperature equal to 10 degrees higher than the maximum temperature observed on the enclosure during the temperature test, but not less than 80 °C (176 °F) or 70 °C (158 °F) if marked in accordance with [75.22](#). The sample is to be conditioned in the oven for 7 hours.

61.3 After the conditioning, the sample shall not show any signs of distortion, deterioration, shrinkage, warping, or softening that would allow access to hazardous live parts.

62 Strength of Handles Test

62.1 A handle, shoulder harness, belt harness, or other supporting means used to lift or carry a power pack shall withstand a force of four times the weight of the power pack without breakage of the handle, its securing means, or that portion of the enclosure to which the handle is attached.

62.2 To determine whether a power pack complies with the requirements in [62.1](#), a force equal to four times the weight of the power pack is to be uniformly applied over a 76-mm (3-inch) width at the center of the handle, the shoulder harness, or belt harness, without clamping. The load is to be started at zero and is to be gradually increased so that the test value will be attained in 5 to 10 seconds and maintained at that value for 1 minute. If more than one handle, shoulder harness, or belt harness is provided on a power pack and the power pack cannot be carried by one handle or the shoulder or belt alone, the force is to be distributed between the handles, shoulder harness, or belt harness. The distribution of forces is to be determined by measuring the percentage of the weight of the power pack sustained by each handle, shoulder harness, or belt harness with the power pack in the normal carrying position. If a power pack is furnished with more than one handle, shoulder harness, or belt and can be carried by one handle, shoulder harness, or belt harness only, each handle, shoulder harness, or belt harness shall withstand the total force.

63 Stability Test

63.1 Under conditions of normal use, a power pack shall not become physically unstable to the degree that it poses a risk of injury to persons.

63.2 A power pack shall not tip over but shall return to its normal at rest position when:

- a) Tipped through an angle of 10° from an at rest position on a horizontal surface; or
- b) Placed on an inclined plane inclined at an angle of 10° from the horizontal.

63.3 A power pack is not to be energized during this test. The test is to be conducted under conditions most likely to cause the power pack to overturn.

63.4 With reference to the requirements in [63.2](#), for a power pack that is constructed so that while being tipped through an angle of 10°, a part or surface of the power pack not normally in contact with the horizontal supporting surface touches the supporting surface before the power pack has been tipped through an angle of 10°, the tipping is to be continued until the surface or plane of the surface of the power pack originally in contact with the horizontal supporting surface is at an angle of 10° from the horizontal supporting surface.

64 Hydrostatic Strength Test

64.1 A hydrostatic strength test is to be conducted by filling the pressure confining portion of the sample with water so as to exclude all air, connecting the sample to a hydraulic pump, gradually increasing the pressure to the specified test value, and holding it for a period of 1 minute. As a result of the test, the pressure confining portion of the sample shall withstand without rupture a test pressure of five times the maximum pressure developed by the system, or five times the rated pressure of the power pack's air compressor, whichever is greater.

64.2 Prior to the test, parts molded of polymeric material are to be conditioned in an air circulating oven for 7 hours at a temperature of 70 °C (158 °F) or 10 °C (18 °F) higher than the maximum temperature measured on the part under normal load, whichever is greater. The samples are to be removed from the oven and allowed to cool to room temperature prior to the test.

65 Rain Test

65.1 The following Rain Test shall be performed on all power packs that are designated outdoor use or temporary outdoor use.

65.2 Enclosures shall have no wetting of a hazardous voltage or hazardous energy live part nor have entrance of water above the lowest hazardous voltage or hazardous energy live part inside the enclosure at the conclusion of this test. Additionally, the power pack shall comply with the Leakage Current Test, Section [49](#), at the conclusion of this test.

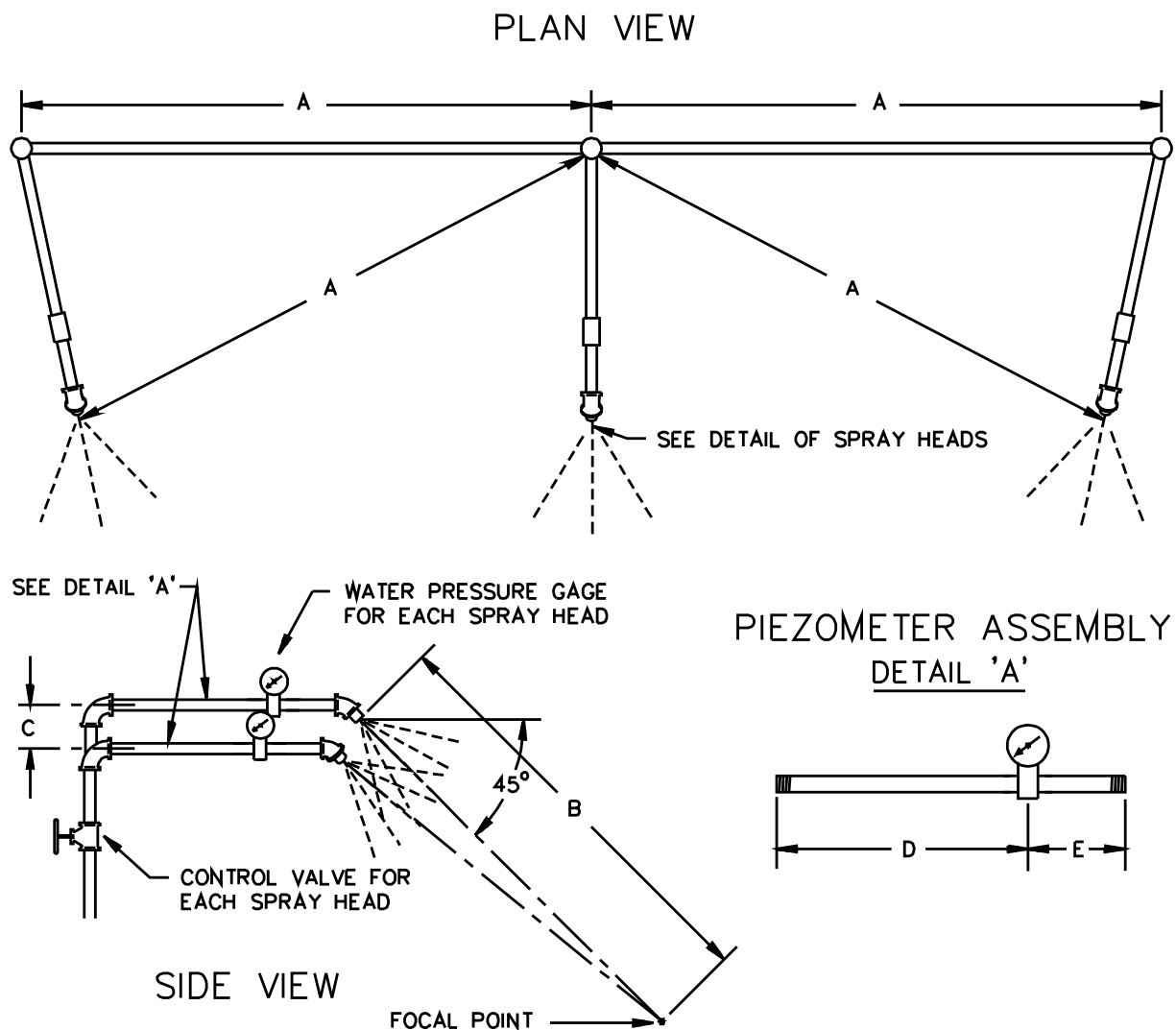
65.3 The product may be marked “rainproof” per [74.1](#) if it complies with [65.2](#). The product may be marked “raintight” per [74.1](#) if no water enters the enclosure of the power pack as specified in [65.6](#).

NOTE: In addition to electric shock hazard, wetting of a hazardous energy live circuit of a power pack may result in a fire hazard of a battery.

65.4 The rain test apparatus is to consist of three spray heads mounted in a water supply pipe rack as illustrated in [Figure 65.1](#). The spray heads are to be constructed in accordance with the details illustrated in [Figure 65.2](#). The water pressure for all tests is to be maintained at 34.5 kPa (5 psig) at each of the spray heads. The product is to be brought into the focal area of the three spray heads in the position intended during use. The spray is to be directed at a 45° angle from the vertical toward the product. The total exposure is to be for 1 hour.

65.5 If fans or other means for ventilation are provided, which could affect the ingress of water, the test shall be conducted with the ventilation means both on and off unless it is evident that one of the modes of operation will produce the more onerous result.

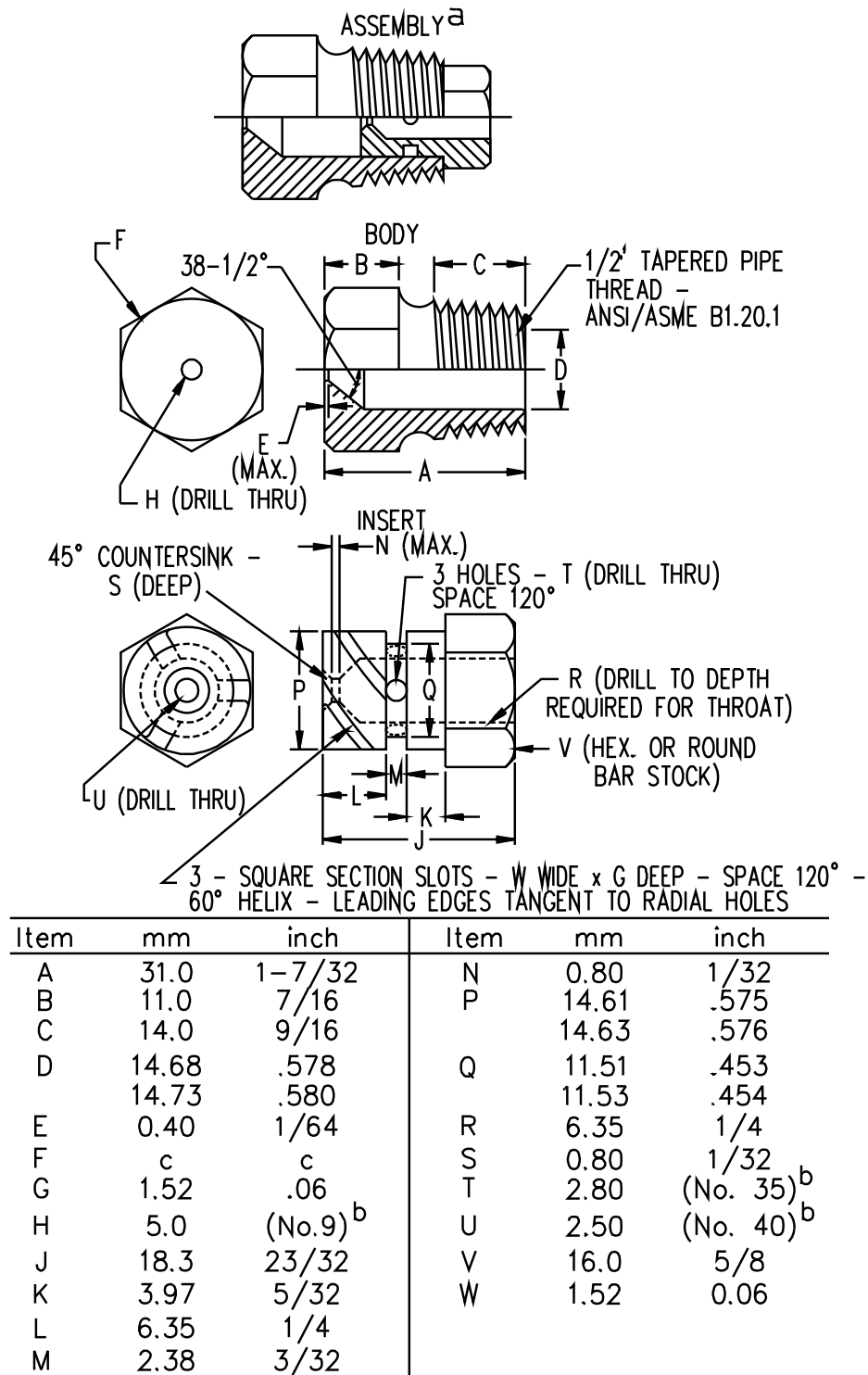
Figure 65.1
Rain Test Spray Head Piping



Item	mm	inch
A	710	28
B	1400	55
C	55	2-1/4
D	230	9
E	75	3

RT101F

Figure 65.2
Rain Test Spray Head



^a Nylon Rain-Test Spray Heads are available from Underwriters Laboratories

^b ANSI B94.11M Drill Size

^c Optional - To serve as a wrench grip.

RT100F

65.6 After the exposure, the outside of the enclosure is to be dried with a towel. A power pack with hazardous voltage circuits shall be subjected to and comply with the Leakage Current Test, Section 49. Finally, the enclosure is to be opened. The product is to be inspected for the presence of water.

66 Tests on Insulating Materials

66.1 As required by 31.11, insulating material shall be subjected to this test and shall withstand without electrical breakdown the potential indicated in 66.2.

Exception No. 1: The insulating material need not be subjected to this test if it is generic material noted in Table 66.1. See 66.3.

Exception No. 2: An insulating system consisting of N multiple layers of any thickness need not be tested in accordance with 66.2 if all possibilities of N-1 [N minus 1] layers withstand double the test potential defined in the Dielectric Voltage Withstand Test, Section 53, and applied as described in 66.2. "N" must be a minimum of 2 layers.

66.2 The insulating material is to be placed between two opposing electrodes. The electrodes are to be cylindrical brass or stainless steel rods 1/4 inch (6.4 mm) in diameter with edges rounded to a 0.8 mm (1/32 inch) radius. The upper moveable electrode is to weigh 50 ±2 grams (1.76 ±0.07 ounces) to exert sufficient pressure on the specimen to provide good electrical contact. The test potential shall be as indicated in the Dielectric Voltage Withstand Test, Section 53, and is to be maintained for 1 second.

66.3 With reference to Exception No. 1 to 66.1, insulation may be of a generic material type specified in Table 66.1 where the layer(s) of each generic material is of a minimum thickness such that all layers collectively are greater than or equal to the minimum thickness required (T):

$$T \leq A_1(EF_1) \pm A_2(EF_2) \pm A_3(EF_3)$$

In which:

A_1 , A_2 , and A_3 denote the total thickness of each generic material type, inches (mm);

EF_1 , EF_2 , and EF_3 denote the equivalency factor specified in Table 66.1 for the generic material type corresponding to A_1 , A_2 , A_3 ;

T is the thickness requirement for electrical grade paper.

Table 66.1
Equivalency Factors for Insulation Materials

Generic material	Equivalency factor (EF)
Electrical grade paper, fiber, or pressboard	1
Impregnated rag paper	1.3
Acetate sheet	1.5
Polyvinyl chloride (PVC)	1.3
Silicone rubber (SIR)	0.5
Impregnated glass or acetate cloth	1.2
Polyester	b

Table 66.1 Continued on Next Page

Table 66.1 Continued

Generic material	Equivalency factor (EF)
Polyethylene terephthalate (PETP)	b
Fluorinated ethylene propylene (FEP)	3
Polytetrafluoroethylene (PTFE)	3
Aramid paper	c
Polyamide (PI)	6
Mica ^a	4.7
NOTE: See 66.3 .	
^a EF applies if not subject to mechanical damage.	
^b To determine equivalence to 0.71 mm (0.028 inch) thick electrical grade paper, EF=4; to determine equivalence to 0.33 mm (0.013 inch) thick electrical grade paper, EF=2.	
^c To determine equivalence to 0.71 mm (0.028 inch) thick electrical grade paper, EF=0.3; to determine equivalence to 0.33 mm (0.013 inch) thick electrical grade paper, EF=1.5.	

67 Accelerated Aging of Gaskets, Sealing Compounds, and Adhesives Test

67.1 The requirements in [67.2](#) – [67.6](#) apply to gaskets and sealing compounds employed to make an enclosure raintight or rainproof as determined by the Rain Test, Section [65](#). The requirements in [67.7](#) apply to an adhesive used to secure a gasket.

67.2 Neoprene and rubber gasket materials shall have physical properties as specified in [Table 67.1](#) before and after accelerated aging under the conditions specified in [Table 67.2](#).

Exception: Foamed materials as specified in [67.3](#) need not have the physical properties specified in [Table 67.1](#).

67.3 Foamed neoprene and foamed rubber gasket materials shall be subjected to accelerated aging under the conditions specified in [Table 67.2](#). The material shall not harden or otherwise deteriorate to a degree that affects its sealing properties.

67.4 Thermoplastic gasket materials shall be subjected to accelerated aging under the conditions specified in [Table 67.2](#). A thermoplastic material shall not deform or melt, or otherwise deteriorate to a degree that will affect its sealing properties. A solid polyvinyl chloride material shall have physical properties as specified in [Table 67.1](#) before and after the accelerated aging.

Table 67.1
Physical Properties for Gaskets

Property	Neoprene or rubber compound		Polyvinyl chloride material	
	Before conditioning	After conditioning	Before conditioning	After conditioning
Recovery – Maximum set when 25.4 mm (1 inch) gage marks are stretched to 63.5 mm (2-1/2 inches), held for 2 minutes, and measured 2 minutes after release	6.4 mm (1/4 inch)	–	Not Specified	
Elongation – Minimum increase in distance between 25.4 mm (1 inch) gage marks at break	250 %	85 %	250 %	75 %
	[25.4 – 88.9 mm (1 to 3-1/2 inches) of original]		[25.4 – 88.9 mm (1 to 3-1/2 inches) of original]	
Tensile Strength – Minimum force at breaking point	5860 kPa (850 psi)	75 % of original	8273 kPa (1200 psi)	90 % of original

67.5 Tensile strength and elongation are to be determined using the test methods and apparatus described in ASTM D412.

67.6 A sealing compound is to be applied to the surface it is intended to seal. For a temperature rise not exceeding 35 °C (63 °F), a representative sample of the surface with the sealing compound applied is to be conditioned for 7 days in an air oven at 87.0 ±1.0 °C (189.0 ±1.8 °F). The sealing compound shall not melt, become brittle, or otherwise deteriorate to a degree that will affect its sealing properties as determined by comparing the aged sample to the unaged sample.

67.7 For a gasket secured by an adhesive and subject to a temperature rise not exceeding 35 °C (63 °F), a sample of the gasket secured to the mounting surface by the adhesive is to be exposed for 72 hours to each of the following conditions:

- a) A temperature of 100.0 ±1.0 °C (212.0 ±1.8 °F);
- b) Immersion in distilled water at a temperature of 23.0 ±1.0 °C (73.0 ±1.8 °F); and
- c) A temperature of minus 10.0 °C (14.0 ±1.8 °F).

The force required to peel the gasket from its mounting surface after exposure shall not be less than 50 % of the value determined using an unconditioned sample, but not less than 0.04 kg/mm (2 pounds per inch) of gasket width.

67.8 The temperature rises specified in this Section correspond to the maximum temperature rise measured on the gasket during the Temperature Test. A material other than those specified in this Section shall be non-absorptive and it, and all materials having a higher temperature rise, shall provide the resistance to aging and temperatures for the application.

Table 67.2
Accelerated aging conditions

Measured temperature rise			
°C	(°F)	Rubber and neoprene	Thermoplastic
35	(63)	Air oven aging for 70 hours at 100 °C (212 °F)	168 hours in an air circulating oven at 87.0 ±1.0 °C (188.6 ±1.8 °F)
50	(90)	Air oven aging for 168 hours at 100 °C (212 °F)	240 hours in an air circulating oven at 100.0 ±1.0 °C (212 ±1.8 °F)
55	(99)	168 hours in an air circulating oven at 113.0 ±1.0 °C (235.4 ±1.8 °F)	168 hours in an air circulating oven at 113.0 ±1.0 °C (235.4 ±1.8 °F)
65	(117)	240 hours in an air circulating oven at 121.0 ±1.0 °C (249.8 ±1.8 °F)	168 hours at 121.0 ±1.8 °C (249.8 ±1.8 °F) or 60 days at 97.0 ±1.0 °C (206.6 ±1.8 °F) in an air circulating oven
80	(144)	168 hours in an air circulating oven at 136.0 ±1.0 °C (276.8 ±1.8 °F)	168 hours in an air circulating oven at 136.0 ±1.0 °C (276.8 ±1.8 °F)

68 Metallic Coating Thickness Test

68.1 As required by [11.15](#), the method for determining the thickness of a protective coating is described in [68.2](#) – [68.9](#).

68.2 The solution used for the test is to be made from distilled water and is to contain 200 grams per liter (26.7 ounces per gallon) of American Chemical Society (ACS) reagent grade chromic acid (CrO₃) and 50 grams per liter (6.7 ounces per gallon) of ACS reagent grade concentrated sulfuric acid (H₂SO₄). The latter

is equivalent to 27 milliliters per liter of ACS reagent grade concentrated sulfuric acid, specific gravity 1.84, containing 96 % of H_2SO_4 .

68.3 The test solution is to be contained in a glass vessel such as a separatory funnel with the outlet equipped with a stopcock and a capillary tube having an inside bore of 0.64 mm (0.025 inch) and a length of 139.7 mm (5.5 inches). The lower end of the capillary tube is to be tapered to form a tip, the drops from which are about 0.025 milliliters each. To preserve an effectively constant level, a small glass tube is to be inserted in the top of the funnel through a rubber stopper and its position is to be adjusted so that, when the stopcock is open, the rate of dropping is 100 ± 5 drops per minute. If desired, an additional stopcock may be used in place of the glass tube to control the rate of dropping.

68.4 The sample and the test solution are to be kept in the test room long enough to acquire the temperature of the room, which is to be noted and recorded. The test is to be conducted at an ambient temperature of $21 - 32^\circ\text{C}$ ($70 - 90^\circ\text{F}$).

68.5 Each sample is to be cleaned before testing. All grease, lacquer, paint, and other nonmetallic coatings are to be removed using solvents. Samples are then to be thoroughly rinsed in water and dried with clean cheesecloth. Care is to be exercised to avoid contact of the cleaned surface with the hands or any foreign material.

68.6 The sample to be tested is to be supported 17.8 – 25.4 mm (0.7 – 1 inch) below the orifice, so that the drops of solution strike the point to be tested and run off quickly. The surface to be tested is to be inclined about 45° from horizontal.

68.7 The stopcock is to be opened and the time in seconds until the dropping solution dissolves the protective metal coating exposing the base metal is to be measured. The end point is the first appearance of the base metal recognizable by a change in color.

68.8 Each sample of the test lot is to be subjected to the test at three or more points, excluding cut, stenciled, and threaded surfaces, on the inside surface and at an equal number of points on the outside surface, at places where the metal coating may be expected to be the thinnest. On an enclosure made from pre-coated sheets, the external corners that are subjected to the greatest deformation are likely to have thin coatings.

68.9 To calculate the thickness of the coating being tested, select from [Table 68.1](#) the thickness factor for the temperature at which the test was conducted and multiply by the time in seconds required to expose base metal as described in [68.7](#).

Table 68.1
Coating thickness factors

Temperature		Thickness factors, 0.0003 mm (0.00001 inch) per second	
$^\circ\text{C}$	$(^\circ\text{F})$	Cadmium platings	Zinc platings
21.1	(70)	1.331	0.980
21.7	(71)	1.340	0.990
22.2	(72)	1.352	1.000
22.8	(73)	1.362	1.010
23.3	(74)	1.372	1.015
23.9	(75)	1.383	1.025

Table 68.1 Continued on Next Page

Table 68.1 Continued

Temperature		Thickness factors, 0.0003 mm (0.00001 inch) per second	
°C	(°F)	Cadmium platings	Zinc platings
24.4	(76)	1.395	1.033
25.0	(77)	1.405	1.042
25.6	(78)	1.416	1.050
26.1	(79)	1.427	1.060
26.7	(80)	1.438	1.070
27.2	(81)	1.450	1.080
27.8	(82)	1.460	1.085
28.3	(83)	1.470	1.095
28.9	(84)	1.480	1.100
29.4	(85)	1.490	1.110
30.0	(86)	1.501	1.120
30.6	(87)	1.513	1.130
31.1	(88)	1.524	1.141
31.7	(89)	1.534	1.150
32.2	(90)	1.546	1.160

68.10 Zinc metallic coating thickness may also be measured as follows:

- a) By x-ray in accordance with ASTM B568;
- b) By eddy-current in accordance with ASTM E376; or
- c) By weight of coating in accordance with ASTM A90.

69 Permanency of Wrapped Hang Tag Marking

69.1 Following the test described in [69.2](#) – [69.5](#), the marking shall be considered permanently affixed when there is no indication of the results shown in (a) – (d). Manipulation of the hang tag, such as straightening by hand, is allowed when determining compliance with these requirements.

- a) Tearing at any point for more than 1.6 mm (1/16 inch);
- b) Movement of the tag more than 12.7 mm (1/2 inch) along the length of the cable;
- c) Shrinkage, wrinkling, cracking, or other deformation that renders the marking illegible; or
- d) Visible curling or loosening around the edges of a tag with an adhesive back.

69.2 Nine samples of a hang tag are to be tested as described in [69.5](#). Each sample is to consist of a length of cable to which the hang tag has been attached in the intended manner. If the hang tag is secured by an adhesive, the test is to be conducted no sooner than 24 hours after application of the hang tag. Three samples are to be tested as received; the additional samples are to be conditioned as described in [69.3](#) and [69.4](#) prior to testing.

69.3 Three samples are to be conditioned for 240 hours in an air-circulating oven maintained at a uniform temperature of 87.0 ± 1.0 °C (188.6 ± 1.8 °F). Following removal from the oven, the samples are to remain at a temperature of 23.0 ± 2.0 °C (73.4 ± 3.6 °F) and a relative humidity of 50 ± 5 % for 30 minutes before testing.

69.4 Three additional samples are to be conditioned for 72 hours at a temperature of 32.0 ± 2.0 °C (89.6 ± 3.6 °F) and a relative humidity of 85 ± 5 %. The samples are to be tested within 1 minute after exposure.

69.5 Each sample cable with attached hang tag is to be tightly suspended and clamped at each end in a vertical plane with the attachment plug or fitting pointing upward. A 22.2 N (5-lb) force is to be applied for 1 minute at the uppermost corner of the tag farthest from the cable and within 6.4 mm (1/4 inch) of the vertical edge of the hang tag. The force is to be applied vertically downward in a direction parallel to the major axis of the cable.

70 Power Pack Ampacity Test

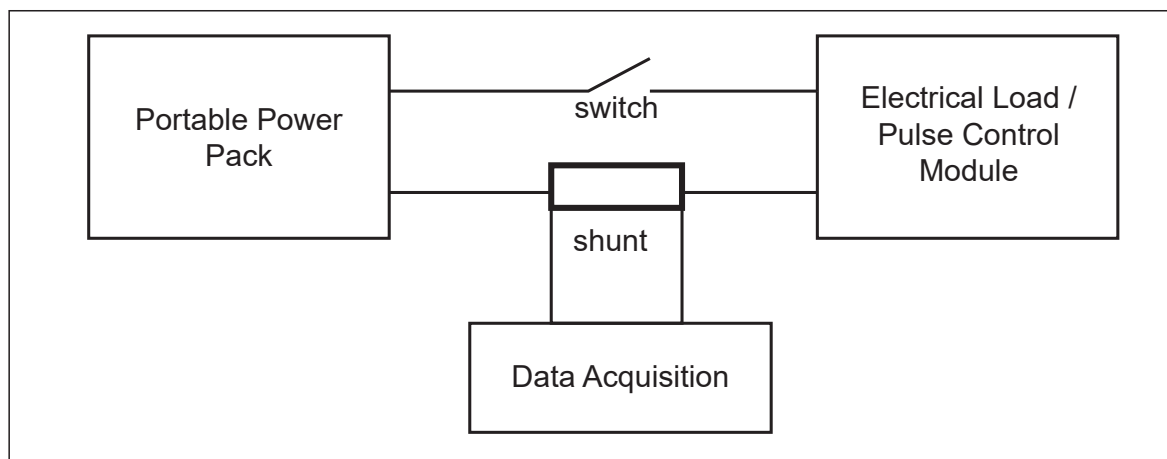
70.1 A power pack with a booster function shall be subjected to the Power Pack Ampacity Test. Lead acid or valve regulated nickel batteries are to be loaded for 10 seconds at the maximum discharge current or power specified by the manufacturer to the discharge cutoff voltage of 7.2 volts. For lithium-ion or sodium-ion powered units with protection circuitry, the Power Pack Ampacity Test is to be conducted for 5 seconds and the battery packs may be loaded at the maximum discharge current or power specified by the manufacturer to the lowest discharge cutoff voltage that the battery pack will allow. The ampacity rating along with the time duration of the ampacity output shall be included in the pack's rating if less than 10 seconds for lead acid or valve regulated nickel powered unit, or less than 5 seconds for lithium-ion or sodium-ion powered units. The ampacity is to be recorded for the duration of the test. At the end of the test, the ampacity is to be recorded and this shall be the power pack's ampacity rating.

70.2 In addition to the maximum discharge current rating, if a peak discharge current rating has been declared by manufacturer for the power pack with boost function, the Power Pack Ampacity Test shall be repeated with the power pack loaded at the output (clamps) at the peak discharge current for a duration of 100 ms or a duration specified by the manufacturer, whichever is longer. The declared peak current rating along with the time duration of the peak current output shall be included in the power pack's rating. The current is to be recorded for the duration of the test. At the end of the test, the current is to be recorded and this shall be the power pack's peak current rating.

70.3 [Figure 70.1](#) gives an example of the test setup for measuring the power pack booster current rating. A typical test setup consists of the following:

- a) Electrical Load or Pulse Control Module capable to initiate a manual jump start for the time required in [70.1](#) and [70.2](#);
- b) Switch interface between the load / pulse control module and the device under test rated for the expected maximum / peak discharge current;
- c) Shunt interface between the load / pulse control module and the device under test, sized for the expected maximum / peak discharge current;
- d) Data acquisition, Oscilloscope, or other means for capturing and recording the maximum / peak discharge current during test.

Figure 70.1
Examples of Test Setup for Power Pack Ampacity Test



su5140

70.4 The Power Pack Ampacity Temperature Test, see [50.3](#), is to be conducted at the same time as the Power Pack Ampacity Test, but it is continued longer to obtain the maximum temperature rise than to obtain the ampacity rating.

71 Back Feed Test

71.1 Under both normal operation and single fault conditions, and 1 second after the power pack is working at stored energy mode, the voltage present at the input connections shall not exceed 60 V dc, or the available current shall not exceed 3.5 mA.

71.2 A power pack with a fully charged battery is to be subjected to the Back Feed Test. The input connection is to be connected to suitable meters to measure the voltage and current available at the input connection point when the power pack is disconnected from the power source. Following these measurements, component faults are to be simulated, one at a time, and the measurements are to be repeated. All component faults required to determine compliance are to be performed. In all cases, the values measured for voltage and current shall not exceed the limits in [71.1](#).

72 Cold Bend Test

72.1 As directed in [13.2.2.4](#), samples of the cable shall be subjected to the Cold Bend Test described in [72.2](#). There shall be no evidence of cracks on the inside or outside surfaces after the test has been completed.

72.2 The specimens and the appropriate mandrel, as specified in [Table 72.1](#), are to be cooled for a period of 4 hours at the manufacturer's recommended lower ambient temperature. After this cooling

period, the specimens are to be wound onto the mandrel for six complete turns. The winding is to be done at a rate of about 3 seconds per turn, and successive turns are to be in contact with one another. The test is to be performed in the cold chamber where space and mounting means are available in the chamber. Where this is not practical, it is appropriate to remove a specimen and a mandrel from the test chamber and perform the test outside the chamber. In either case, the winding is to be completed within 30 seconds of the time that the cold chamber is opened.

Table 72.1
Mandrel Diameters

Size of conductor AWG	Mandrel diameter	
	mm	(inches)
10	14.30	(0.563)
8	17.48	(0.688)
6	31.75	(1.250)
4	34.93	(1.375)
2	39.70	(1.563)
1	68.28	(2.688)
—	—	—
1/0	73.03	(2.875)
2/0	76.20	(3.000)
3/0	82.55	(3.250)
4/0	88.90	(3.500)

73 Clamp Tests

73.1 General

73.1.1 The applicable clamp tests shall be performed as described in [13.2.3.4](#), [13.2.3.5](#), and [13.2.3.6](#).

73.2 Cold drop test

73.2.1 Following exposure to this test, there shall not be significant deterioration of physical properties of the integrally coated insulation as determined by a visual examination for the presence of cracks, peeling, deformation, eroding, excessive wear, or other imperfections of the insulating material that result in exposing the surface of the metal clamp.

73.2.2 Three samples of the insulated clamp are to be subjected to a low-temperature exposure for one hour consisting of a conditioning temperature based on its lower ambient rating as indicated in [13.2.3](#). The samples are then to be dropped 1500 mm (5 feet) onto a concrete surface. Following this exposure, the samples are to be subjected to the Dielectric Voltage-Withstand Test in [73.3](#).

73.3 Dielectric voltage-withstand test

73.3.1 Samples of the insulated clamp are to be subjected to this test with no indication of dielectric breakdown.

73.3.2 A 500 volt, 60 Hz potential is to be applied between:

- a) The connector and foil wrapped around the handle of the clamp;

- b) The connector and the assembly rivet; and
- c) The connector and the clamp spring.

Exception No. 1: The test potential of 500 Vac specified may be replaced by a dc voltage of 707 V.

Exception No. 2: Assembly rivets or coil springs recessed against contact are not required to be tested.

73.4 Secureness test

73.4.1 The connection between cable and clamp shall be intact with no broken insulation after the test of [73.4.2](#).

73.4.2 The clamp is to be secured in a manner that allows the cable to hang freely in a downward position. A weight equal to 2 times the weight of the power pack is to be applied to the cable and supported by the cable for 10 minutes. At the end of the 10 minutes, the connection and insulation is to be observed.

MARKINGS

Advisory Note: In Canada, there are two official languages, English and French. Annex A provides translations in French of the English safety markings specified in this Standard. Markings required by this Standard may have to be provided in other languages to conform with the language requirements of the country where the product is to be used.

74 General

74.1 A product shall be legibly and permanently marked with the following:

- a) The manufacturer's name, trade name, trademark, or other descriptive marking by which the organization responsible for the product may be identified;
- b) A distinctive model number or the equivalent;
- c) The electrical rating of voltage, current or power, frequency if applicable, for all input and output;
- d) For portable power pack with booster function, the maximum ampacity and peak ampacity if applicable, and the discharge/cranking duration per [70.1](#) and [70.2](#) if applicable;
- e) The operating temperature range as specified by the manufacturer;
- f) The date or other dating period of manufacture not exceeding any three consecutive months; and
- g) The "Rainproof" or "Raintight" rating for an enclosure for temporary outdoor use or outdoor use products.

Exception: The date of manufacture may be abbreviated; or may be in a nationally accepted conventional code or in a code affirmed by the manufacturer, provided that the code:

- a) Does not repeat in less than 10 years; and*
- b) Does not require reference to the production records of the manufacturer to determine when the product was manufactured.*

74.2 When a manufacturer produces or assembles a power pack at more than one factory, each finished power pack shall have a distinctive marking, to identify it as the product of a particular factory.

74.3 With reference to [74.1](#) and the Cautionary Markings, Section [75](#), the markings shall be legible and permanent. These markings shall be:

- a) Molded, die-stamped, paint stenciled, stamped or etched metal that is permanently secured;
- b) Pressure sensitive label complying with the applicable requirements for indoor and outdoor use labels in UL 969 and CSA C22.2 No. 0.15, at the indicated temperature rating, up to 70 °C (158 °F) on the specified surface;
- c) A hang tag applied to a hose or cord, provided the hang tag complies with the requirements in the Permanency of Wrapped Hang Tag Marking Test, Section [69](#); or
- d) A hang tag applied to a hose or cord, complying with the applicable requirements for indoor and outdoor use flag labels or flag tags in UL 969A at the indicated temperature rating on the specified cable. The hang tag shall be either of the following forms:
 - 1) A flag tag having a hole to permit securement to the cord or hose by a plastic strap or equivalent means. The strap shall not be removable without cutting.
 - 2) A flag-type with an adhesive back. The tag is to be wrapped tightly once around and is to adhere to the supply cord. The ends of the tag are to adhere to each other and project as a flag. The required markings are to be positioned on the projecting flag portion of the tag.

74.4 Power packs intended to be used or marketed for use in repair facilities, the power pack shall be marked as follows: "Suitable for use in a repair facility."

75 Cautionary Markings

75.1 A product having a hidden or unexpected risk of injury to persons shall be marked to inform the user of the risk.

75.2 A cautionary marking shall be permanent and legible, in accordance with [74.3](#), and shall be located in a visible location on a permanent part of the product.

75.3 A cautionary marking intended to instruct the operator shall be legible and visible from the position normally assumed by the operator when starting the product or from the position normally assumed or the specific operation involved. Other such markings for servicing or making settings and adjustments shall be legible and visible to the individual when such work is being done.

75.4 A marking intended to inform the user of a risk of injury to persons shall be prefixed by a signal word "CAUTION," "WARNING," or "DANGER." The marking shall be in letters not less than 2.4 mm (3/32 inch) high. The signal word shall be more prominent than any other required marking on the product.

75.5 The literature accompanying a package containing a basic product and attachments intended to be marketed as a complete unit shall indicate what attachments are intended for use with the basic product if use of such attachments may expose the user to a risk of injury.

75.6 An attachment that is packaged and marketed separately from the basic product and recommended by the manufacturer for use on the basic product shall be marked to identify the basic product with which it is intended to be used. The identification shall appear in at least one of the following locations:

- a) On the attachment;
- b) On the package housing the attachment;
- c) In the instruction book for the basic product; or

d) In information furnished with the attachment.

75.7 For components or assemblies used in power packs, such as air compressors, internal batteries, and the like, specific cautionary markings may exist on the enclosure or body of these components or assemblies and which may not be visible to the user once installed in the power pack. If the risk is not removed or reduced due to the installation of this component or assembly in the power pack, then the cautionary marking appearing on the component or assembly shall be repeated on the power pack.

75.8 A compartment involving a risk of electric shock and housing no user serviceable parts shall be marked where readily visible during any approach to servicing. The marking shall consist of the word "CAUTION" and the following or equivalent: "Risk of Electric Shock. Do not remove cover. No user serviceable parts inside. Refer servicing to qualified service personnel."

75.9 The locations and type designations of user serviceable components shall be marked on the power pack where the marking will be readily visible during servicing of the components, unless replacement of the component by a different type cannot result in a risk of fire, electric shock, or injury to persons.

75.10 There shall be a replacement marking adjacent to a user accessible fuse or fuseholder if the fuse is used to reduce the risk of fire or electric shock. The marking shall be located where it will be readily visible during replacement of the fuse, and shall consist of the word "CAUTION" and the following or equivalent: "For continued protection against risk of fire, replace only with same type and rating of fuse." If the fuse is soldered in place and is perceptible during user servicing, the marking shall, in addition to the above, include the following wording or the equivalent: "Refer replacement to qualified service personnel".

75.11 A power pack that contains a lead-acid or valve regulated nickel battery that is to be charged shall be marked with the word "WARNING" and the following or equivalent:

- a) "Do not overcharge the internal battery. See Instruction Manual."
- b) "Do not smoke, strike a match, or cause a spark in the vicinity of the power pack."
- c) "Only charge the internal battery in a well ventilated area."

75.12 A power pack provided with an AC plug with an earthing conductor shall be plainly marked with the word "CAUTION" and the following or the equivalent: "Risk of Electric Shock. Connect only to properly grounded outlets."

75.13 A power pack shall be plainly marked with the word "CAUTION" and the following or the equivalent: "Risk of Injury To Persons. Do not use this product if the power cord or the battery cables are damaged in any way."

75.14 For polarity identification, one clamp shall be black and the other clamp shall be a contrasting color, other than white at each end of the battery booster cable set. In addition, the contrasting color clamps shall be marked "+" and "POS" or "POSITIVE", the black clamps shall be marked "-" and "NEG" or "NEGATIVE."

75.15 Instructions pertaining to the proper use and connection of the battery cables shall appear on the power pack and as written instructions in the Operating Instructions provided with the product.

75.16 Power packs which incorporate an interlock device complying with Section 28, Interlocks, shall be marked where readily visible by any personnel attempting to defeat the interlock. The marking shall include the word "WARNING" and the following or the equivalent: "Risk of Electric Shock. This cover is provided with an interlock. Do not defeat its purpose or attempt to service without removing cover completely." The general location of the interlock shall also be indicated.

75.17 With reference to [6.2](#), the power pack shall be marked with the word “WARNING” and the following or the equivalent: “Risk of Explosion. This equipment has arcing or sparking parts which should not be exposed to flammable vapors. This equipment should be located at least 457 mm (18 inches) above the floor when used in a repair facility.”

75.18 With reference to [6.2](#), a power pack that is not intended to be used in a repair facility shall be marked with the word “CAUTION” and the following or the equivalent: “This device is not intended for use in a commercial repair facility.”

75.19 With reference to [6.3](#), temporary outdoor use power packs shall be marked with the word “CAUTION” and the following or the equivalent: “This device is intended to be stored indoors when not in use. This device shall not be stored or left outdoors when not in use.”

75.20 With reference to [6.3](#), temporary outdoor use power packs shall be marked with the word “CAUTION” and the following or the equivalent: “This device is intended for temporary use outdoors and reasonable care should be exercised when using this device in wet conditions.”

75.21 With reference to [6.3](#), indoor use only devices shall be marked “DANGER” and the following or the equivalent: “This device is intended to be used indoors only. Do not use outdoors.”

75.22 With reference to [6.4](#) and [7.3.3](#), products with an enclosure RTI rating less than 80 °C shall be marked with the word “WARNING” and the following or the equivalent: “Risk of Electric Shock and Risk of Fire. This device is not to be stored in a vehicle.

75.23 A portable power pack with rated capacity larger than 1 kWh shall be marked “WARNING” and the following or equivalent: “Risk of Fire. This device contains energy higher than 1 kWh. Do not use in a sleeping room or any habitable space of a dwelling unit.”

75.24 With reference to [13.3.2\(a\)\(2\)](#), a portable power pack shall be marked with the word “CAUTION” and the following or the equivalent: “Risk of Electric Shock. Never use portable power pack to supply power tools to cut or access live parts or live wirings.”

INSTRUCTIONS

76 General

76.1 A product shall be provided with legible installation instructions, operation instructions, and instructions pertaining to a risk of fire, electric shock, or injury to persons associated with the use of the product. Also, user maintenance instructions and moving and storage instructions associated with the use of the product by the end user shall be included.

76.2 The instructions mentioned in [76.1](#) shall be:

- a) In separate manuals; or
- b) Combined in one or more manuals when the instructions pertaining to a risk of fire, electrical shock, or injury to persons are separated in format and emphasized to distinguish them from the rest of the text.

76.3 An illustration is allowed with a required instruction to clarify the intent but shall not replace the written instruction.

76.4 The following items shall be entirely in upper case letters or shall be emphasized to distinguish them from the rest of the text:

- a) The headings for the installation, operation, user maintenance, and moving and storage instructions;
- b) The heading for the instructions pertaining to a risk of fire, electric shock, or injury to persons; and
- c) The opening and closing statements of the instructions specified in [77.3](#) – “IMPORTANT SAFETY INSTRUCTIONS” and “SAVE THESE INSTRUCTIONS,” or the equivalent.

76.5 Unless otherwise indicated, the text of the instructions in [77.3](#) and [77.4](#) shall be in the words specified or words that are equivalent, clear, and understandable. Substitution of the signal word “DANGER” for “WARNING” is allowed when the risk associated with the product is such that a situation exists which, if not avoided, will result in death or serious injury.

77 Instructions Pertaining to Risk of Fire, Electric Shock, or Injury to Persons

77.1 Instructions pertaining to a risk of fire, electric shock, or injury to persons shall warn the user of reasonably foreseeable risks and state the precautions to be taken to reduce such risks. Such instructions shall be preceded by the heading “INSTRUCTIONS PERTAINING TO RISK OF FIRE, ELECTRIC SHOCK, OR INJURY TO PERSONS” or the equivalent.

77.2 Numbering of the items in the list in [77.3](#) and including other instructions pertaining to a risk of fire, electric shock, or injury to persons that the manufacturer determines to be necessary and that do not conflict with the intent of the instructions are acceptable.

77.3 The instructions pertaining to a risk of fire, electric shock, or injury to persons shall include those items in the following list that are applicable to the product. The statement “IMPORTANT SAFETY INSTRUCTIONS” or the equivalent shall precede the list, and the statement “SAVE THESE INSTRUCTIONS” or the equivalent shall either precede or follow the list. The word “WARNING” shall be entirely in upper case letters or shall be emphasized to distinguish it from the rest of the text.

IMPORTANT SAFETY INSTRUCTIONS

WARNING – When using this product, basic precautions should always be followed, including the following:

- a) Read all the instructions before using the product.
- b) To reduce the risk of injury, close supervision is necessary when the product is used near children.
- c) Do not put fingers or hands into the product.
- d) Use of an attachment not recommended or sold by power pack manufacturer may result in a risk of fire, electric shock, or injury to persons.
- e) To reduce risk of damage to the electric plug and cord, pull the plug rather than the cord when disconnecting the power pack.
- f) Do not use a battery pack or appliance that is damaged or modified. Damaged or modified batteries may exhibit unpredictable behavior resulting in fire, explosion or risk of injury.
- g) Do not operate the power pack with a damaged cord or plug, or a damaged output cable.
- h) Do not disassemble the power pack, take it to a qualified service person when service or repair is required. Incorrect reassembly may result in a risk of fire or electric shock.

i) To reduce the risk of electric shock, unplug the power pack from the outlet before attempting any instructed servicing.

j) WARNING – RISK OF EXPLOSIVE GASES.

1) WORKING IN VICINITY OF A LEAD ACID/VALVE REGULATED NICKEL BATTERY IS DANGEROUS. LEAD-ACID/VALVE REGULATED NICKEL BATTERIES GENERATE EXPLOSIVE GASES DURING NORMAL BATTERY OPERATION. FOR THIS REASON, IT IS OF THE UTMOST IMPORTANCE THAT YOU FOLLOW THE INSTRUCTIONS EACH TIME YOU USE (THE BOOSTER FUNCTION OF) THE POWER PACK.

2) To reduce risk of battery explosion, follow these instructions and those published by battery manufacturer and manufacturer of any equipment you intend to use in vicinity of the battery. Review cautionary marking on these products and on engine.

NOTE: Item (j) applies to a power pack with a lead-acid or valve regulated nickel battery. However, a power pack with a booster function should include this instruction since the power pack will boost a lead-acid battery in the vehicle.

k) PERSONAL PRECAUTIONS

1) Consider having someone close enough by to come to your aid when you work near a lead-acid or valve regulated nickel battery.

2) Have plenty of fresh water and soap nearby in case battery acid contacts skin, clothing, or eyes.

3) Wear complete eye protection and clothing protection. Avoid touching eyes while working near battery.

4) If battery acid contacts skin or clothing, wash immediately with soap and water. If acid enters eye, immediately flood eye with running cold water for at least 10 minutes and get medical attention immediately.

5) NEVER smoke or allow a spark or flame in vicinity of battery or engine.

6) Be extra cautious to reduce risk of dropping a metal tool onto battery. It might spark or short-circuit battery or other electrical part that may cause explosion.

7) Remove personal metal items such as rings, bracelets, necklaces, and watches when working with a lead-acid or valve regulated nickel battery. A lead-acid or valve regulated nickel battery can produce a short-circuit current high enough to weld a ring or the like to metal, causing a severe burn.

l) When charging the internal battery, work in a well ventilated area and do not restrict ventilation in any way.

NOTE: Item (l) applies to a power pack with a lead-acid or valve regulated nickel battery.

m) Under abusive conditions, liquid may be ejected from the battery; avoid contact. If contact accidentally occurs, flush with water. If liquid contacts eyes, additionally seek medical help. Liquid ejected from the battery may cause irritation or burns.

n) Do not expose a power pack to fire or excessive temperature. Exposure to fire or temperature above 130 °C may cause explosion. The temperature of 130 °C can be replaced by the temperature of 265 °F.

NOTE: The temperature of 130 °C applies to a common lithium-ion battery, the value can be adjusted for a special lithium-ion battery and battery with other chemistry according to the instructions of the manufacturer.

o) Have servicing performed by a qualified repair person using only identical replacement parts. This will ensure that the safety of the product is maintained.

p) Attach output cables to a battery and chassis as indicated below. Never allow the output clamps to touch one another.

1) Instructions shall include step by step directions for the proper use of the booster function detailing the correct steps to connect and disconnect the booster cables to the battery.

2) Each step shall be a different numbered item.

3) For a booster cable that has the potential of arcing when connecting or disconnecting the battery terminal, the following or equivalent warning shall be added:

“Always attach the booster output negative jaw to chassis away from battery terminals area after the positive terminal has been connected. Always disconnect the booster output negative jaw from chassis before disconnecting the positive terminal.”

NOTE: Item (p) applies to a power pack with booster cables.

q) Risk of Electric Shock. Never use power pack to supply power tools to cut or access to live parts or live wirings, or materials that may contain live parts or live wirings inside, such as building walls, etc.

NOTE: Item (q) applies to a portable power pack with construction in accordance with [13.3.2\(a\)](#).

r) For a portable power pack provided with a quick release system for a shoulder harness as required in Section [41](#), instructions regarding the operation of the quick release system.

SAVE THESE INSTRUCTIONS

77.4 The instructions pertaining to a risk of fire, electric shock, or injury to persons, or the installation instructions shall include the following items if applicable. If the following instructions are included in the installation instructions, a reference to these instructions shall be included in the list mentioned in [77.3](#) as a separate item. The headings and the word “WARNING” shall be entirely in upper case letters or shall be emphasized to distinguish it from the rest of the text.

GROUNDING INSTRUCTIONS

This product must be grounded. If it should malfunction or breakdown, grounding provides a path of least resistance for electric current to reduce the risk of electric shock. This product is equipped with a cord having an equipment grounding conductor and a grounding plug. The plug must be plugged into an outlet that is properly installed and grounded in accordance with all local codes ordinances.

WARNING – Improper connection of the equipment grounding conductor is able to result in a risk of electric shock. Check with a qualified electrician if you are in doubt as to whether the product is properly grounded. Do not modify the plug provided with the product – if it will not fit the outlet, have a proper outlet installed by a qualified electrician.

78 Installation Instructions

78.1 Installation instructions shall contain all the information needed to install the product for use as intended, and shall be preceded by the heading “INSTALLATION INSTRUCTIONS” or the equivalent.

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78.2 The installation instructions shall contain complete instructions concerning the proper selection of the cord set if a cord set is not provided with the power pack.

Exception: The instruction requirement in 78.2 may appear in the Instructions Pertaining To Risk of Fire, Electric Shock, or Injury to Persons, Section 77, instead of the Installation Instructions.

79 Operating Instructions

79.1 Operating instructions shall contain all the information needed to operate the product as intended, and shall be preceded by the heading "OPERATING INSTRUCTIONS" or the equivalent.

79.2 Instructions in relation to operating that appear in the instructions pertaining to a risk of fire, electric shock, or injury to persons, are not required to be repeated here; but a reference to those instructions shall be included here.

79.3 With reference to 6.2, the Operating Instructions for a power pack shall contain statements informing the user not to place the power pack on the floor, or at a height less than 457 mm (18 inches) above the floor, during use in a repair facility.

79.4 The instruction manual shall contain the following information:

a) Instructions regarding battery charging, temperature limits for appliance and battery use and storage, and the recommended temperature range for charging.

b) A warning shall be provided against modifying or attempting to repair the power pack except as indicated in the instructions for use and care.

79.5 Temporary outdoor use equipment shall be provided with instructions stating the following:

a) The device shall be stored indoors and protected from the elements;

b) The unit shall not be charged outdoors;

c) When in use, steps should be taken to reduce the exposure to rain, sleet, snow, and the like.

79.6 Indoor use only power packs shall be provided with instructions stating that the device may only be used indoors and that no outdoor use is allowed under any conditions.

79.7 With reference to 7.3.3, products that are provided with a nonmetallic enclosure with an RTI rating less than 80 °C (176 °F) shall be provided with instructions stating that the pack is not to be stored or kept in a vehicle.

80 User Maintenance Instructions

80.1 Instructions for user maintenance shall include explicit instructions for all cleaning and servicing that are intended to be performed by the user, and shall be preceded by the heading "USER MAINTENANCE INSTRUCTIONS" or the equivalent.

81 Moving and Storage Instructions

81.1 If moving or storage of the product is able to result in damage to the product that could result in a risk of fire, electric shock, or injury to persons during subsequent use, the instructions shall describe the proper moving and storage procedure, and shall be preceded by the heading "MOVING AND STORAGE INSTRUCTIONS" or the equivalent.

Annex A – Safety Marking Translations (Normative for Canada and Informative for the US)

This Annex includes the markings required to be translated and suggested French translations. For Canada, this Annex is a normative (mandatory) part of this Standard. For the US, this Annex is an informative (non-mandatory) part of the Standard.

Reference	English	French
75.8	CAUTION: Risk of Electric Shock. Do not remove cover. No user serviceable parts inside. Refer servicing to qualified service personnel.	MISE EN GARDE : Risque de décharge électrique. Ne pas enlever le couvercle. Aucune des pièces à l'intérieur ne peut être réparée par l'utilisateur. L'entretien courant doit être effectué par un personnel d'entretien qualifié.
75.10	CAUTION: For continued protection against risk of fire, replace only with same type and rating of fuse.	MISE EN GARDE : Pour une protection continue contre le risque d'incendie, remplacer seulement par des fusibles du même type et calibre.
75.10	Refer replacement to qualified service personnel.	Le remplacement doit être effectué par un personnel d'entretien qualifié.
75.11	WARNING	AVERTISSEMENT
75.11	Do not overcharge the internal battery. See Instruction Manual.	Ne pas surcharger la batterie interne. Consulter le manuel d'utilisation.
75.11	Do not smoke, strike a match, or cause a spark in the vicinity of the power pack.	Il ne faut pas fumer, allumer une allumette ou produire des étincelles à proximité du bloc d'alimentation.
75.11	Only charge the internal battery in a well ventilated area.	Charger la batterie uniquement dans un endroit bien aéré.
75.12	CAUTION: Risk of Electric Shock. Connect only to properly grounded outlets.	MISE EN GARDE : Risque de décharge électrique. Brancher l'unité uniquement à des prises de courant correctement mises à la masse.
75.13	CAUTION: Risk of Injury To Persons. Do not use this product if the power cord or the battery cables are damaged in any way.	MISE EN GARDE : Risque de blessure aux personnes. Ne pas utiliser ce produit si le cordon d'alimentation ou les câbles de batterie sont endommagés de quelque façon.
75.16	WARNING: Risk of Electric Shock. This cover is provided with an interlock. Do not defeat its purpose or attempt to service without removing cover completely.	AVERTISSEMENT : Risque de décharge électrique. Le couvercle est muni d'un interverrouillage. Il ne faut pas mettre en échec ce dispositif ou tenter de procéder à un entretien sans retirer le couvercle complètement.
75.17	WARNING: Risk of Explosion. This equipment has arcing or sparking parts which should not be exposed to flammable vapors. This equipment should be located at least 457 mm (18 inches) above the floor when used in a repair facility.	AVERTISSEMENT : Risque d'explosion. Cet équipement comporte des pièces susceptibles de produire des arcs ou des étincelles qui ne doivent pas être exposées à des vapeurs inflammables. Cet équipement doit être situé à au moins 457 mm (18 po) au-dessus du sol lorsqu'il est utilisé dans un atelier de réparation.
75.18	CAUTION: This device is not intended for use in a commercial repair facility.	MISE EN GARDE : Le dispositif n'est pas destiné à être utilisé dans un atelier de réparation commercial.
75.19	CAUTION: This device is intended to be stored indoors when not in use. This device shall not be stored or left outdoors when not in use.	MISE EN GARDE : Le dispositif est destiné à être rangé à l'intérieur lorsqu'il n'est pas utilisé. Le dispositif ne doit pas être rangé ou laissé à l'extérieur lorsqu'il n'est pas utilisé.
75.20	CAUTION: This device is intended for temporary use outdoors and reasonable care should be exercised when using this device in wet conditions.	MISE EN GARDE : Le dispositif est destiné à une utilisation à l'extérieur temporaire et il faut faire preuve d'une prudence raisonnable lorsqu'il est utilisé dans des conditions humides.
75.21	DANGER: This device is intended to be used indoors only. Do not use outdoors.	DANGER : Le dispositif est destiné à être utilisé à l'intérieur seulement. Ne pas l'utiliser à l'extérieur.
75.22	WARNING: Risk of Electric Shock and Risk of Fire. This device is not to be stored in a vehicle.	AVERTISSEMENT : Risques de décharge électrique et d'incendie. Le dispositif ne doit pas être rangé dans un véhicule.

Reference	English	French
75.23	WARNING: Risk of Fire. This device contains energy higher than 1 kWh. Do not use in a sleeping room or any habitable space of a dwelling unit.	AVERTISSEMENT : Risque de d'incendie. Cet appareil contient une énergie supérieure à 1 kWh. Ne pas utiliser dans une chambre à coucher ou tout espace habitable d'une unité d'habitation.
75.24	CAUTION: Risk of Electric Shock. Never use portable power pack to supply power tools to cut or access live parts or live wirings.	MISE EN GARDE : Risque de décharge électrique. N'utilisez jamais de bloc d'alimentation portable pour fournir des outils électriques pour couper ou accéder à des pièces sous tension ou à des câblages en direct.



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